



# Visa Smart Debit/Credit Acquirer Device Validation Toolkit User Guide

Version 6.0

June 2010





## Contents

|                                                    |     |
|----------------------------------------------------|-----|
| 1. Disclaimer.....                                 | 1   |
| 2. Introduction.....                               | 3   |
| 3. Overview .....                                  | 5   |
| 3.1. Objective .....                               | 5   |
| 3.2. Audience .....                                | 5   |
| 3.3. Structure.....                                | 6   |
| 3.4. Components.....                               | 7   |
| 3.5. Usage .....                                   | 7   |
| 3.6. Scope .....                                   | 10  |
| 3.7. Future Enhancements.....                      | 11  |
| 3.8. Related Documents .....                       | 11  |
| 3.9. Summary of Changes.....                       | 11  |
| 4. Test Cases .....                                | 17  |
| 4.1. Pre-requisites .....                          | 17  |
| 4.2. Instructions .....                            | 19  |
| 4.3. Test Case Summary.....                        | 23  |
| 5. Test Cases .....                                | 27  |
| 6. Test Card Profiles .....                        | 65  |
| 6.1. Baseline Card.....                            | 66  |
| 6.2. Test Card 1.....                              | 74  |
| 6.3. Test Card 2 (Previously Test Card 21) .....   | 76  |
| 6.4. Test Card 3.....                              | 78  |
| 6.5. Test Card 4.....                              | 82  |
| 6.6. Test Card 5 (Previously Test Card 22) .....   | 83  |
| 6.7. Test Card 6.....                              | 89  |
| 6.8. Test Card 7 (Previously Test Card 23) .....   | 93  |
| 6.9. Test Card 8 (Previously Test Card 26) .....   | 94  |
| 6.10. Test Card 9 (Previously Test Card 30) .....  | 95  |
| 6.11. Test Card 10 (Previously Test Card 31) ..... | 96  |
| 6.12. Test Card 11 .....                           | 97  |
| 6.13. Test Card 12.....                            | 102 |
| 6.14. Test Card 13.....                            | 103 |
| 6.15. Test Card 14.....                            | 107 |
| 6.16. Test Card 15.....                            | 108 |
| 6.17. Test Card 16.....                            | 110 |
| 6.18. Test Card 17.....                            | 112 |
| 6.19. Test Card 18 (Previously Test Card 49) ..... | 114 |
| 6.20. Test Card 19 (Previously Test Card 50) ..... | 118 |
| 6.21. Test Card 20.....                            | 121 |
| 6.22. Test Card 21 (Previously Test Card 32) ..... | 122 |
| 6.23. Test Card 22 (Previously Test Card 33) ..... | 123 |
| 6.24. Test Card 23 (Previously Test Card 39) ..... | 125 |

|                                                              |            |
|--------------------------------------------------------------|------------|
| 6.25. Test Card 24 (Previously Test Card 41) .....           | 127        |
| 6.26. Test Card 25 (Previously Test Card 43) .....           | 129        |
| 6.27. Test Card 26 (Previously Test Card 44) .....           | 130        |
| 6.28. Test Card 27 (Previously Test Card 45) .....           | 131        |
| 6.29. Test Card 28 (Previously Test Card 46) .....           | 133        |
| 6.30. Test Card 29 (Previously Test Card 47) .....           | 136        |
| 6.31. Test Card 30 (Previously Test Card 48) .....           | 137        |
| <b>Appendix A: Visa CA Test Public Keys for VSDC .....</b>   | <b>138</b> |
| A.1: 1152 Bit VSDC TEST Key .....                            | 138        |
| A.2: 1408 Bit VSDC TEST Key .....                            | 139        |
| A.3: 1984 Bit VSDC TEST Key .....                            | 140        |
| <b>Appendix B: Terminal Action Code (TAC) Settings.....</b>  | <b>141</b> |
| B.1: Terminal Action Code (TAC) settings for Terminals ..... | 141        |
| <b>Appendix C: VSDC Stand-in Processing Conditions .....</b> | <b>143</b> |
| <b>Appendix D: Compliance Report .....</b>                   | <b>147</b> |
| D.1: Terminal Information .....                              | 147        |
| D.2: ADVT Test Results .....                                 | 152        |
| D.3: ADVT Detailed Test Results Sheet (Optional).....        | 155        |
| <b>Appendix E: List of Acronyms .....</b>                    | <b>157</b> |

# 1.Disclaimer

The Acquirer Device Validation Toolkit described herein provides a means for a Visa Acquirer (or agent) implementing a chip program to test their terminals before they are deployed. The tests prescribed here do not supersede the requirement for the terminals to undergo type approval testing at an accredited EMVCo laboratory.

The Acquirer Device Validation Toolkit tests must be included in a Visa Acquirer's chip migration project plan as they provide additional testing and review methods particularly important after the terminal has been re-configured to suit the Acquirer's requirements.

The Acquirer Device Validation Toolkit test cards and test scripts to be used with terminals are designed to determine whether the terminal can process certain card profiles that are currently known to cause acceptance issues. Visa reserves the right to add or remove tests and test requirements in its sole discretion.

The Acquirer Device Validation Toolkit is provided as a service to Acquirers to assist them in eliminating or reducing card acceptance problems. Visa does not warrant the Toolkit or any Toolkit test results for any purpose whatsoever, and expressly disclaims any and all warranties relating to the Toolkit. No vendor or other third party may refer to a product, service or facility as "Visa-approved", nor otherwise state or imply that Visa has, in whole or part, approved any aspect of a vendor or its products, services or facilities, except to extent and subject to the terms and restrictions expressly set forth in a written agreement with Visa or in an approval letter provided by Visa. All other references to "Visa approval" are strictly prohibited by Visa.

All references to Visa operating regulations in this document are deemed to be references to both Visa International Operating Regulations and/or Visa Europe Operating Regulations, as appropriate.



## 2. Introduction

Visa Smart Debit/Credit (VSDC) provides a global chip-based payment service that allows Members to strategically and competitively position themselves for the future. The program is based on specifications developed by Europay, MasterCard, and Visa (EMV) working collaboratively to ensure that all chip-based debit and credit cards can be accepted in any EMV chip reading terminal worldwide.

From an acquiring perspective, chip introduces many new features and complexities to the card acceptance process. During a chip-based transaction, the card and terminal proceed through a series of steps to determine the final outcome of the transaction. These steps require additional data and processing capabilities at the terminal level.

Terminals deployed in one country or region can experience acceptance problems when being used with cards from other countries and regions, even though both the cards and terminals would have been EMV or Payment Scheme approved. These issues may often be the result of incorrect terminal configuration, inadequate integration testing or misunderstandings about EMV and Visa requirements.

To help in ensuring that the terminals Acquirers deploy do not contribute to interoperability problems, Visa has developed the Acquirer Device Validation Toolkit—a set of test cards and test cases to be used on terminals to ensure correct terminal configuration, to assist with integration testing and to ensure that Visa's terminal requirements are being met.

In addition to ensuring card acceptance, these tests also enable the User Interface of live terminals to be tested. This is necessary to make sure that user prompts such as error messages, Application Selection menus and PIN Entry messages are appropriate and readily comprehensible to the cardholder and merchant.

Acquirers are required to run these tests on all terminals prior to deployment (including all variations of hardware, software, and parameter settings) and Visa recommends that Acquirers run these tests on terminals already deployed in the field. Acquirers are required to fill out a compliance report (see Appendix D: Compliance Report) and submit it to their Visa representative once the tests are completed.

Although the ADVT is a global product, in some cases it is supported and distributed by Visa's regional offices. For more information on the ADVT, you may contact your Visa representative using one of the following email addresses according to your geographical location:

|              |                     |
|--------------|---------------------|
| Asia Pacific | CADSupport@visa.com |
| Canada       | CADSupport@visa.com |

|                                         |                     |
|-----------------------------------------|---------------------|
| Central Europe, Middle East, and Africa | CEMEAChip@visa.com  |
| Europe                                  | ADVTK_EU@visa.com   |
| Latin America and Caribbean             | CADSupport@visa.com |
| United States of America                | PMF@visa.com        |
| Visa Global Office                      | CADSupport@visa.com |

## 3. Overview

This section provides an overview of the Acquirer Device Validation Toolkit and its associated User's Guide (this document).

### 3.1. Objective

The objective of this document is to define a toolkit that provides Visa Acquirers with a high level of confidence that the chip terminals they are deploying will not contribute to interoperability problems.

### 3.2. Audience

The audience for this document is Visa Acquirers or their agent(s) responsible for deploying terminals in their marketplace that accept Visa Smart Debit/Credit (VSDC) cards. It shall not be shared with or distributed to any other parties.

NOTE: The term Acquirer in this document is used generically to represent the entity in the marketplace responsible for terminal deployment. Depending on the marketplace, it could represent the Acquirer, merchant, a Value Added Network (VAN), or a vendor providing terminal deployment services on behalf of an Acquirer, merchant, or VAN.

### 3.3. Structure

This document contains the following sections:

- Chapter 1: Disclaimer
- Chapter 2: Introduction—This section provides background information highlighting the need for an Acquirer Device Validation Toolkit.
- Chapter 3: Overview—This section provides an overview of the document including objective, audience, structure, components, usage, scope, related documents and summary of changes in different versions of the document.
- Chapter 4: Introduction to Test Cases
- Chapter 5: Test Cases—This section outlines the test cases and associated test cards.
- Chapter 6: Test Card Profiles—This section provides the test card profiles that were used to create the test cards outlined in Chapter 4.
- Appendix A: Visa Certificate Authority (CA) Public Test Keys for Visa Smart Debit Credit (VSDC)—These test keys need to be loaded into the terminal to support the tests associated with Static and Dynamic Data Authentication.
- Appendix B: Terminal Action Code (TAC) Settings—The TACs need to be loaded into the terminal for it to operate properly.
- Appendix C: VSDC Stand-in Processing Conditions—When an acquirer is connected online to the Visa Certification Management System (VCMS), or the Visa Member Test System (VMTS) the transaction is processed in Stand-in. When the transaction is processed in Stand-in, the VSDC Stand-in Conditions can be helpful in determining the reason(s) VCMS/VMTS approved/declined the transaction. The same considerations apply when a Visa-confirmed third party supplied host simulator is used instead of VCMS/VMTS.
- Appendix D: Compliance Report—This appendix provides an example of a compliance report for Acquirers to complete and submit to their Visa regional representatives after running the test cases on their terminals.
- Appendix E: List of Acronyms – This appendix provides a list of commonly used acronyms in this User's Guide and in the EMV environment.

### 3.4. Components

The toolkit consists of:

- Test Cards—Cards configured with specific settings in order to make certain conditions visible.
- Test Cases—Cases outlining the appropriate cards to use along with the expected results.
- Documentation—Documentation providing background information about the tests and forms that Acquirers can use to track and document their test results.
- Compliance Report—A sample of the kind of report that Acquirers must fill out and submit to their Visa regional representative after completing the Acquirer Device Validation Toolkit test cases.

Acquirers can obtain additional toolkits (including test cards) from their Visa regional representative (see section 2 for email addresses).

### 3.5. Usage

An Acquirer must utilize the Acquirer Device Validation Toolkit (ADVT) prior to deploying a new chip card acceptance device or after upgrading an existing device. As described in the Visa operating regulations, an Acquirer that fails to utilize the ADVT on a device that causes a chip interoperability issue, may be subject to penalties as defined in the Visa Chip Interoperability Compliance Program.

Acquirers are required to use the toolkit prior to initial terminal deployment (including all variations of hardware, software, and parameter settings) to ensure that the terminal has been set up and configured correctly. It is expected that Acquirers will run every applicable test to gain the full benefit of the toolkit. When the Acquirer's test result does not match the expected outcome of the test, it is anticipated that the Acquirer will work with their terminal vendor (and Visa, if necessary) to correct the problem. The Acquirer will continue to perform the test until the problem is resolved and the Acquirer's result matches the expected outcome.

In addition, it is strongly recommended that Acquirers use the toolkit on terminals previously deployed in order to ascertain if there are potential acceptance problems with terminals in the field.

NOTE: Visa Acquirers shall also use a subset of the test cards in the toolkit to conduct online transactions through a connection to the VisaNet Certification Management Service (VCMS) / Visa Member Test System (VMTS) or a Visa-confirmed third party supplied host simulator. The online cards are defined within the document..

The following guidelines are intended to provide a more detailed outline of the specific cases that will govern use of the ADVT. Where ADVT usage is required,

the latest version of the toolkit shall always be used. If this is not possible due to upgrade schedules, etc., ADVT users must consult with their Visa Representative to determine regional policies regarding replacements of earlier version of the toolkit.

**ADVT use is mandatory in the following cases:**

- Deployment of a new EMV card accepting device, containing any of the following:
  - o New EMV kernel
  - o New version of payment application
  - o New communications interface
- Modification or reconfiguration of an existing device to make any of the following changes:
  - o Major changes to the EMV-approved kernel (as defined in EMV Bulletin 11)
  - o Changes to the payment component of the terminal application, affecting EMV processing.
  - o Changes to the Cardholder Verification Method (CVM) capabilities
- Changes to a Merchant's or Acquirer's network architecture. For example, in a case where a Merchant has switched Acquirers, even though their terminal configuration might remain the same.
- Introduction of a new model<sup>1</sup> of terminal hardware
- In some instances, as requested by Visa International or a Visa Regional office, based on evidence of an acceptance or interoperability problem affecting the device or connectivity to VisaNet.

**ADVT use is strongly recommended in the following cases:**

- Introduction of Dynamic Currency Conversion (DCC) functionality.
- A strong suspicion by Visa International, any Visa Regional offices or an Acquirer of the presence of an acceptance or interoperability problem affecting the device or connectivity to VisaNet.

**ADVT use is recommended in the following cases:**

- Minor modifications or reconfiguration of existing terminals for any of the following:
  - o Change of Language Support
  - o New communications interface (e.g. from Dial-up to high-speed)
  - o Change of supported Currency Code/Country Code
- Upgrades or modifications to the Acquirer Host systems which affect the transmission of chip data (ADVT Online validation should be performed from at least

---

<sup>1</sup> It is possible to have "families" of terminals which are identical from a payment point of view. Here a new "model" is taken to mean a change which may affect card acceptance. This includes the user interface presented to either the cardholder or merchant.

---

one EMV Chip-reading device)

ADVT use is not required in the following cases:

- Minor changes to the EMV-approved kernel (as defined in EMV Bulletin 11). Note that replacing the IFM with another approved module is defined as a minor change.
- Change to software that does not affect payment processing, e.g. screen layout, and report generation on a POS terminal, advertising graphics on an ATM.
- Addition of a new peripheral device not requiring changes to the existing code, e.g. a new printer or cash dispenser module.
- Addition of a new Online PIN-only PED.
- A change to the terminal-to-host protocol which does not affect authorization messages.
- Change to CA Public Keys used for Offline Data Authentication – ADVT testing does not use live keys.
- Introduction of a new version of ADVT by Visa International provided the device has already undergone successful validation using an earlier version of ADVT in accordance with these guidelines.

Please note, however, that some Visa Regional offices may apply additional policies governing the period by which earlier versions of the ADVT must be phased out and replaced by the most recent version.

---

NOTE: An acquirer or their agent (including processors or national/regional/global acquiring networks) can request waiving of the ADVT testing requirement if they can attest that the deployed application has already been tested on the same acquiring network. The deployed application would be recognized by concatenation of all identifiers:

- Kernel id - as submitted to EMV and listed on the EMVCo website
- Acquiring network - as identified by the acquiring network or regional or global body
- Application identifier - as identified by the application developer or system integrator

If the device deployer wishes to see the ADVT test results recognized in multiple regions, they will need to request this. Granting the request is at the sole discretion of Visa, and may not be allowed under regional policies. If the request is accepted, the compliance report can then be forwarded to Visa headquarters for retention and access by other regional personnel.

### 3.5.1. ADVT Version

On release of a new version of the ADVT, Acquirers will be given a six month grace period to upgrade to the new version. During this grace period, testing will still be allowed with their existing version of the toolkit. However, on expiration of the grace period, it is expected that Acquirers would have completed their upgrade to the latest version of the toolkit and results from earlier versions will no longer be accepted.

Please note, however, that some Visa Regional offices may apply more stringent policies governing the period by which earlier versions of the ADVT must be phased out and replaced by the most recent version.

### 3.6. Scope

| Within Scope                       | Outside of Scope                    | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Terminal testing.                  | Acquirer host certification.        | <p>The toolkit focuses on helping to ensure terminals deployed in the field are configured in a way that promotes the best potential for global interoperability.</p> <p>While some of the cards in this toolkit are to be used for online testing, this toolkit is not specifically designated as a host certification toolkit. Acquirers will continue to perform host system certification using the current set of test cards and scripts. Please see your Visa regional representative to obtain the test kit for Acquirer host certification.</p> |
| Complement to EMV Level 2 testing. | Replacement of EMV Level 2 testing. | <p>It is assumed that Acquirers and/or terminal vendors will perform these tests on terminals that have already passed EMV Level 1 and Level 2 testing. These tests will complement EMV testing to ensure that terminals have been configured correctly prior to deployment.</p>                                                                                                                                                                                                                                                                        |

### 3.7. Future Enhancements

The Acquirer Device Validation Toolkit may be expanded in the future to include additional device and/or host system tests.

### 3.8. Related Documents

This section lists documents that may be read and/or referred to in conjunction with this document:

- Europay, MasterCard, Visa (EMV), (latest version).
- Visa operating regulations (latest version).
- Transaction Acceptance Device Requirements (TADR) – Requirements (latest version).
- Transaction Acceptance Device Guide (TADG) – Requirements and Best Practices (latest version).

### 3.9. Summary of Changes

This section provides a summary of changes made in different versions of the Acquirer Device Validation Toolkit document.

| Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.0     | First official release of the document                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2.1     | <ol style="list-style-type: none"> <li>1. Two new Sections added: <ul style="list-style-type: none"> <li>– Section 3.9 – Summary of Changes</li> <li>– Appendix B – List of Acronyms</li> </ul> </li> <li>2. Wording changes for clarification in various Sections:</li> <li>3. Corrections to Typographic errors in various Sections:</li> <li>4. Card Personalization change: <ul style="list-style-type: none"> <li>– Section 5.11 – Test Card 10: Changed Application Effective Date from “50 01 01” to “49 01 01”.</li> </ul> </li> </ol>                                        |
| 2.1.1   | <ol style="list-style-type: none"> <li>1) Realigned tables in Chapter 5 that were distorted</li> <li>2) Updated Appendix B: List of Acronyms</li> <li>3) Corrections and Typographic errors in various Sections:</li> <li>4) Wording changes for clarification in various Sections</li> </ol>                                                                                                                                                                                                                                                                                         |
| 2.1.2   | <ol style="list-style-type: none"> <li>1) Correction of Typographic errors in various Sections:</li> <li>2) Wording changes/additions for clarification in various Sections:</li> <li>3) Additions to support and strengthen ADVT Online requirements as follows: <ul style="list-style-type: none"> <li>– Section 8: Changed STIP Condition # 16 from ‘Decline’ to ‘Approve’</li> </ul> </li> <li>4) Changes throughout the document to ensure consistency in use of “Acquirer Device Validation Toolkit”</li> <li>5) New Appendix A.3 – ADVT Detailed Test Results Sheet</li> </ol> |
| 3.0     | <ol style="list-style-type: none"> <li>1) Five new cards added as follows: <ul style="list-style-type: none"> <li>– Card # 43: Card without a PAN Sequence Number</li> <li>– Card # 44: Card with a PAN Sequence Number = 11</li> <li>– Card # 45: Card with an IPK Certificate based on a 1016-bit IPK</li> <li>– Card # 46: Card containing an Issuer URL and Issuer Discretionary Data</li> </ul> </li> </ol>                                                                                                                                                                      |

| Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <ul style="list-style-type: none"> <li>– Card # 47: Card with a Blocked VSDC Application</li> </ul> <p>2) Application Version Number updated to correctly reflect VSDC Applet version used (00 8C).</p> <p>3) Data element changes to specific cards to accommodate the following:</p> <ul style="list-style-type: none"> <li>– Card # 3: Additional functionality to T= 1 card</li> <li>– Card # 16: Unique BIN used for iCVV testing &amp; Offline Plaintext PIN with 6-digits</li> <li>– Card # 17: Unique PAN used for online differentiation</li> <li>– Card # 18: Reduced PIN Try Limit from “127” to “15”</li> <li>– Card # 21: Correction of UDKs on 19-digit card</li> <li>– Card # 24: Triggering DDA failure in a different way</li> <li>– Card # 25: Unique PAN used for online differentiation</li> <li>– Card # 29: Reduced PIN Try Limit from “127” to “15”</li> <li>– Card # 34: Reduced PIN Try Limit from “127” to “15”</li> <li>– Card # 35: Unique PAN used for online differentiation</li> <li>– Card # 36: Reduced PIN Try Limit from “127” to “15”</li> <li>– Card # 38: Reduced PIN Try Limit from “127” to “15”</li> <li>– Card # 39: Reduced PIN Try Limit from “127” to “15”</li> <li>– Card # 41: Unique PAN used for online differentiation</li> </ul> <p>4) Correction of ICC Key Modulus value for Card # 27</p> <p>5) Minor Typo error corrections.</p> |
| 3.1     | <p>1) Modification to Card # 46 to accommodate an Application Expiration Date = December 31, 2025 (Sections; 4.3: Test Case Summary, 4.4: Test Case 46 &amp; 5.47: Test Card 46)</p> <p>2) Corrections to minor typographic errors in Sections; 5.45, 5.46, 5.47 &amp; 5.48</p> <p>3) Card # 7 (Section 5.8): Changed Application Priority Indicator from ‘01’ to ‘81’ to allow Application Preferred Name to be displayed.</p> <p>4) Corrections to Track 1 data coding on all cards:</p> <ul style="list-style-type: none"> <li>– ‘00’ before the CVV</li> <li>– ‘000000’ after the CVV</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 3.2     | <p>1) Card # 41: Correction to Signed Static Application Data (Tag 93)</p> <p>2) Section 4.4 - Test Case 30: Wording changes for clarification.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3.2.1   | <p>1) Corrected all 25 minor documentation errors as defined in the ADVT Known Issues List – Version 3.2 (March 29, 2005) document</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3.2.2   | <p>1) Corrected a minor documentation error as defined in the ADVT Known Issues List – Version 3.2.1 (April 29, 2005) for Section 4.4 Test Case 46</p> <p>2) Card # 18: Updated Data element incorrectly titled “Short File Identifier (SFI)” which was corrected to “Application File Locator (AFL)”</p> <p>3) Corrected a minor documentation error in the Test Purpose and Description for Section 4.4 Test Case 42</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <ol style="list-style-type: none"> <li>4) Corrected a minor documentation error in the Card Conditions for Section 4.4 Test Case 45</li> <li>5) Added a notation to Section 4.4 Test Case 47</li> <li>6) Section 4.4 Test Case 4 is for “informational purposes only” given that the 896-bit CA Public Key has now reached the end of its life</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3.2.3   | <ol style="list-style-type: none"> <li>1) Data Element: PIN Try Limit in Section 5.1.2 – corrected the DGI from “11 01” to “80 10/90 10”</li> <li>2) Data Element: Issuer Private Key Exponent in Section 5.1.2 - removed the DGI value of “02 02”</li> <li>3) Data Element: ICC Public Key Exponent and ICC Public Key Remainder in Section 5.1.2 - corrected the DGI value from “02 05” to “81 03”</li> <li>4) Data Elements with DGI values of “02 05” updated to “02 05 (02 02)”</li> <li>5) Card # 28: Changed notation from “(SDA with 1152 key)” to “(SDA with 1152-bit CA key and 1152-bit Issuer Key)”</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4.0     | <ol style="list-style-type: none"> <li>1) Wording changes for clarification in test cases 1, 3, 9, 11, 12, 16, 17, 18, 19, 20, 21, 22, 25, 26, 28, 30, 31, 32, 33, 34, 35, 37, 39, 40, 41, 43, 44, 45, 46, 47</li> <li>2) Test Card #4 – Removed from Test Deck</li> <li>3) Test Card #5 – Removed from Test Deck</li> <li>4) Test Card #7 – Removed from Test Deck</li> <li>5) Test Card #8 – Removed from Test Deck</li> <li>6) Three new cards added as follows: <ul style="list-style-type: none"> <li>– Card # 48: Card with 1408 bit Test Keys</li> <li>– Card # 49: Card with 1984 bit Test Keys and supports Japanese CVM List</li> <li>– Card # 50: Card supports the Visa RID with the Plus PIX</li> </ul> </li> <li>7) Data element changes to specific cards to accommodate the following: <ul style="list-style-type: none"> <li>– Card # 3: Updated IAC Denial</li> <li>– Card # 13: Added Proprietary Tag Data</li> <li>– Card # 18: Corrected VLP Personalization</li> <li>– Card # 22: Support card requirements related to cardholder confirmation and acceptance of a card containing a non-ASCII Application Preferred Name</li> <li>– Card # 32: Updated PIN Try Limit to 00</li> <li>– Card # 33: Updated PIN Try Limit to 00</li> <li>– Card # 46: Corrected Issuer Application Data, Updated CVM List, Updated for VPay and IAC Denial</li> <li>– Card # 47: Removed Data Elements for Application Block</li> </ul> </li> <li>8) Business Justifications added to all Test Cases</li> <li>9) Removed Component Values for 896-bit VSDC Test Key in Section 6.0</li> <li>10) Added Component Values for 1408-bit and 1984-bit VSDC Test Keys in Section 6.0</li> <li>11) Test Card #32 and Test Card #33 – Not applicable for ATMs</li> </ol> |
| 5.0     | <ol style="list-style-type: none"> <li>1) Cards removed: <ul style="list-style-type: none"> <li>- Test Card #2</li> <li>- Test Card #9</li> <li>- Test Card #25 - Functionality combined with Test Card #1 (T=0) and Test Card #3 (T=1) for Issuer Authentication</li> </ul> </li> <li>2) Data element changes to specific cards to accommodate the following: <ul style="list-style-type: none"> <li>– Card # 1: Updated with Issuer Authentication as mandatory</li> <li>– Card # 3: Updated DDA ICC 1152-bit key, Corrected Issuer Authentication Data</li> </ul> </li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <ul style="list-style-type: none"> <li>– Card # 6: Added qVSDC with cryptogram 10</li> <li>– Card # 11: Added qVSDC with cryptogram 17</li> <li>– Card # 13: Changed proprietary tag in the application data to C3</li> <li>– Card # 16: Added zero length tag (ICC PK Remainder)</li> <li>– Card # 17: CDOL2 updated to include the Terminal Verification Results</li> <li>– Card # 20: Updated Application Preferred Name to “Electron de Visa” and changed all data to zero in mag stripe data except Expiry Date and Service Code</li> <li>– Card # 27: Added double length tag (ICC PK Remainder)</li> <li>– Card # 49: Updated ATR paramaters</li> </ul> <p>Note: The following changes are not made to test cards</p> <ul style="list-style-type: none"> <li>– Card # 29: Updated DGI for ICC Public Key Remainder and Exponent</li> <li>– Card # 37: Updated DGI for Cardholder Verification Method</li> <li>– Card # 50: Corrected “Application File Locator (AFL)” to value of 08 01 01 00 18 01 02 00</li> </ul> <p>3) Wording changes for clarification in test cases : 13, 18, 20, 21, 24, 27, 29, 30, 31, 37, 41, 50</p>                                                                                         |
| 5.1     | <p>1) Card # 4: Reintroduced with the following features:</p> <ul style="list-style-type: none"> <li>– Terminal Risk Management bit is not set (0) in the Application Interchange Profile</li> <li>– Floor Limit Exceeded bit set in the IAC – Denial</li> </ul> <p>2) Included VSDC Applet Version with each card profile in Section 5</p> <p>3) Deleted data and corresponding tables related to test cases for cards removed from the toolkit</p> <p>4) Minor editorial updates throughout the document</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.1.1   | <p>1) Update to Section 1: Disclaimer clarifying document references to Visa operating regulations</p> <p>2) Updates to Usage section (Section 2.5) – new sub-section added for ‘ADVT Version’</p> <p>3) Update to Test Case 19, Expected Results stating that ‘fallback to magnetic stripe in an acceptable result’</p> <p>4) Update to Test Case 34, Expected Results clarifying that for ATM Devices the transaction should result in an offline decline.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6.0     | <p>1. The following card changes were made in this version:</p> <ul style="list-style-type: none"> <li>• Upgrade all cards due to expire on December 31, 2010 to extend their Application Expiration Dates to December 2015</li> <li>• Replace CVVs on the chip with iCVVs for all cards</li> <li>• Removal of 14 cards (10, 18, 19, 24, 27, 28, 29, 34, 35, 36, 37, 38, 40 &amp; 42) now considered redundant or unnecessary</li> <li>• Updates some cards with an 1152-bit CA PK Certificate</li> <li>• Reassigned numbering of some cards in the toolkit to eliminate gaps</li> <li>• Specific card changes (these are the new card numbers): <ul style="list-style-type: none"> <li>- Card # 1: Unique PAN introduced to allow card identification online</li> <li>- Card # 8: Introduced a unique PAN for online card identification</li> <li>- Card # 11: Intrduced 3 applications for multi-application testing</li> <li>- Card # 13: Unique PAN introduced to allow card identification online and proprietary application data added to the card. Six digit Reference PIN also added to this card.</li> <li>- Card # 15: Moved the data element with a zero length from Card # 16 to this card</li> </ul> </li> </ul> |

| Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <ul style="list-style-type: none"> <li>- Card # 16: Introduced 2 applications with unique suffixes for multi-application testing</li> <li>- Card # 22: Introduced 5 applications with unique suffixes for multiapplication testing.<br/>The first 3 applications are expired to trigger a decline</li> <li>- Card # 45: Updated with a IPK Certificate based on an 1144-bit IPK</li> </ul> <p>2. The following documentation changes were made in this version:</p> <ul style="list-style-type: none"> <li>• Section 2.5 – Usage: Updates made to the Usage Guidelines</li> <li>• Appendix B1: Updates to TAC – Online and TAC - Default</li> <li>• Appendix B1: Removal of references to Early Option Acquiring TACs</li> </ul> |



## 4. Test Cases Introduction

This section outlines the test cases that Acquirers are required to perform on their terminals. It also highlights the specific test card to be used for each test case.

### 4.1. Pre-requisites

Prior to running the Acquirer Device Validation Toolkit test cases, the Acquirer must ensure the following:

#### 4.1.1. Terminal Capabilities

Before beginning any of the tests, it is important to understand the capabilities of your terminal. This will help you ensure you are performing the tests correctly for your specific terminal.

- Terminal Type—Determine if your terminal is an Automated Teller Machine (ATM) Cash machine, standalone Point of Sale (POS) device, integrated POS device, or Cardholder Activated Terminal.
- Cardholder Verification Methods—Determine the cardholder verification methods that your terminal supports (Online Personal Identification Number (PIN), Offline Enciphered PIN, Offline Plaintext PIN, Signature, No CVM Required—this CVM allows you to accept a card without any verification of the cardholder). This is important, as the success criteria associated with some of the tests is specific to the cardholder verification method.
- Offline Data Authentication—Determine if your terminal supports Static Data Authentication and/or Dynamic Data Authentication. This is important, as some of the tests are specific to these capabilities.
- Floor Limit—Determine the floor limit of your terminal. Always use a below the floor limit amount during testing unless the test case specifically states that it must go online.

#### 4.1.2. Terminal Log

It is very useful to the testing process for the terminal to have the ability to make the values of certain data objects (such as the Terminal Verification Results and Transaction Status Information) generated during the transaction available to the tester. This could take the form of a log file or some means of printing this information on a receipt or displaying it on the screen. In some cases, a log produced through online interaction with a host can be used.

#### 4.1.3. Visa CA Test Public Keys

During use of the Acquirer Device Validation Toolkit, terminals must be configured with the Visa CA Test Public Keys. These test keys are located in Appendix A: Visa CA Test Public Keys for VSDC.

NOTE: Prior to deployment, the Visa CA Test Public Keys must be removed from the terminals and the Acquirer must ensure that the production Visa CA Public Keys are installed in the terminal.

#### 4.1.4. Terminal Action Codes (TACs)

Visa supports one set of TACs for full data option Acquirers. Acquirers must ensure that the TAC settings are correct. The TAC settings are provided in Appendix B: TAC Settings. See also, Terminal Acceptance Device Requirements (latest version).

#### 4.1.5. Configured for Operational Use

The terminals must be configured for operational use. For example, the terminal must include the Visa AIDs (for Visa Credit/Debit and Visa Electron, where appropriate), terminal country code, correct date/time, and floor limits.

#### 4.1.6. EMVCo Level 1 and 2 Approval

Terminals, prior to deployment, must have passed the EMVCo Level 1 and Level 2 approval process (this requirement does not apply to terminals deployed prior to the EMVCo approval process).

## 4.2. Instructions

### 4.2.1. Self-Administered Tool

In the first instance, the ADVT is a self-administered tool. Users must work to fix the problems on their own whenever possible and only use Visa assistance for problems that cannot be resolved between the terminal vendor and Acquirer technical team.

### 4.2.2. Initially Deployed Terminals

For terminals being initially deployed, the intent is for Acquirers to run each applicable test and make modifications to the terminal configuration until the terminal meets the expected outcome of the test. Acquirers need to run these tests on each terminal type as well as each terminal hardware and/or software configuration. After running all tests and making the appropriate terminal configuration modifications, Acquirers need to submit their results to Visa.

NOTE: See "For Information Gathering Purposes Only Tests" for the test scenarios that do not require Acquirer action.

### 4.2.3. Previously Deployed Terminals

For terminals that have already been deployed, the intent is for Acquirers to run the test, gather the results in the provided forms and submit the results to their Visa region using the Compliance Report provided in Appendix D.

### 4.2.4. For Information Gathering Purposes Only Tests

Some tests outlined in this toolkit are for information gathering purposes only. If a terminal fails these tests, no Acquirer action to upgrade the terminal is necessary. There are some instances, however, where it is strongly recommended to update the terminal if it fails one of these tests. In most cases, this is because the functionality, although currently optional, will later become mandatory. In all cases, the Acquirer must submit the test result to Visa.

#### 4.2.5. Changes to Terminals

If changes are made to terminal configuration or settings, the Acquirer/tester must re-run the Acquirer Device Validation Toolkit tests as described in the ADVT Usage Guide, Section 3.5.

#### 4.2.6. Decline Responses vs. Other Errors

A decline response is different from an error message. In some cases, a decline response by the terminal is an acceptable outcome of the test case. Error messages, where the terminal is unable to complete the transaction (e.g., unable to perform a complete EMV transaction from Application Selection to Completion), are generally unacceptable and can indicate a problem with the terminal or an incorrect terminal setting/configuration. Testers should not be alarmed if decline responses occur (as long as a decline is allowed in the success criteria) but must investigate error messages (such as “Card Error” and “Not Accepted” or the equivalent). For further information on these errors, please refer to EMV 4.0, Section 7.2: Standard Messages.

#### 4.2.7. Online Testing

General: In the Test Cases section, some tests are designated for online testing. For these tests, the transaction must be sent online to VCMS/VMTS or a host simulator for validation of the ARQC and/or CVV data. If the test is not designated as an “online test,” it may be performed as an offline transaction if this is within the capabilities of the terminal.

Visa Acquirers: Visa Acquirers are required to use the test cards designated for online testing to conduct online tests by connecting their terminal to their test host system and generating transactions through to the VisaNet Certification Management Service (VCMS), Visa Member Test System (VMTS) in the Visa Europe region, or a Visa-confirmed third party supplied test host which mimics VCMS/VMTS. The test cards are configured with test keys that are set up in VCMS/VMTS allowing it to validate and generate the online cryptograms. When cryptograms are successfully validated by VCMS/VMTS and successfully sent to the test card, it helps to ensure that all the components involved in the transaction are integrated properly. For the online tests, Card Authentication (the validation of the Authorization Request Cryptogram) shall be performed and must be successful (unless otherwise noted in the test case).

The Test Case Summary table in section 4.3 identified the cards to be used for online testing.

To help you in determining the reason VCMS/VMTS (or the third party test host) approved/declined the online transaction, please refer to Appendix C: VSDC Stand-in Processing Conditions.

NOTE: Although some cards are specifically designated for online tests, any test card that is not personalized to decline offline may be used for online testing.

NOTE: Access to the VisaNet Certification Management Service or Visa Member Test System is provided to Visa Members or Clients only.

#### 4.2.8. Compliance Report

Once Acquirers complete the test cases in this section, they need to fill out a Compliance Report and submit it to their Visa regional office. The Visa region will review the Compliance Report and contact the Acquirer, if necessary.

##### 4.2.8.1 Chip Compliance Reporting Tool

For a more convenient means of reporting the ADVT test results, Visa recently developed the Chip Compliance Reporting Tool (CCRT), a solution aimed at providing an alternative to the manual methods currently used for submission of ADVT test results. CCRT is a web-based, user-friendly tool that allows chip acquirers or their processors (users) to complete and submit the mandatory compliance reports via a global automated online system. Hosted on Visa Online (VOL), CCRT is designed in accordance with Visa's three-tier architectural requirements and provides a high-level of application and data security.

CCRS allows users to:

- Submit new compliance reports
- Review and update draft reports
- Check on the status of pending reports submitted to Visa
- Track approved reports

It benefits users by:

- Providing a convenient, secure online solution for ADVT results reporting.

- Reducing potential for errors in manual entry by guiding users to choose from applicable options and providing mandatory information requirements.
- Allowing the "re-use" of reports as a starting point for new reporting, reducing time spent completing the reports.
- Supporting online status review and automated management of reports submitted to Visa, expediting communication between Visa and clients.

For more details on CCRT please contact your local Visa Representative.

#### 4.2.9. Test Cards

Acquirers will use the test cards provided to run the test cases. One card is used for each test and, for ease of use, the test card number matches the test case number (e.g., for Test Case 1, the Acquirer will use Test Card 1). After completing the test cases, the Acquirer must return the Compliance Report to their Visa regional representative as per instructions specified by Visa.

#### 4.2.10. Transaction Amount

To expedite the transactions in the toolkit, it is recommended that an amount below the floor limit be used (unless otherwise specified in the test).

#### 4.2.11. PIN-Based Transactions

For Offline PIN or Online PIN, a PIN of '1234' must be used except for Test Case 13 which uses a PIN of '123412'.

Note: When PIN is used for the transaction, the signature line does not need to be printed on the receipt (if applicable) nor obtained from the cardholder (unless the combination CVM of Offline PIN and signature applies).

#### 4.2.12. Additional Toolkits

Acquirers can obtain additional toolkits (including test cards) from their Visa representative.

### 4.3. Test Case Summary

This section provides a brief description of each test case included in the toolkit.

Test cases labelled as “Offline” may either be performed as either offline or online transactions (depending on card behaviour and terminal capabilities). Test cases labelled as “online” must be performed as online transactions.

| Current Test Case | Test Case in Previous Versions | Card Description                                                                                                                                                                                                                                                                                        | Offline | Online |
|-------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|
| Test Case 1       |                                | Card is a basic VSDC card with a unique PAN and supporting mandatory Issuer Authentication.                                                                                                                                                                                                             | ✓       | ✓      |
| Test Case 2       | Test Case 21                   | Card has a 19 digit Primary Account Number.                                                                                                                                                                                                                                                             | ✓       | ✓      |
| Test Case 3       |                                | Card supports the T = 1 protocol, Issuer Authentication as mandatory, Dynamic Data Authentication (DDA) with an 1152-bit ICC key and enciphered Offline PIN.                                                                                                                                            | ✓       | ✓      |
| Test Case 4       |                                | Card personalized without Terminal Risk Management and configured to decline when Terminal Floor Limit is Exceeded.                                                                                                                                                                                     | ✓       |        |
| Test Case 5       | Test Case 22                   | Multi-application card containing five applications, each with a unique suffix and an Application Preferred Name containing non-ASCII characters. The first three applications are expired to trigger an offline decline, and Applications 4 & 5 both have a unique PAN for transaction identification. | ✓       |        |

| Current Test Case | Test Case in Previous Versions | Card Description                                                                                                                                                                                                                                                                                                                                      | Offline | Online |
|-------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|
| Test Case 6       |                                | Dual interface card supporting the following:<br><br>Contact interface: An extended length PDOL (45 bytes) and Language Preferences (Japanese, Korean & Chinese) codes supported.<br><br>Contactless interface: supporting both MSD and qVSDC (CVN 10) contactless transactions.                                                                      | ✓       |        |
| Test Case 7       | Test Case 23                   | Test ensures the Terminal Action Codes (TACs) for are correctly set up in the terminal.                                                                                                                                                                                                                                                               | ✓       |        |
| Test Case 8       | Test Case 26                   | Card created to allow magnetic stripe fallback testing, where necessary.                                                                                                                                                                                                                                                                              | ✓       |        |
| Test Case 9       | Test Case 30                   | Card contains an unrecognized method code in the CVM List ('Reserved for Future Use'), with instructions to apply the next CVM when the CVM fails.                                                                                                                                                                                                    | ✓       |        |
| Test Case 10      | Test Case 31                   | Card contains an unrecognized method code in the CVM List ('Reserved for Future Use'), with instructions to stop CVM processing when the CVM fails.                                                                                                                                                                                                   | ✓       |        |
| Test Case 11      |                                | Dual interface card supporting the following:<br><br>Contact interface: 3 payment applications; 1 <sup>st</sup> with unknown application ID, 2 <sup>nd</sup> with a blocked application and 3 <sup>rd</sup> with a valid application and a unique PAN.<br><br>Contactless interface: supporting both MSD and qVSDC (CVN 17) contactless transactions. | ✓       |        |
| Test Case 12      |                                | Card is restricted to domestic transactions through the use of the card's internal Geographic Restrictions feature.                                                                                                                                                                                                                                   | ✓       |        |

| Current Test Case | Test Case in Previous Versions | Card Description                                                                                                                                                                                            | Offline | Online |
|-------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|
| Test Case 13      |                                | Card contains contains a PSE, and has proprietary tag data within PSE. Also contains proprietary data within the application. There is also a 6-digit Offline PIN used on this card.                        | ✓       | ✓      |
| Test Case 14      |                                | Card requests a long string of data (0 x 64 bytes) in Processing Options Data Object List (PDOL).                                                                                                           | ✓       |        |
| Test Case 15      |                                | Card with a record length of 2 bytes (IPK Certificate). As a negative test, it also contains a data element (IPK Remainder) where its length is zero bytes.                                                 | ✓       |        |
| Test Case 16      |                                | Card contains two applications. The first application (Visa Credit) requires cardholder confirmation, while the second application (Visa Debit) does not require cardholder confirmation.                   | ✓       |        |
| Test Case 17      |                                | Card supports the minimum set of VSDC data elements (Magnetic Stripe Image) and with a Cryptogram Version Number of 12.                                                                                     | ✓       | ✓      |
| Test Case 18      |                                | Card supports the T=1 protocol and contains an Issuer Public Key Certificate signed by Visa's 1984-bit CA test key.                                                                                         | ✓       |        |
| Test Case 19      |                                | Card containing the Visa RID (A000000003) with the Plus PIX (8010) and a suffix of '01'.                                                                                                                    |         | ✓      |
| Test Case 20      |                                | Card is a Visa Electron card with a non-usable mag stripe.                                                                                                                                                  | ✓       | ✓      |
| Test Case 21      | Test Case 32                   | Card contains a CVM List with Offline PIN as the first method in the list. The PIN Try Limit is exceeded and the CVM List provides instructions to apply the next CVM (signature) when the first CVM fails. | ✓       |        |

| Current Test Case | Test Case in Previous Versions | Card Description                                                                                                                                                                                                                                                                                    | Offline | Online |
|-------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|
| Test Case 22      | Test Case 33                   | Card contains a CVM List with Offline PIN as the first method in the list. The PIN Try Limit is exceeded and the CVM List provides instructions to fail cardholder verification, and stop CVM processing when the first CVM fails. The IACs require an offline decline when PIN Try Limit exceeded. | ✓       |        |
| Test Case 23      | Test Case 39                   | Card contains a CVM List where the first CVM is the combination CVM of Signature and Offline PIN.                                                                                                                                                                                                   | ✓       |        |
| Test Case 24      | Test Case 41                   | Card with a 16-digit account number padded with hexadecimal "Fs" up to maximum account number length.                                                                                                                                                                                               | ✓       | ✓      |
| Test Case 25      | Test Case 43                   | Card without a PAN Sequence Number                                                                                                                                                                                                                                                                  | ✓       | ✓      |
| Test Case 26      | Test Case 44                   | Card with a PAN Sequence Number = 11.                                                                                                                                                                                                                                                               | ✓       | ✓      |
| Test Case 27      | Test Case 45                   | Card with an IPK Certificate based on an 1144-bit IPK.                                                                                                                                                                                                                                              | ✓       |        |
| Test Case 28      | Test Case 46                   | Card contains a PSE, with an Issuer URL in both the PSE and application data, extra Issuer Application Data in Tag 9F 10, an Application Expiration Date = December 31, 2025 and a CVM List which does not contain Signature.                                                                       | ✓       | ✓      |
| Test Case 29      | Test Case 47                   | Card that is Blocked from use.                                                                                                                                                                                                                                                                      | ✓       |        |
| Test Case 30      | Test Case 48                   | Card supports Static Data Authentication (SDA) with an IPK Certificate associated with a 1408-bit IPK.                                                                                                                                                                                              | ✓       |        |

## 5. Test Cases

This section provides the test cases.

NOTE: Please be sure to read Section 4.1, Pre-requisites and Section 4.2, Instructions prior to beginning the tests. These sections contain critical information.

Each test case is outlined in a table with the following information:

- **Test Case**—This section provides a reference number to the test case. There is a single card associated with each test case so that Test Card 1 is used with Test Case 1, etc.
- **Specific Terminal Conditions**—This section highlights information related to the terminal. Although most of the tests apply to all terminals, there are some tests that only apply to specific terminals (such as terminals supporting Offline PIN or terminals supporting Dynamic Data Authentication).
- **Online Testing**—Specific cards may be used for online testing. Please refer to Section 4.2.7, Online Testing for further details.
- **Test Purpose and Description**—This section provides a description of the test case.
- **Expected Results**—This section outlines the success/failure criteria for the test.
- **Card Conditions**—This section highlights the configuration of the test card used for the test case.
- **Reference (Specifications/Rules)**—This section references the specification or rule that Acquirers may refer to for background information on the test. This information is especially important in the event that the Acquirer fails the test.
- **Business Justification**—This section provides a business-oriented description of why each test is required.

### Test Case 1

**Specific Terminal Conditions:** This test applies to all terminals (POS, ATMs, etc.).

**Online Testing:** In order to conform to the ADVT mandate, it is necessary to perform this test online. See Section 4.2.7: Online Testing for additional information. An online transaction must be initiated to VCMS/VMTS/approved host simulator, Online Card Authentication must pass and a cryptogram associated with Online Issuer Authentication must be provided in the response, received by the terminal, and forwarded to the card.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                            | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure acceptance of a basic VSDC card. This card contains most commonly used VSDC features.</p> <p>Note: This is a T=0 test card and the card is personalized without the Payment System Environment. If the terminal has difficulty with these tests, ensure your terminal can accept cards supporting the T=0 protocol and personalized without the PSE.</p> | <p>This test has multiple steps:</p> <p>1a) All Devices: Perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test. For terminals supporting SDA, the terminal log must show that the Transaction Status Information (TSI) byte 1, bit 8 is set to '1' (Offline Data Authentication performed), the Terminal Verification Results, byte 1, bit 8 is set to '0' (Offline Data Authentication was performed), and byte 1, bit 7 is set to '0' (Offline Static Data Authentication did not fail).</p> <p>If the application name is displayed and the terminal supports the Issuer Code Table Index of 01, the terminal must display the Application Preferred Name of Credito de Visa. For these devices, the Visa AID (A0000000031010) must be printed on the receipt and it is strongly recommended that the Application Preferred Name (Credito de Visa) also be printed on the receipt.</p> <p>If the application name is displayed and the terminal does not support the Issuer Code Table Index of 01, the Application Label of Visa Credit must be displayed. For these devices, the Visa AID (A0000000031010) must be printed on the receipt and it is strongly recommended that the Application Label (Visa Credit) also be printed on the receipt.</p> <p>Note: It is a Visa Best Practice to print the application name (either Application Preferred Name or Application Label depending on support for the Issuer Code Table Index) on the receipt. Please refer to the Terminal Acceptance Device Guide..</p> <p>The transaction must be approved online. An offline decline is not acceptable.</p> |

## Test Case 1 (continued)

| Test Purpose & Description | Expected Results (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Continued                  | <p>1b) Applicable to Non-Zero Floor Limit Devices: Perform an online transaction (above the floor limit) to help ensure that the floor limit is set up correctly. The terminal must attempt to send the transaction online. The TVR, byte 4, bit 8 must be set to '1' (transaction exceeds floor limit). Note: You will not be able to perform this test if you do not successfully pass part 1a.</p> <p>Since the transaction is above the floor limit, the transaction must be sent online. If connected to VCMS/VMTS/approved host simulator, the transaction must be approved online. If conducting the tests in an offline mode (e.g., no connectivity to VCMS/VMTS/approved host simulator) the transaction must be declined offline after attempting to go online (due to the IAC and TAC-default for Floor Limit Exceeded).</p> <p>1c) Applicable to Readers that Have Separate Insertion Areas for Chip and Magnetic Stripe Transactions (i.e., Not Applicable to Combined Readers such as ATMs where the card is inserted into a single slot for both chip and magnetic-stripe transactions): Attempt to read the card via the magnetic stripe. Ensure that the terminal prompts the user to insert the card into the chip reader. This ensures that the terminal does not allow EMV chip cards to be processed as magnetic stripe (except where fallback criteria are met).</p> <p>1d) Applicable to Online Tests:<br/>The transaction must be sent online to VCMS/VMTS/approved host simulator where VCMS/VMTS/approved host simulator will respond with an Issuer Authentication cryptogram (Authorization Response Cryptogram—ARPC). The terminal must be able to receive the cryptogram in the response data and forward it to the card. If the online transaction results in a decline, the user has failed the test (indicating that the device either did not forward the cryptogram to the card or incorrectly forwarded the cryptogram to the card).</p> |
| Card Conditions            | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <p>Test card is a T=0 card, card does not contain the PSE; card contains the Application Label of Visa Credit and the Application Preferred Name of Credito de Visa.</p> <p>For the online test, the card is configured to generate an online Card Authentication cryptogram (referred to as the Authorization Request Cryptogram) and an Online Issuer Authentication cryptogram (referred to as the Authorization Response Cryptogram) must be provided in the response message.</p> <p>This card is personalized for Issuer Authentication as mandatory.</p> | <p>EMV 4.1, Book 1, Section 12.3.2: Using the Payment Systems Environment.</p> <p>Terminal Acceptance Device Requirements.</p> |
| <b>Business Justification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                |
| <p>This represents a card containing the most commonly used VSDC features. For this reason, it is important to ensure universal acceptance of this card.</p>                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                |

## Test Case 2 (Previously Test Case 21)

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

Online Testing: In order to conform to the ADVT mandate, this test must be performed online. See Section 4.2.7: Online Testing for additional information.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure acceptance of a card with a 19 digit account number.</p> <p>Note: It is now a Visa operating regulation for new and existing chip reading devices to have the ability to read Visa account numbers up to and including 19 digits. At present, there is no operating regulation or mandate requiring that the acquiring host system must be able to process 19-digit account numbers. However, ensuring that your acquiring system can handle up to 19-digit account numbers will future-proof your system.</p> <p>Please note that if you are an Acquirer who has agreed to accept V PAY cards you must have the capability to process 19-digit PANs.</p> | <p>For Offline tests, the terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>In addition, because the Primary Account Number is 19 digits, the PAN field on the chip is padded with 1 F (as per EMV). This 'F' must not be printed on the receipt. The terminal fails this test if it prints the 'F' as part of the Primary Account Number on the receipt.</p> <p>For Online tests, the transaction must be sent to VCMS/VMTS/approved host simulator and be approved. If the transaction is declined, it is not necessarily indicative of terminal failure. It could be that the acquiring host system is not capable of accepting 19-digit account numbers. If this is the case, please include comments in the Test Results section within the Compliance Report in Appendix A.</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Card contains a 19 digit account number.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Visa operating regulations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| In addition to ensuring 19 digit account number acceptance by chip reading devices, the purpose of this card is also to gather information on general acceptance of 19-digit Visa PANs in the Visa Acquiring environment. It is recommended that Acquiring systems have the capability to accept cards with 19-digit PANs.                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

### Test Case 3

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.)

Online Testing: In order to conform to the ADVT mandate, it is necessary to perform this test online. See Section 4.2.7: Online Testing for additional information. An online transaction must be initiated to VCMS/VMTS/approved host simulator, Online Card Authentication must pass and a cryptogram associated with Online Issuer Authentication must be provided in the response, received by the terminal, and forwarded to the card.

| Test Purpose & Description                                                                                                                                                                                                                      | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure acceptance of a card that supports the T=1 protocol and supports Issuer Authentication as "Mandatory".</p> <p>Note: While cards can support either the T=0 or T=1 protocols, terminals must support both.</p>                      | <p>Terminal must perform a complete transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The transaction must be sent online to VCMS/VMTS/approved host simulator where VCMS/VMTS/approved host simulator will respond with an Issuer Authentication cryptogram (Authorization Response Cryptogram—ARPC). The terminal must be able to receive the cryptogram in the response data and forward it to the card. If the online transaction results in a decline, the user has failed the test (indicating that the device either did not forward the cryptogram to the card or incorrectly forwarded the cryptogram to the card).</p> <p>The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> |
| Card Conditions                                                                                                                                                                                                                                 | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Card supports the T=1 protocol.                                                                                                                                                                                                                 | EMV 4.1, Book 1, Section 9: Transmission Protocols.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Business Justification                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| In some countries, Visa Issuers prefer the use of the T=1 communications protocol. There are several million T=1 protocol cards in circulation. As such, Visa needs to ensure all terminals are capable of accepting cards using this protocol. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## Test Case 4

Specific Terminal Conditions: This test applies to POS terminals.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure the terminal correctly performs terminal risk management – specifically Floor Limit Checking - in accordance with Visa rules, even when the card is not personalized to request this feature.</p> <p>Note: EMV only requires a terminal to perform Terminal Risk Management (TRM) if the “TRM is to be performed” bit is set in the card’s Application Interchange Profile (AIP). However, Visa requires POS terminals to always perform TRM, even when this AIP bit is not set.</p> | <p>Terminal must perform a complete transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>On entering a transaction amount that exceeds the terminal floor limit, the transaction must be declined offline. An offline or online approval is not acceptable and indicates a failure of the test.</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p>Card is personalized without Terminal Risk Management being set (AIP Byte 1, Bit 4 = 0) and ‘Floor Limit Exceed’ bit being set in the Issuer Action Code – Denial (Byte 4, Bit 8 = 1)</p>                                                                                                                                                                                                                                                                                                      | <p>EMV 4.2, Book 3, Section 10.6: Terminal Risk Management.</p> <p>VIS – Terminal Specification – Section 2.1.6</p>                                                                                                                                                                                                                                                                                                                                                                                                             |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>Visa rules state that Terminal Risk Management should always be performed, irrespective of whether or not Terminal Risk Management is personalized on the card. This card is intended to test the terminal’s compliance with this rule.</p>                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## Test Case 5 (Previously Test Case 22)

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.). Refer to the following table for expected results details.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                          | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>This test has the following objectives:</p> <ol style="list-style-type: none"><li>1. Ensure acceptance of a card that contains multiple (five) applications, each distinguished by a unique suffix appended to the Visa AID.</li><li>2. Ensure acceptance of a card containing a non-ASCII Application Preferred Name.</li></ol> | <p>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>Please see the table below for details of expected results in accordance with the specific terminal scenarios.</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                     | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                               |

Card contains five applications (3 x Visa Credit and 2 x Visa Debit) each with a unique suffix appended to the AID:

- Application #1: Visa Credit is the first priority application. It contains an Application Preferred Name in Cyrillic code (i.e. Виса Кредит) and an Issuer Code Table Index of 05. This application is expired (i.e. its Application Expiration Date is personalized with 31 December 2005) and its IAC – Denial is set to decline transactions based on the expired application.
- Application #2: Visa Debit is the second priority application. It contains an Application Preferred Name in Cyrillic code (i.e. Виса Дебет) and an Issuer Code Table Index of 05. This application is expired (i.e. its Application Expiration Date is personalized with 31 December 2005) and its IAC – Denial is set to decline transactions based on the expired application.
- Application #3: Visa Credit is the third priority application. It contains an Application Preferred Name in Cyrillic code (i.e. Виса Кредит) and an Issuer Code Table Index of 05. This application is expired (i.e. its Application Expiration Date is personalized with 31 December 2005) and its IAC – Denial is set to decline transactions based on the expired application.
- Application #4: Visa Debit is the fourth priority application. It contains an Application Preferred Name in Cyrillic code (i.e. Виса Дебет) and an Issuer Code Table Index of 05. The application has a unique PAN to allow easier identification of online transactions.
- Application #5: Visa Credit is the fifth priority application. It contains an Application Preferred Name in Cyrillic code (i.e. Виса Кредит) and an Issuer Code Table Index of 05. The application has a unique PAN to allow easier identification of online transactions.

EMV 4.1, Book 1, Section 12.3.1: Matching Terminal Applications to ICC Applications.

Terminal Acceptance Device Requirements.

EMV 4.1, Book 1, Section 12.4: Final Selection.

EMV 4.1, Book 4, Section 11.1: Language Selection.

EMV 4.1, Book 4, Section 11.3: Application Selection.

### Business Justification

1. As multi-application cards become more popular, it is important to ensure that terminals are able to correctly identify and select appropriate applications on the card and that the user interface is appropriate for the environment (i.e., the user interface must not confuse the merchant or the cardholder). According to the Terminal Acceptance Device Requirements, "Application Selection Indicators for Visa AIDs must indicate support for Partial selection."
2. For cardholder convenience, Issuers may choose to have the name of the application presented to the cardholder for selection in the cardholder's language (this is the Application Preferred Name). If the terminal supports the relevant alphabet ("Issuer Code Table Index"), it will display the Application Preferred Name rather than the Application Label. Otherwise, the terminal must ignore this feature and display the application name to the cardholder in the format specified in the Application Label.

|                     | Cardholder Selection Supported | Issuer Code Table Index 05 Supported | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------|--------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Terminal Scenario 1 | Yes                            | No                                   | <p>All five payment applications must be displayed to the cardholder in priority order using their Application Label. Since the first three applications are expired, either the fourth (Visa Debit – Priority 4) or fifth (Visa Credit – Priority 5) should be selected. The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> <p>The Visa AID must be printed on the receipt and it is strongly recommended that the Application Label be printed as well.</p> |

|                     | Cardholder Selection Supported | Issuer Code Table Index 05 Supported | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|--------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Terminal Scenario 2 | Yes                            | Yes                                  | <p>All five payment applications must be displayed to the cardholder in priority order using their Application Preferred Name. Since the first three applications are expired, either the fourth (Visa Debit – Priority 4) or fifth (Visa Credit – Priority 5) should be selected. The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> <p>The Visa AID must be printed on the receipt and it is strongly recommended that the Application Preferred Name (i.e. either Виса Кредит or Виса Дебет) be printed as well.</p> |
| Terminal Scenario 3 | No                             | N/A                                  | <p>In accordance with Visa rules, since the terminal does not support the displaying of mutually supported applications or Cardholder Selection, the highest priority application should be selected for the transaction. The transaction will be declined offline because the highest priority application is personalized with an expired application.</p> <p>The Visa AID must be printed on the receipt and it is strongly recommended that the Application Label (Visa Credit) be printed as well.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## Test Case 6

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure acceptance of a Dual Interface card containing the following within the contact payment application:</p> <ul style="list-style-type: none"> <li>• A long PDOL (45 bytes)</li> <li>• Language Preference field with Japanese, Korean and Chinese language codes specified</li> </ul> <p>Note: The Language Preference field is an optional data element that issuers may include on their cards. If included on the card, the terminal must be able to handle the data element field.</p>                                                                        | <p>Terminal must perform a complete transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>In addition, if the terminal supports Japanese, Korean, or Chinese, the terminal must display any cardholder messages in that language (e.g., when prompting the cardholder to agree to the amount of the transaction, the terminal must display messages such as "Amount OK" to the cardholder in one of the above languages).</p> <p>The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/MTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p>Card contains the Language Preference field with three languages in the following priority: Japanese, Korean, and Chinese. Also contains a 45-byte PDOL.</p>                                                                                                                                                                                                                                                                                                                                                                                                              | <p>EMV 4.1, Book 1, Section 11.3.4: Data Field Returned in the Response Message.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>For cardholder convenience, Issuers may use the Language Preference feature to allow cardholders to be presented with terminal messages in their language of choice. The terminal needs to ensure one of the following:</p> <ul style="list-style-type: none"> <li>• If it does not support any of the preferred languages identified on the card, it continues to execute the transaction using the language it supports.</li> <li>• If it does support one of the preferred languages, all terminal displays are presented in the highest priority language.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## Test Case 7 (Previously Test Case 23)

Specific Terminal Conditions: This test applies to all terminal types (POS, ATM, etc.).

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure Terminal Action Codes (TACs) are correctly configured (refer to Chapter 7: Terminal Action Code (TAC) Settings for the TACs that must be loaded into the device).</p> <p>Note: In this test, the Application Usage Control on the card indicates that the card cannot be used for international transactions. This will cause the terminal to set the “service not allowed for card product” bit in the Terminal Verification Results which must result in a declined transaction.</p> | <p>The terminal must decline the transaction offline and the terminal log must show that the Terminal Verification Results, byte 2, bit 5 is set to ‘1’ (Requested Service Not Allowed For Card Product). The terminal log must show that the terminal requests an AAC in the GENERATE AC command and the Authorization Response Code is set to ‘Z1.’</p> <p>The transaction must be declined offline. The terminal fails the test if the transaction is terminated with an error message, approved offline, or sent online for authorization.</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p>Card contains Application Usage Control indicating that the card cannot be used for international transactions and the Issuer Action Code settings contain all zeroes.</p>                                                                                                                                                                                                                                                                                                                       | <p>Terminal Acceptance Device Requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p>For risk management and acceptance purposes, Visa has defined and specified a set of values (referred to as Terminal Action Codes) that must be used on Chip Card Acceptance Devices accepting Visa cards. It is therefore important to ensure these values are being correctly applied. Note: TAC values are mandated by Visa for all devices. The values can be found in the Terminal Acceptance Device Requirements or in Chapter 7 of this document.</p>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### Test Case 8 (Previously Test Case 26)

**Specific Terminal Conditions:** This test applies to all terminals (POS, ATMs, etc.) that support magnetic stripe fallback. Note: Magnetic stripe fallback is NOT mandated at a Visa global level. However, Visa regional offices may apply regional or domestic policies on fallback. Please consult with your Visa regional representative to determine if regional or domestic policies apply.

| Test Purpose & Description                                                                                                                                                                                                              | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure that the terminal properly allows fallback.</p> <p>Note: Because regional and/or domestic rules govern the policy on fallback, check with your Visa regional representative to determine if fallback is allowed.</p>       | <p>The terminal must attempt to read the chip, realize it is faulty, and allow the magnetic stripe to be read.</p> <p>Applicable to Readers that Have Separate Insertion Areas for Chip and Magnetic Stripe Transactions:<br/>The terminal must clearly indicate during the attempt to read the chip that the 'chip cannot be read'. To indicate that fallback is supported, the terminal must provide a message such as "Swipe Magnetic Stripe".</p> <p>Combined Reader (Readers, such as ATMs, where there is a single insertion point for both magnetic stripe and chip transactions): In these devices, fallback to magnetic stripe is transparent to the user. However, the user must ensure that the device properly allows fallback (i.e., a magnetic-stripe transaction). The terminal fails this test when the terminal does not allow the magnetic stripe to be read and/or when the receipt contains the Visa AID (A0000000031010).</p> <p>Note 1: Some fallback procedures allow for more than one attempt to read the chip card.</p> <p>Note 2: This card will not fail until the Get Processing Options command is sent. Some implementations of fallback will not work at this stage, although it is a Visa recommendation that fallback be possible at any point in the transaction (up to and including the Second Generate AC).</p> |
| Card Conditions                                                                                                                                                                                                                         | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Card contains a faulty chip.                                                                                                                                                                                                            | <p>Visa operating regulations.<br/>Terminal Acceptance Device Guide.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Business Justification                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Some Visa regional offices have defined rules around magnetic stripe fallback following failure of chip-based transactions. This card may be used to ensure correct rules are being applied and that the user interface is appropriate. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## Test Case 9 (Previously Test Case 30)

Specific Terminal Conditions: This test applies to POS terminals.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                              | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure that the terminal correctly processes a card containing a CVM that the terminal does not recognize and the CVM is not on the list of CVMs that must be recognized by the terminal (i.e., the first CVM in the list is a Reserved For Future Use CVM, with instructions to apply the next CVM if CVM processing fails).</p> | <p><b>POS Devices Only:</b></p> <p>Terminal must perform a complete transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>When encountering a new CVM (represented by a "Reserved For Future Use" CVM value in the CVM List), the terminal must set the Terminal Verification Results, byte 3, bit 7 to '1' (Unrecognized CVM)</p> <p>Since this CVM list indicates that the Reserved For Future Use CVM must only be performed when supported by the terminal, the terminal must proceed to the remaining CVMs in the CVM list. For POS devices supporting signature, signature must be requested. The Terminal Verification Results, byte 3, bit 8 must be set to '0' (Cardholder Verification Successful).</p> <p>The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/MTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                         | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p>Card contains a CVM value in the "Reserved For Future Use" range.</p>                                                                                                                                                                                                                                                                | <p>EMV 4.1, Book 3, Section 10.5: Cardholder Verification.</p> <p>Terminal Acceptance Device Requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Business Justification                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p>The CVM List of a Visa chip card may contain a method not recognizable by the terminal. If the terminal encounters such a method, it must follow the CVM rules and proceed with the transaction. This card is designed to ensure correct terminal behavior.</p>                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### Test Case 10 (Previously Test Case 31)

Specific Terminal Conditions: This test applies to POS terminals.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                           | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure that the terminal correctly processes a card containing a CVM that the terminal does not recognize and the CVM is not on the list of CVMs that must be recognized by the terminal (i.e., the first CVM in the list is a “Reserved For Future Use CVM” with instructions to stop CVM processing when the CVM fails).</p> | <p><b>POS Devices Only:</b><br/>Terminal must perform a complete transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>When encountering a new CVM (represented by a “Reserved For Future Use” CVM value in the CVM List), the terminal must set the Terminal Verification Results, byte 3, bit 7 to ‘1’ (Unrecognized CVM).</p> <p>Since this CVM list indicates that the Reserved For Future CVM must always be performed and CVM processing must fail if this CVM is not successful, the terminal must set the Terminal Verification Results, byte 3, bit 8 to ‘1’ (Cardholder Verification Failed).</p> <p>The transaction must be declined offline (the card is configured to decline offline for cardholder verification failure).</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                      | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>Card contains a CVM value in the “Reserved For Future Use” range.</p>                                                                                                                                                                                                                                                             | <p>EMV 4.1, Book 3, Section 10.5: Cardholder Verification.</p> <p>Terminal Acceptance Device Requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Business Justification                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>The CVM List of a Visa chip card may contain a method not recognizable by the terminal. If the terminal encounters such a method, it must follow the CVM rules and proceed with the transaction. This card is designed to ensure correct terminal behavior.</p>                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## Test Case 11

Specific Terminal Conditions: This test applies to all terminals that support Cardholder Confirmation (POS, ATMs, etc.).

| Test Purpose & Description                                                                                                                                                                                                                                                                                                | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure correct acceptance of a card containing multiple applications, but with only one application valid for use.</p>                                                                                                                                                                                              | <p>Terminal must perform a complete transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion.</p> <p>The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> <p>This test is applicable to all terminals, irrespective of whether or not 'Cardholder Application Selection' is supported. Only one application is valid for use and therefore should be the one selected.</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                           | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>Card contains three applications, with the last one as the only usable application:</p> <ul style="list-style-type: none"> <li>• Application # 1 – contains an unknown AID</li> <li>• Application # 2 – Blocked</li> <li>• Application # 3 - Valid</li> </ul>                                                          | <p>EMV 4.2, Book 1, Section 12.4: Final Selection.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Business Justification                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>As multi-application cards become more popular, it is important to ensure that terminals are able to correctly identify and select appropriate applications on the card and that the user interface is appropriate for the environment (i.e., the user interface must not confuse the merchant or the cardholder).</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Test Case 12

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

| Test Purpose & Description                                                                                                                                                                                                                                                                           | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure the terminal correctly handles a “Conditions of Use Not Satisfied” (6985) response to the GET PROCESSING OPTIONS command.</p>                                                                                                                                                           | <p>Terminal must send the GET PROCESSING OPTIONS command to the card. The card is personalized to perform the Geographic Restrictions check and respond with “Conditions of Use Not Satisfied” (6985). This must prompt the terminal to return to Application Selection and conclude that there are no applications to use for the transaction. At this time, the terminal must display a “Not Accepted” message or its equivalent (specific message content is based on best practice only and is not mandated). If the terminal accepts the card and completes the transaction, it fails this test.</p> <p>Note that fallback to magnetic stripe processing is an acceptable result.</p> <p>Combined Reader (Readers, such as ATMs, where there is a single insertion point for both magnetic stripe and chip transactions). If the transaction completes in a combined reader, the user must verify that the transaction did not take place using the chip (i.e., ensure that the transaction took place via fallback using the magnetic stripe). The user can ensure this by either checking the logs to ensure that the transaction was magnetic stripe or ensuring that the AID (A0000000031010) does not appear on the receipt.</p> |
| Card Conditions                                                                                                                                                                                                                                                                                      | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p>Card supports the Geographic Restrictions check and is restricted to domestic transactions.</p>                                                                                                                                                                                                   | <p>Terminal Acceptance Device Requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Business Justification                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p>As part of their risk management requirements, an Issuer may choose to restrict use of VSDC cards to domestic environments only. It is therefore important to ensure that if a terminal encounters such a card in an international situation, the appropriate terminal behavior is performed.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## Test Case 13

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

| Test Purpose & Description                                                                                                                                                                                                 | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure acceptance of a card containing proprietary data. The test also ensures correct processing of card with a 6-digit (Offline on Online) PIN.</p>                                                                | <p>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> |
| Card Conditions                                                                                                                                                                                                            | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>Card contains proprietary data. It contains the proprietary tags C2 with a value of "Sample" and DF99 with a value of "80 80" in the PSE. Also contains a proprietary tag "C2" in a record in the application data.</p> | <p>EMV 4.1, Book 3, Section 7.0: Files for Financial Transaction Interchange.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Business Justification                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p>An Issuer may choose to include Discretionary Data in the card. It is important to ensure that terminals encountering cards that contain such data do not react negatively to its presence.</p>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Test Case 14

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                                                                           | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To ensure acceptance of a card where the PDOL requests a long string of data.                                                                                                                                                                                                                                                                                                                                                        | <p>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>In addition, the terminal must send 97 zeroes followed by the Transaction Date in the GET PROCESSING OPTIONS command.</p> <p>The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                                                      | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Card contains a PDOL that requests a long string of data.                                                                                                                                                                                                                                                                                                                                                                            | EMV 4.1, Book 3, Section 5.4: Rules for Using a Data Object List.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Cases have been noted in the past, where (often through personalization discrepancies) the length of a terminal-based data object requested by the card in a Data Object List (DOL) may differ from the actual length of the data object. EMV has specified rules to address this situation. Cards must not be rejected due to this situation. This card is intended to ensure that the specified rules are being correctly applied. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Test Case 15

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure acceptance of a card where a less than 128-byte record has a length that is two bytes. Also ensure acceptance of a card containing a data element with zero length.</p> <p>Note: As per EMV, a data element can have a length field of two bytes even though the data value is less than 128 bytes in length. Usually, the length is one byte when the data value is less than 128 bytes in length, and it is 2 bytes when the data value is greater than 128 bytes in length. Issuers, however, can use a length of 2 bytes even when the data value is less than 128 bytes in length.</p> | <p>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Card contains a data element where the length of a record is two bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | EMV 4.1, Book 3, Annex B.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cases have been noted in the past, where (often through personalization discrepancies) the length field of a data record in the card is formatted as 2-bytes even though the actual record length may be less than 127 bytes (usually if a data record length is 2 bytes, the record contains more than 127 bytes). EMV has specified rules to address this situation. Cards must not be rejected due to this situation. This card is intended to ensure that the specified rules are being correctly applied.                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Test Case 16

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

| Test Purpose & Description                                                                                                                                                                                                                                                                          | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>This test has the following objectives:</p> <ol style="list-style-type: none"> <li>1. Ensure acceptance of a card that contains two applications distinguished by suffixes added to the Visa AID.</li> <li>2. Ensure support of card requirements related to cardholder confirmation.</li> </ol> | <p>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> <p>Devices <u>not</u> supporting Cardholder Selection/Confirmation:<br/>First, the terminal attempts to select the Visa Credit application (this is the highest priority application). Upon recognizing that this application requires cardholder confirmation, the device terminates the transaction and begins processing the second application (Visa Debit)—the second highest priority application. Since Visa Debit does not require cardholder confirmation, the transaction proceeds to completion using the Visa Debit application.</p> <p>Note: Devices that do not support cardholder confirmation must use the Visa Debit application for this test; they must not select and process the transaction using the Visa Credit application. In the event the device erroneously selects the Visa Credit application, the transaction will be declined offline because this application is expired. Use of the Visa Credit application for the transaction and/or an offline decline constitutes a failure of this test.</p> <p>Devices supporting Cardholder Selection/Confirmation:<br/>Both applications should be displayed to the cardholder in priority order (Visa Credit first, followed by Visa Debit). The cardholder should select the "Visa Debit" application (second) for the transaction, since the "Visa Credit" application has expired.</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Note: Since the objective of this test is to ensure that the desired application, as selected by the cardholder, is the one used for the transaction (not the one with the highest priority), an erroneous selection of the Visa Credit application by the terminal will result in a decline. A transaction using the "Visa Credit" application will be declined offline because this application is has expired. Use of the "Visa Credit" application for this transaction and/or an offline decline constitutes a failure of this test.</p> <p>The Visa AID must be printed on the receipt and it is strongly recommended to print the Application Label (Visa Debit) as well.</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>Card contains two applications (Visa Credit and Visa Debit) each with an AID that has a unique suffix:</p> <ul style="list-style-type: none"> <li>• Visa Credit application is the first priority application and requires cardholder confirmation. It has an expired application and the IACs indicate to decline offline for expired application.</li> <li>• Visa Debit application is the second priority application and does not require cardholder confirmation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>EMV 4.1, Book 1, Section 12.3.1: Matching Terminal Applications to ICC Applications.</p> <p>Terminal Acceptance Device Requirements.</p> <p>EMV 4.1, Book 1, Section 12.4: Final Selection.</p> <p>EMV 4.1, Book 4, Section 11.3: Application Selection.</p>                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>3. As multi-application cards become more popular, it is important to ensure that terminals are able to correctly identify and select appropriate applications on the card and that the user interface is appropriate for the environment (i.e., the user interface must not confuse the merchant or the cardholder). According to the Terminal Acceptance Device Requirements, "Application Selection Indicators for Visa AIDs must indicate support for Partial selection.</p> <p>The first application requires cardholder confirmation (through cardholder selection or cardholder confirmation). If the device does not support cardholder selection or cardholder confirmation, it must NOT proceed with a transaction using the first application (Visa Credit). It must stop processing the Visa Credit application and proceed to application selection for the second application (Visa Debit).</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

### Test Case 17

**Specific Terminal Conditions:** This test applies to all terminals (POS, ATMs, etc.).

**Online Testing:** In order to conform to the ADVT mandate, this test must be performed online. See Section 4.2.7: Online Testing for additional information.

| Test Purpose & Description                                                                                                                                                               | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To ensure acceptance of a card containing the minimum set of VSDC data elements and functions (i.e., Magnetic Stripe Image).                                                             | <p>Terminal must perform a complete transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The transaction must be sent online to VCMS/VMTS/approved host simulator and be approved. The transaction must contain the TVR settings for ICC Data is not Missing (byte 1, bit 6 is '0') and Offline Data Authentication Not Performed (byte 1, bit 8 is '1').</p> |
| Card Conditions                                                                                                                                                                          | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Card supports minimum set of VSDC data elements and functions (e.g., Magnetic Stripe Image where neither SDA nor DDA is supported) and the CDOLs contain the minimum set of data.        | <p>EMV 4.1.</p> <p>Terminal Acceptance Device Requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Business Justification                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Issuers may choose to support simple VSDC cards (i.e., cards that support minimum VSDC features and data). This test ensures that terminals accept and successfully process these cards. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Test Case 18 (Previously Test Case 49)

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                                                                | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To ensure acceptance of a T= 1 card, supporting SDA with a certificate of length 1984.                                                                                                                                                                                                                                                                                                                                    | <p>Applicable to Terminals Supporting SDA:<br/>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The terminal log must show that the Transaction Status Information (TSI) byte 1, bit 8 is set to '1' (Offline Data Authentication performed), the Terminal Verification Results, byte 1, bit 8 is set to '0' (Offline Data Authentication was performed), and byte 1, bit 7 is set to '0' (Offline Static Data Authentication did not fail).</p> <p>The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                                           | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <ul style="list-style-type: none"> <li>Card supports T=1 protocol</li> <li>Card supports SDA and contains a certificate that has been signed by the Visa CA Test Key of 1984 bits.</li> </ul>                                                                                                                                                                                                                             | <p>EMV 4.1, Book 3, Section 10.3: Offline Data Authentication</p> <p>Terminal Acceptance Device Requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Visa will shortly be providing Issuer Public Key Certificates to Issuers based on a 1984-bit Visa Certificate Authority Public Key. Concerns were raised regarding some terminals' ability to support keys of this length, particularly terminals that were deployed in the earlier stages of chip migration. This card is intended for use in ensuring that the terminal is capable of supporting an IPK of this length. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

### Test Case 19 (Previously Test Case 50)

**Specific Terminal Conditions:** This test applies to ATMs that support Plus.

**Online Testing:** In order to conform to the ADVT mandate, this test must be performed online. See Section 4.2.7: Online Testing for additional information.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To monitor acceptance of a card with Plus AID, with a Suffix to ensure correct Partial Name Selection behavior.</p> <p>Note: Because regional and/or domestic rules govern the policy on Plus, check with your Visa regional representative for current local rules and regulations.</p>                                                               | <p>Application to ATMs accepting Plus Cards only:</p> <p>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The AID must be printed on the receipt. Note, this should also include any Suffix that is present, since this is part of the AID (A0 00 00 00 03 80 10 01). It is also strongly recommended that the Application Label (Plus) is printed on the receipt as well.</p> <p>The transaction must be sent online and be approved.</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                           | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Card containing the Visa RID with the Plus PIX and a Suffix (A0 00 00 00 03 80 10 01).                                                                                                                                                                                                                                                                    | Visa Global ATM Member Guide, Appendix A: Acquirer Participation Requirements<br>Terminal Acceptance Device Requirements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Business Justification                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| This card is included to assess general acceptance of the Visa RID with the PLUS PIX at ATMs. Plus is a deposit access product that offers worldwide cash access and other around-the-clock financial services through the Visa Global ATM Network. The PLUS Program can be added to any banking card and complements the utility of other Visa products. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## Test Case 20

Specific Terminal Conditions: This test applies to terminals supporting Visa Electron.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure acceptance of a Visa Electron card with non-usable mag stripe data.</p> <p>Note: ATMs must support the Visa Electron AID (A0 00 00 00 03 20 10). Refer to the Terminal Acceptance Device Guide for more details.</p> <p>Merchants signed to accept Visa Electron must support the Visa Electron AID in their terminals (A0 00 00 03 20 10). For other merchants, if the terminal supports the Visa AID, it is required to support the Visa Electron AID unless the merchant specifically chooses to exclude it.</p> <p>To accept Visa Electron cards, the only activity that is required is to add the Visa Electron AID to the terminal. No other activities (coding, adding keys, etc.) are required as terminals that support Visa Electron use the same code and keys as required for the Visa AID.</p>                                                       | <p>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>It is strongly recommended that the Application Label (VISA ELECTRON) or the Application Preferred Name (ELECTRON DE VISA) where appropriate, be printed on the receipt.</p> <p>The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> <p>Note: Consult with your Regional representative for current local rules and regulations related to Visa Electron acceptance.</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p>Card is a Visa Electron card (AID = A0 00 00 00 03 20 10).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Terminal Acceptance Device Requirements. EMV 4.1, Book 4, 6.6.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>This card ensures that the rules governing acceptance of the Electron AID are being applied and that, where a combined reader is used, the terminal does not perform unnecessary processing on the mag stripe data which may hinder chip acceptance. If the Visa AID is supported by the terminal, the Electron AID must also be supported, unless the merchant has specifically chosen to exclude it and ATMs must support the Visa Electron AID.</p> <p>When the magnetic stripe of a card is read and the service code begins with a 2 or a 6, indicating that a chip is present, the terminal must process the transaction using the chip and ignore any other features of the mag stripe data. Failure to apply these rules may lead to acceptance problems with chip-based Electron cards and/or chip-only products which do not have meaningful mag stripe data.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## Test Case 21 (Previously Test Case 32)

Specific Terminal Conditions: This test applies to any terminal supporting Offline PIN.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                  | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure that the terminal correctly processes a card where the PIN Try Limit is exceeded and the card is personalized to proceed to signature or Online PIN when Offline PIN processing fails, or is not supported by the terminal.</p>                                                                                                                                | <p>Applicable to Devices Supporting Offline PIN only:</p> <p>Terminal must perform a complete transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The terminal must set Terminal Verification Results, byte 3, bit 6 to '1' (PIN Try Limit Exceeded) and proceed to the next CVM in the CVM list. It must set the TVR, byte 3, bit 1 to '0' for Cardholder Verification successful. By continuing to process the CVM list, the terminal will verify the cardholder through signature or Online PIN depending on the methods it supports.</p> <p>Note: If the terminal supports Offline Plaintext PIN, but not Signature or Online PIN, then cardholder verification will fail and the transaction will be declined offline. The terminal must set the TVR, byte 3, bit 1 to '1' for Cardholder Verification failed and since the corresponding card IAC is set to decline offline, transaction must be declined offline.</p> <p>If using the tool in an offline mode (no connectivity to VCMS/MTS/approved host simulator) and the amount is above the floor limit, the transaction must attempt to go online and then decline when online is unavailable (due to the IAC and TAC for Floor Limit Exceeded).</p> <p>If using the tool in an online mode (with connectivity to VCMS/MTS/approved host simulator), the transaction must be sent online to VCMS/MTS/approved host simulator. VCMS/MTS/approved host simulator will decline the transaction due to the VCMS/MTS/approved host simulator STIP response for PIN Try Limit Exceeded.</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                             | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>Card supports Offline PIN, the PIN Try Limit is exceeded, and the card is configured to support signature or Online PIN when the PIN try limit is exceeded.</p>                                                                                                                                                                                                          | <p>EMV 4.1, Book 3, Section 10.5.1: Offline PIN Processing.</p> <p>Terminal Acceptance Device Requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Business Justification                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>Cards may have their PIN Try Limit exceeded and still be usable. Issuers may even issue cards with the PIN Try limit already exceeded. It is important that terminals appropriately handle this situation according to EMV and do not perform additional processing which contradicts EMV such as rejecting the card or displaying incorrect or misleading messages.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Test Case 22 (Previously Test Case 33)

Specific Terminal Conditions: This test only applies to terminals supporting Offline PIN.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                  | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure that the terminal correctly processes a card where the PIN Try Limit is exceeded and the card is personalized not to proceed.</p>                                                                                                                                                                                                                              | <p>Applicable to Devices Supporting Offline PIN Only:</p> <p>A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The terminal must set Terminal Verification Results, byte 3, bit 6 to '1' (PIN Try Limit Exceeded) and byte 3, bit 8 to '1' (CVM failed).</p> <p>The transaction must be declined offline (the card is configured to decline offline when the PIN Try Limit is exceeded, so it will return an AAC irrespective of device type or capabilities).</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                             | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>Card supports Offline PIN, the PIN Try Limit is exceeded, and the IAC indicates to decline offline for this condition.</p>                                                                                                                                                                                                                                               | <p>EMV 4.1, Book 3, Section 10.5.1: Offline PIN Processing.</p> <p>Terminal Acceptance Device Requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Business Justification                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p>Cards may have their PIN Try Limit exceeded and still be usable. Issuers may even issue cards with the PIN Try limit already exceeded. It is important that terminals appropriately handle this situation according to EMV and do not perform additional processing which contradicts EMV such as rejecting the card or displaying incorrect or misleading messages.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Test Case 23 (Previously Test Case 39)

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

| Test Purpose & Description                                                                                                                                                                                                             | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure that the terminal correctly processes a card containing a CVM List that supports the combination CVM of signature and Offline PIN.</p> <p>Note: The Offline PIN value is: "1234".<br/>The Online PIN value is "1234".</p> | <p>Terminal must perform a complete transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>If the device supports both Offline PIN and Signature then, by default, it supports the combination CVM of Offline PIN and Signature.. If this is the case, the device must validate the Cardholder's Offline PIN and print the signature line on the receipt.</p> <p>For ATM transactions, online PIN must be used.</p> <p>For devices supporting Online PIN and signature or Online PIN only, online PIN must be used.</p> <p>For devices supporting Offline PIN but not Online PIN, Offline PIN must be used. For devices that only support signature, signature must be used.</p> <p>The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator). In this scenario, the terminal must attempt to send the transaction online and then decline offline when online is not available (due to the IAC and TAC-Default for Floor Limit Exceeded).</p> |
| Card Conditions                                                                                                                                                                                                                        | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p>Card contains a CVM List that supports the combination CVM of Signature/Offline PIN.</p>                                                                                                                                            | <p>EMV 4.1, Book 3, Section 10.5: Cardholder Verification.</p> <p>Terminal Acceptance Device Requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Business Justification                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>Although a combination CVM (i.e., Signature plus Offline PIN) is not commonly used by Visa Issuers, it is important to ensure all terminals accept such a method.</p>                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## Test Case 24 (Previously Test Case 41)

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

Online Testing: In order to conform to the ADVT mandate, it is necessary to perform this test online. See Section 4.2.7: Online Testing for additional information. For this test, ensure that the Terminal Verification Results field in the online message is set to the appropriate value as listed in the success criteria.

| Test Purpose & Description                                                                                                                                                                                                                                                                 | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To determine whether the terminal can handle transactions from a card that contains a 16-digit account number padded with hexadecimal "Fs" to the maximum account length.                                                                                                                  | <p>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The terminal must <u>not</u> print the padded Fs and the full Primary Account Number on the receipt.</p> <p>The transaction must be sent online to VCMS/VMTS/approved host simulator and be approved.</p> |
| Card Conditions                                                                                                                                                                                                                                                                            | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Card is personalized with a 16-digit account number and the PAN field is padded with Fs to the maximum account length.                                                                                                                                                                     | EMV 4.1, All Books, Section 4.3: Data Element Format Conventions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Business Justification                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| There have been cases where Issuers have used the maximum length of the Primary Account Number field by padding the unused portion with 'Fs'. It is important to ensure that all terminals accept any card configured in this way and that the padded 'Fs' are not printed on the receipt. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Test Case 25 (previously Test Case 43)

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

Online Testing: In order to conform to the ADVT mandate, this test must be performed online. See Section 4.2.7: Online Testing for additional information.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To ensure acceptance of a card without a PAN Sequence Number.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The main objective of this test is to ensure that the transaction is forwarded online without a PAN Sequence Number (or with a PAN Sequence Number of all zeros) and Online Card Authentication passes (Field 44.8 = 2). To accomplish this, the transaction must be sent online to VCMS/VMTS/approved host simulator and be approved.</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Card is personalized without a PAN Sequence Number.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Terminal Acceptance Device Requirements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>The PAN Sequence Number is an optional data element that Issuers may use to differentiate card applications having the same Primary Account Number. If the Issuer chooses not to include this data element, it is important to ensure that terminals and the Acquirer Host System have recognized this omission and not erroneously included this data element in the online message.</p> <p>Note: The PAN Sequence Number, if present, must come from the card; the terminal or acquirer must never populate the PAN Sequence Number field in the online or clearing message with a static value or a value from a terminal or acquirer-system table.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

Test Case 26 (previously Test Case 44)

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

Online Testing: In order to conform to the ADVT mandate, this test must be performed online. See Section 4.2.7: Online Testing for additional information.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| To ensure acceptance of a card with a PAN Sequence Number = 11.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  | <p>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The main objective of this transaction is to ensure that the transaction is forwarded online with a PAN Sequence Number of 11 and Online Card Authentication passes (Field 44.8 = 2). To accomplish this, the transaction must be sent online to VCMS/VMTS/approved host simulator and approved.</p> |  |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
| Card is personalized with a PAN Sequence Number of '11'.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  | Terminal Acceptance Device Requirements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| The PAN Sequence Number is an optional data element that Issuers may use to differentiate card applications having the same Primary Account Number. In most cases, when this data element is used, its value is less than '10'. There have been interoperability problems, however, when the value is over 10 because Acquirers have formatted this binary value as hex. The incorrect formatting of this field leads to erroneous Online Card Authentication failures which may lead to declines. This test ensures that a PAN Sequence Number greater than 10 is formatted correctly as a binary value. |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |

**Test Case 27 (Previously Test Case 45)**

**Specific Terminal Conditions:** This test applies to terminals supporting SDA. ATMs and other online-only terminals may be excluded.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                        | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure acceptance of a card with an IPK Certificate based on a 1144-bit Issuer Public Key.</p>                                                                                                                                                                                                                                                                              | <p>Applicable to Devices Supporting SDA only:<br/>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator).</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                   | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>Card is personalized with an Issuer Public Key Certificate based on a 1144-bit Issuer Public Key.</p>                                                                                                                                                                                                                                                                          | <p>EMV 4.1, Book 2, Section 6.1: Keys and Certificates</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Business Justification                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>It has been discovered that there are some faulty RSA cryptographic engines that are unable to handle key lengths not evenly divisible by 16, 8 or 4. With this in mind, a card with an IPK Certificate based on an 1144-bit (i.e. 143 bytes) IPK was proposed. This test ensures that terminals can support cards with IPKs that are not evenly divisible by 16, 8, or 4.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Test Case 28 (Previously Test Case 46)

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

Online Testing: In order to conform to the ADVT mandate, this test must be performed online. See Section 4.2.7: Online Testing for additional information.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                                                   | Expected Results                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure acceptance of a card with the following features:</p> <ul style="list-style-type: none"> <li>• an Issuer URL in the FCI Issuer Discretionary Data</li> <li>• extra Issuer Application Data</li> <li>• an Application Expiration Date = December 31, 2025</li> <li>• a CVM List with no Signature</li> <li>• specific IAC-Denial settings related to PIN activity</li> </ul>                     | <p>Terminal must complete all VSDC application functions from Application Selection through to the 'Amount Entry' prompt. A rejection of the card before the 'Amount Entry' prompt is a failure to the test.</p> <p>A terminal which supports PIN (offline or online) must request PIN entry. The transaction must be sent online to VCMS/VMTS/approved host simulator and be approved.</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                              | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                              |
| Card is personalized with an Issuer URL, Issuer Discretionary Data and Application Expiration Date = December 31, 2025.                                                                                                                                                                                                                                                                                      | EMV 4.1                                                                                                                                                                                                                                                                                                                                                                                     |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                             |
| <p>The Issuer URL was introduced to allow Issuers to specify the location of their Library Servers for Internet service. There are a few known cases where terminals react negatively to cards containing an Issuer URL. This test ensures that terminal can accept a card containing an Issuer URL.</p> <p>The "No Signature" CVM List has been known to cause acceptance problems with some terminals.</p> |                                                                                                                                                                                                                                                                                                                                                                                             |

### Test Case 29 (Previously Test Case 47)

Specific Terminal Conditions: This test applies to all terminals (POS, ATMs, etc.).

| Test Purpose & Description                                                                                                                                                                            | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure correct terminal behavior for a card that is blocked from use.</p> <p>Note: The payment industry best practice recommends that a blocked card must not be accepted through fallback.</p> | <p>Terminal must not accept the card. The card must be rejected immediately after insertion with a message such as "Card Blocked". The terminal fails this test if it accepts the card.</p> <p>Note: Some regions may have regional or domestic fallback rules in place. In these cases, fallback may not be permitted for this test case. Please check with your Visa regional representative for existence of any rules related to fallback.</p> |
| Card Conditions                                                                                                                                                                                       | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Card that is blocked from use.                                                                                                                                                                        | EMV 4.1, Book 1, Section 12.4: Final Selection                                                                                                                                                                                                                                                                                                                                                                                                     |
| Business Justification                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| This card is included to gather information on terminal behavior associated with a blocked card.                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### Test Case 30 (Previously Test Case 48)

**Specific Terminal Conditions:** This test applies to terminals supporting SDA. ATMs and other Online-only terminals may be excluded.

| Test Purpose & Description                                                                                                                                                                                                                                                                                                                                                                                                       | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To ensure acceptance of a card supporting SDA with a certificate of length 1408.</p>                                                                                                                                                                                                                                                                                                                                          | <p>Applicable to Terminals Supporting SDA:<br/>Terminal must perform a complete VSDC transaction without error. A complete transaction is defined as the performance of all selected VSDC functions from Application Selection through to Completion. Error messages (such as Not Accepted or Card Error) are not acceptable and indicate failure of the test.</p> <p>The terminal log must show that the Transaction Status Information (TSI) byte 1, bit 8 is set to '1' (Offline Data Authentication performed), the Terminal Verification Results, byte 1, bit 8 is set to '0' (Offline Data Authentication was performed), and byte 1, bit 7 is set to '0' (Offline Static Data Authentication did not fail).</p> <p>The transaction must be approved offline or approved online. An offline decline is not acceptable and indicates failure of the test. The only situation where a decline is an acceptable response is when both the amount is above the floor limit and tests are being conducted in an offline mode (i.e., no connectivity to VCMS/VMTS/approved host simulator).</p> |
| Card Conditions                                                                                                                                                                                                                                                                                                                                                                                                                  | Reference (Specification/Rule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p>Card supports SDA and contains a certificate that has been signed by the Visa CA Test Key of 1408 bits.</p>                                                                                                                                                                                                                                                                                                                   | <p>EMV 4.1, Book 3, Section 10.3: Offline Data Authentication.</p> <p>Terminal Acceptance Device Requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Business Justification                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>Visa will shortly be providing Issuer Public Key Certificates to Issuers based on a 1408-bit Visa Certificate Authority Public Key. Concerns were raised regarding some terminals' ability to support keys of this length, particularly terminals that were deployed in the earlier stages of chip migration. This card is intended for use in ensuring that the terminal is capable of supporting an IPK of this length.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |



## 6. Test Card Profiles

This section provides details of the test card profiles. This information can be used to create additional test cards. It includes a baseline card profile, followed by a description of variations from the baseline to create individual card profiles.

For an overview of all the test card profiles, please refer to Chapter 4: Test Cases, Section 4.3: Test Case Summary.

## 6.1. Baseline Card

This section outlines the profile for the baseline card. All other cards will use this profile as the basis, with specific parameter changes in accordance with each test case.

### 6.1.1. Magnetic Stripe - Track Data

All cards must contain a personalized magnetic stripe and the magnetic stripe must be encoded with both Track 1 and Track 2 data.

| Data Element                                                                                                                                                                                                  | Track 1 | Track 2 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|
| PAN: 47 61 73 90 01 01 00 10                                                                                                                                                                                  | ✓       | ✓       |
| Last Name: VISA ACQUIRER TEST                                                                                                                                                                                 | ✓       |         |
| First Name: CARD XX (where XX is the card number and changes for each card, e.g., 01, 12, 22, etc.)                                                                                                           | ✓       |         |
| Expiration date: December 2015 (YYMM = 1512)                                                                                                                                                                  | ✓       | ✓       |
| Service Code: 201                                                                                                                                                                                             | ✓       | ✓       |
| Discretionary Data: 11 43 80 05 75 00 00 00 <ul style="list-style-type: none"> <li>PVV Index = 1</li> <li>PVV = 1438</li> <li>Visa Reserved = 00</li> <li>CVV = 575</li> <li>Zero Filling = 000000</li> </ul> | ✓       | ✓       |

Track 1 Example: B4761739001010010^VISA ACQUIRER TEST CARD 01^15122011143800575000000

Track 2 Example: 4761739001010010=15122011143857589

The clear CVV test keys are:

|       |                  |
|-------|------------------|
| CVKA: | 0131517010204061 |
| CVKB: | 91B0D0F180A1C1E0 |

The clear PVV test keys are:

|       |                  |
|-------|------------------|
| PVKA: | 2315208C9110AD40 |
| PVKB: | 15EA4CA20131C2FD |

### 6.1.2. Chip Data

This section outlines the chip data to be encoded on the baseline card.

NOTE: Payment Systems Environment: The baseline test card does not contain the Payment Systems Environment (PSE). Each test card must only be personalized with PSE when explicitly outlined in the individual card profile.

NOTE: Applet Version: When using a Visa applet on a GlobalPlatform card for the test cards, the applet version must be the most recent version available (unless otherwise specified in the test case). The current version uses is VSDC 2.7.1.

| Data Element                    | Tag   | Length | Value                                                                                                                                                                               | DGI   |
|---------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Application Identifier (AID)    | 4F    | 0x 07  | A0 00 00 00 03 10 10<br><br>(AID is added during install time not perso time)                                                                                                       | NA    |
| Application Label               | 50    | 0x 0B  | 56 49 53 41 20 43 52 45 44 49 54<br><br>VISA CREDIT<br><br>(Label contains a space)                                                                                                 | 91 02 |
| Application Preferred Name      | 9F 12 | 0x 0F  | 43 52 45 44 49 54 4F 20 44 45 20 56 49 53 41<br><br>CREDITO DE VISA<br><br>(Preferred Name contains spaces)                                                                         | 91 02 |
| Issuer Code Table Index         | 9F 11 | 0x 01  | 01                                                                                                                                                                                  | 91 02 |
| Application Priority Indicator  | 87    | 0x 01  | 01                                                                                                                                                                                  | 91 02 |
|                                 |       |        |                                                                                                                                                                                     |       |
| Application Interchange Profile | 82    | 0x 02  | 5C 00<br><br>Offline Static Data Authentication supported<br>Cardholder Verification is supported<br>Terminal Risk Management to be performed<br>Issuer Authentication is supported | 91 04 |
| Application File Locator        | 94    | 0x 0C  | 08 01 01 00 10 01 03 00 18 01 02 01                                                                                                                                                 | 91 04 |
|                                 |       |        |                                                                                                                                                                                     |       |
| Cardholder Name                 | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54<br>45 53 54 20 43 41 52 44 20 30 31<br><br>Note: This value changes with each test card.                                               | 01 01 |

| Data Element                                                                                                                                                                                                                                                                                                                                                             | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DGI   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Track 2 Equivalent Data                                                                                                                                                                                                                                                                                                                                                  | 57    | 0x 11  | 47 61 73 90 01 01 00 10 D1 51 22 01 11 43 80<br>44 89                                                                                                                                                                                                                                                                                                                                                                                                                         | 01 01 |
| Track 1 Discretionary Data                                                                                                                                                                                                                                                                                                                                               | 9F 1F | 0x 10  | 31 31 34 33 38 30 30 30 34 34 30 30 30 30 30<br>30                                                                                                                                                                                                                                                                                                                                                                                                                            | 01 01 |
| Static Data<br>Authentication                                                                                                                                                                                                                                                                                                                                            |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |
| <p>Note To Vendors Regarding SDA-Related Data: The SDA-related data elements outlined in this section do not have to be used on the card. These data elements are provided as sample data. If the vendor wants to generate their own data, they may do so as long as the SDA data is valid test data. The sample data was created using the Modulus Exponent method.</p> |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |
| Issuer Public Key<br>Certificate<br><br>(Issuer Public Key of 1152<br>bits signed by the Visa CA<br>Test Key of 1152 bits)<br><br>(for CA index 95)                                                                                                                                                                                                                      | 90    | 0x 90  | 8B 39 01 F6 25 30 48 A8 B2 CB 08 97 4A 42 45<br>D9 0E 1F 0C 4A 2A 69 BC A4 69 61 5A 71 DB<br>21 EE 7B 3A A9 42 00 CF AE DC D6 F0 A7 D9<br>AD 0B F7 92 13 B6 A4 18 D7 A4 9D 23 4E 5C<br>97 15 C9 14 0D 87 94 0F 2E 04 D6 97 1F 4A 20<br>4C 92 7A 45 5D 4F 8F C0 D6 40 2A 79 A1 CE<br>05 AA 3A 52 68 67 32 98 53 F5 AC 2F EB 3C<br>6F 59 FF 6C 45 3A 72 45 E3 9D 73 45 14 61 72<br>57 95 ED 73 09 70 99 96 3B 82 EB F7 20 3C 1F<br>78 A5 29 14 0C 18 2D BB E6 B4 2A E0 0C 02    | 02 01 |
| Issuer Public Key Modulus<br>(length of 1152 bits)<br><br>(This is provided for<br>information only; it is not<br>personalized on the card)                                                                                                                                                                                                                              | N/A   | 0x 90  | A6 87 AF 61 9B 88 CB AD 37 19 03 C8 95 79<br>B5 89 0D 60 5F 90 5B 09 3C 1F 85 68 01 AE 33<br>C1 2E 65 D0 2B 64 45 4D 99 21 46 82 83 ED 39<br>78 35 90 9B CB B2 F6 59 46 08 33 BA AC 1C<br>75 34 3F F6 71 EB 93 F0 49 53 C6 AE F4 28 F0<br>7E E2 8F C9 AB FB 65 CF 6A 96 1B 4A 08 5A<br>F2 97 CD 14 53 CF 47 19 86 88 83 D2 0A 8F 62<br>4E 45 92 0B A3 C9 33 F5 E4 44 7D 4A 32 E5 93<br>6E 5A 13 39 32 9B B4 E8 DD 8B F0 04 4C E4<br>42 8E 24 D0 86 6F AE FD 23 48 80 9D 71    |       |
| Issuer Public Key<br>Exponent                                                                                                                                                                                                                                                                                                                                            | 9F 32 | 0x 01  | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 02 02 |
| Issuer Private Key<br>Exponent<br><br>(This is provided for<br>information only; it is not<br>personalized on the card)                                                                                                                                                                                                                                                  | N/A   |        | 6F 05 1F 96 67 B0 87 C8 CF 66 02 85 B8 FB<br>CE 5B 5E 40 3F B5 92 06 28 15 03 9A AB C9<br>77 D6 1E EE 8A C7 98 2E 33 BB 6B 84 57 02<br>9E 26 50 23 B5 BD 32 77 4E E6 2E B0 22 7C 72<br>BD A3 78 2A A4 4B F2 62 A0 30 E2 84 74 A2<br>C4 E1 C2 B9 50 76 44 D0 79 DD 12 6E 89 F9<br>F5 67 4E BC 47 0D B5 57 53 DE 45 1F 2D 09<br>54 42 3A 47 00 81 4F AE 3F 0D 99 84 45 6D 7C<br>B0 62 35 73 45 7E 0B 7E 85 CC 97 AA D0 AD<br>4A 54 D2 52 35 5C 4B A2 43 51 43 CD EF BB<br>DC 2B |       |
| Issuer Public Key<br>Remainder                                                                                                                                                                                                                                                                                                                                           | 92    | 0x 24  | 33 F5 E4 44 7D 4A 32 E5 93 6E 5A 13 39 32 9B<br>B4 E8 DD 8B F0 04 4C E4 42 8E 24 D0 86 6F<br>AE FD 23 48 80 9D 71                                                                                                                                                                                                                                                                                                                                                             | 02 02 |
| Certification Authority<br>Public Key Index                                                                                                                                                                                                                                                                                                                              | 8F    | 0x 01  | 95<br><br>(Visa CA Test Key of 1152 bits)                                                                                                                                                                                                                                                                                                                                                                                                                                     | 02 02 |
| Certificate Expiration Date<br><br>(for information only)                                                                                                                                                                                                                                                                                                                | N/A   |        | 12 15<br>December 2015                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |

| Data Element                                                                                                                                                                                                                                               | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | DGI   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Signed Static Application Data<br><br>Note: The Signed Application Data is created using the PAN and PAN Sequence Number only. This allows the same Signed Application Data to be used on cards with different data elements (e.g., different IACs, etc.). | 93    | 0x 90  | 91 9D 6C 21 0B 39 81 D1 C9 9B 3A D5 5E DF<br>36 A1 38 FF AD 54 D8 38 FA 40 62 2A B9 70 46<br>E0 5E A6 E6 23 0A B8 9D 5B E8 71 11 4E B5<br>43 1B 97 40 3B 8C 3D 2D 4C A9 BB 62 5A C1<br>3F D8 C6 B8 25 43 36 56 CB 56 55 7A AC 39<br>6D 94 5F 6D 40 14 FB 6E 71 E8 DB EA 74 B2<br>85 E9 CF 3F CE AB DF A6 1D 5A 4B E1 6D AA<br>A4 33 F7 F2 64 4B 17 8A 7D D9 3D A9 8B B9<br>D1 0E 84 29 8B DB 6B 6A E0 2D 04 E6 E5 55<br>8C 77 E7 9F 82 C9 E0 46 DF 82 1D D0 27 7A<br>B9 D0 0C                                                                                                                                                                                                              | 02 03 |
| Application Primary Account Number (PAN) (Signed)                                                                                                                                                                                                          | 5A    | 0x 08  | 47 61 73 90 01 01 00 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 03 01 |
| Application PAN Sequence Number (Signed)                                                                                                                                                                                                                   | 5F 34 | 0x 01  | 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 03 01 |
| Application Currency Code                                                                                                                                                                                                                                  | 9F 42 | 0x 02  | 08 40                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 03 02 |
| Application Effective Date                                                                                                                                                                                                                                 | 5F 25 | 0x 03  | 09 07 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 03 02 |
| Application Expiration Date                                                                                                                                                                                                                                | 5F 24 | 0x 03  | 15 12 31                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 03 02 |
| Application Version Number                                                                                                                                                                                                                                 | 9F 08 | 0x 02  | 00 8C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 03 02 |
| Issuer Country Code                                                                                                                                                                                                                                        | 5F 28 | 0x 02  | 08 40                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 03 02 |
| Service Code                                                                                                                                                                                                                                               | 5F 30 | 0x 02  | 02 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 03 02 |
| Application Usage Control                                                                                                                                                                                                                                  | 9F 07 | 0x 02  | FF 00 <ul style="list-style-type: none"> <li>Valid for domestic cash transactions</li> <li>Valid for international cash transactions</li> <li>Valid for domestic goods</li> <li>Valid for international goods</li> <li>Valid for domestic services</li> <li>Valid for international services</li> <li>Valid at ATMs</li> <li>Valid at terminals other than ATMs</li> </ul>                                                                                                                                                                                                                                                                                                                 | 03 02 |
| Cardholder Verification Method List (CVM)                                                                                                                                                                                                                  | 8E    | 0x 0E  | 0000 0000 0000 0000 1E03 0203 1F00 <ul style="list-style-type: none"> <li>Amount X = 00000000</li> <li>Amount Y = 00000000</li> <li>CVM Code 1 '1E03'               <ul style="list-style-type: none"> <li>Signature, if terminal supports CVM</li> <li>Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>CVM Code 2 '0203'               <ul style="list-style-type: none"> <li>Online PIN, if terminal supports CVM</li> <li>Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>CVM Code 3 '1F00'               <ul style="list-style-type: none"> <li>No CVM Required, Always</li> <li>(Cannot fail CVM).</li> </ul> </li> </ul> | 03 02 |

| Data Element                                    | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                     | DGI   |
|-------------------------------------------------|-------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Card Risk Management Data Object List 1 (CDOL1) | 8C    | 0x 15  | 9F 02 06 9F 03 06 9F 1A 02 95 05 5F 2A 02 9A 03 9C 01 9F 37 04<br><br><ul style="list-style-type: none"> <li>Amount, Authorized</li> <li>Amount, Other</li> <li>Terminal Country Code</li> <li>Terminal Verification Results</li> <li>Transaction Currency Code</li> <li>Transaction Date</li> <li>Transaction Type</li> <li>Unpredictable Number</li> </ul>                                                              | 03 02 |
| Card Risk Management Data Object List 2 (CDOL2) | 8D    | 0x 17  | 8A 02 9F 02 06 9F 03 06 9F 1A 02 95 05 5F 2A 02 9A 03 9C 01 9F 37 04<br><br><ul style="list-style-type: none"> <li>Authorization Response Code</li> <li>Amount, Authorized</li> <li>Amount, Other</li> <li>Terminal Country Code</li> <li>Terminal Verification Results</li> <li>Transaction Currency Code</li> <li>Transaction Date</li> <li>Transaction Type</li> <li>Unpredictable Number</li> </ul>                   | 03 02 |
| Issuer Action Code – Default                    | 9F 0D | 0x 05  | F0 40 00 88 00<br><br><ul style="list-style-type: none"> <li>Offline data authentication not performed</li> <li>Offline Static Data Authentication failure</li> <li>Chip data missing</li> <li>PAN on terminal exception file</li> <li>Expired application</li> <li>Transaction exceed floor limit</li> <li>Merchant forced transaction online</li> </ul>                                                                 | 03 02 |
| Issuer Action Code – Denial                     | 9F 0E | 0x 05  | 00 10 00 00 00<br><br><ul style="list-style-type: none"> <li>Service not allowed for card product</li> </ul>                                                                                                                                                                                                                                                                                                              | 03 02 |
| Issuer Action Code – Online                     | 9F 0F | 0x 05  | F0 40 00 98 00<br><br><ul style="list-style-type: none"> <li>Offline data authentication not performed</li> <li>Offline Static Data Authentication failure</li> <li>Chip data missing</li> <li>PAN on terminal exception file</li> <li>Expired application</li> <li>Transaction exceeds floor limit</li> <li>Transaction selected randomly for online transmission</li> <li>Merchant forced transaction online</li> </ul> | 03 02 |
|                                                 |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                           |       |
| Issuer Application Data                         | 9F 10 | 0x 07  | Details Below:                                                                                                                                                                                                                                                                                                                                                                                                            | 92 00 |
| Length                                          |       | 0x 01  | 06                                                                                                                                                                                                                                                                                                                                                                                                                        | 92 00 |
| Derivation Key Index                            |       | 0x 01  | 01                                                                                                                                                                                                                                                                                                                                                                                                                        | 92 00 |
| Cryptogram Version No.                          |       | 0x 01  | 0A                                                                                                                                                                                                                                                                                                                                                                                                                        | 92 00 |

| Data Element                                                                              | Tag   | Length | Value                                                                                                                                     | DGI   |
|-------------------------------------------------------------------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Card Verification Results (CVR)                                                           |       | 0x 04  | 03 00 00 00                                                                                                                               | 92 00 |
| Master MDK A<br>(Note: The same master key is used for ARQC, MAC, ENC)                    |       |        | 2315 208C 9110 AD40<br><br>Note: This MDK is stored in the Visa Certification Management System (VCMS) or Visa Member Test system (VMTS). | 91 03 |
| Master MDK B<br>(Note: The same master key is used for ARQC, MAC, ENC)                    |       |        | 2315 208C 9110 AD40<br><br>Note: This MDK is stored in the Visa Certification Management System (VCMS) or Visa Member Test system (VMTS). | 91 03 |
| UDK A (for ARQC, MAC & ENC)                                                               |       |        | 52D0BBC3 46840A36                                                                                                                         | 80 00 |
| UDK B (for ARQC, MAC & ENC)                                                               |       |        | E8FD234D DE28DEF0                                                                                                                         | 80 00 |
| Application Default Action                                                                | 9F 52 | 0x 04  | 00 00 00 00                                                                                                                               | 0E 01 |
| Issuer Authentication Indicator<br>(0=optional 1=mandatory)                               | 9F 56 | 0x 01  | 00                                                                                                                                        | 0E 01 |
| Geographic Indicator<br>bit 8 = 1:valid for Domestic<br>bit 7 = 1:valid for International | 9F 55 | 0x 01  | C0                                                                                                                                        | 0E 01 |

The following tags are not found personalized on the baseline test card, but some of the other images may require one or more of these tags.

| Data Element                                         | Tag   | Length | Value                                                                                | DGI        |
|------------------------------------------------------|-------|--------|--------------------------------------------------------------------------------------|------------|
| Processing Options Data Object List (PDOL)           | 9F 38 | VAR    | 9F 1A 02                                                                             | 91 02      |
| File Control Information – Issuer Discretionary Data | BF 0C | 0x 0D  | D1 03 31 32 33 C2 06 53 41 4D 50 4C 45                                               | 91 02      |
| Issuer Country Code                                  | 9F 57 | 0x 02  | 08 40                                                                                | 0D 01      |
| Lower Consecutive Offline Limit                      | 9F 58 | 0x 01  | Not used in this document                                                            | 0D 01      |
| Upper Consecutive Offline Limit                      | 9F 59 | 0x 01  | Not used in this document                                                            | 0D 01      |
| Consecutive Transaction Limit International          | 9F 53 | 0x 01  | Not used in this document                                                            | 0D 01      |
| Cumulative Total Transaction Amount Limit            | 9F 54 | 0x 06  | Not used in this document                                                            | 0D 01      |
| PIN Try Limit                                        | --    | 0x 01  | 7F                                                                                   | 80 10/9010 |
| PIN Try Counter                                      | 9F 17 | 0x 01  | 03                                                                                   | 90 10      |
| Reference PIN                                        | --    | 0x 08  | 24 12 34 FF FF FF FF FF<br><br>(Shows the Reference PIN block.<br>The Pin is = 1234) | 80 10      |

The following fields are “Internal Card Data”. These fields are setup by the application during personalization. No external data is provided to the application for the personalization of these values. These fields are used by the application during a transaction.

| Data Element                                  | Tag   | Length | Value                                                                                           | DGI |
|-----------------------------------------------|-------|--------|-------------------------------------------------------------------------------------------------|-----|
| Last Transaction Incomplete Indicator         |       | 1 bit  | Used during transaction. Application allocates during personalization with data input.          | NA  |
| Online Requested by Card Indicator            |       | 1 bit  | Used during transaction. Application allocates during personalization without data input.       | NA  |
| Offline Decline Requested by Card Indicator   |       | 1 bit  | Used during transaction. Application allocates during personalization without perso data input. | NA  |
| Issuer Authentication Failure Indicator       |       | 1 bit  | Used during transaction. Application allocates during personalization without perso data input. | NA  |
| Static Data Authentication Failure Indicator  |       | 1 bit  | Used during transaction. Application allocates during personalization without perso data input. | NA  |
| Dynamic Data Authentication Failure Indicator |       | 1 bit  | Used during transaction. Application allocates during personalization without perso data input. | NA  |
| Application Transaction Counter               | 9F36  | 0x 02  | Used during transaction. Application allocates during personalization without perso data input. | NA  |
| Issuer Script Command Counter                 |       | 4 bits | Used during transaction. Application allocates during personalization without data input.       | NA  |
| Last Online ATC Register                      | 9F 13 | 0x 02  | Used during transaction. Application allocates during personalization without perso data input. | NA  |
| Consecutive Transaction Counter International |       | 0x 01  | Used during transaction. Application allocates during personalization without perso data input. | NA  |
| Cumulative Total Transaction Amount           |       | 0x 06  | Used during transaction. Application allocates during personalization without perso data input. | NA  |

## 6.2. Test Card 1

This section outlines the profile for Test Card 1 – Baseline card with a Unique Primary Account Number and mandatory Issuer Authentication.

VSDC Applet Version: 2.4.0

Changes to make from baseline card:

Note: This must be a T=0 card and it must not contain the Payment System Environment.

### 6.2.1. Magnetic Stripe - Track Data

Magnetic Stripe Track 1:  
B4761739001010119^VISA ACQUIRER TEST CARD  
01^151220117589005400000000

Magnetic Stripe Track 2:  
4761739001010119=15122011758954089

### 6.2.2. Chip Data:

| Data Element                                                                                                                                                                                                                                           | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | DGI   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                                                                                                                                                                                                                                        | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45<br>53 54 20 43 41 52 44 20 30 31<br><br>(VISA ACQUIRER TEST CARD 01)                                                                                                                                                                                                                                                                                                                                                       | 01 01 |
| Track 2 Equivalent Data                                                                                                                                                                                                                                | 57    | 0x 11  | 47 61 73 90 01 01 01 19 D1 51 22 01 17 58 98 93<br>89                                                                                                                                                                                                                                                                                                                                                                                                                      | 01 01 |
| Track 1 Discretionary Data                                                                                                                                                                                                                             | 9F 1F | 0x 10  | 31 37 35 38 39 30 30 38 39 33 30 30 30 30 30 30                                                                                                                                                                                                                                                                                                                                                                                                                            | 01 01 |
|                                                                                                                                                                                                                                                        |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| Application Primary Account Number (Signed)                                                                                                                                                                                                            | 5A    | 0x 0A  | 47 61 73 90 01 01 01 19                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 03 01 |
|                                                                                                                                                                                                                                                        |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| Signed Static Application Data<br>Note: The Signed Application Data is created using the PAN and PAN Sequence Number only. This allows the same Signed Application Data to be used on cards with different data elements (e.g., different IACs, etc.). | 93    | 0x 90  | 8F 48 E6 91 40 34 94 05 76 88 B2 2B 23 7E F0<br>EE 40 23 85 39 BB 9D E9 9A 97 DC 2C 47 B3 42<br>7F 29 26 51 BF 53 8B B8 9C 04 6F 86 CE 05 C5<br>57 8C C1 20 07 F3 D4 F8 43 68 47 66 2D F7 8C<br>B3 85 AF B8 15 B7 E2 80 97 C0 A5 20 F6 7D 42<br>67 A3 53 1E 6C 7C EB 76 10 B1 13 A3 69 C0 D5<br>89 25 15 FE 06 2B F7 BA 16 DA 57 C0 40 95 24<br>50 07 E1 B3 8B 78 23 B9 AB 4A 51 77 A1 83 48<br>AD 4C E7 A8 E9 9F 44 04 C2 56 B4 06 12 86 79<br>4D 8B 41 B2 CF 42 E4 02 2B | 02 03 |
|                                                                                                                                                                                                                                                        |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| Application Default Action Code                                                                                                                                                                                                                        | 9F 52 | 0x 04  | 60 00 00 00<br><br><ul style="list-style-type: none"> <li>If Issuer Authentication performed and failed, decline transaction</li> <li>If Issuer Authentication is mandatory and no ARPC received, decline transaction</li> </ul>                                                                                                                                                                                                                                           | 0E 01 |

|                                 |       |        |                                                                                                                                                                                                                                                    |       |
|---------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|                                 |       |        |                                                                                                                                                                                                                                                    |       |
| Issuer Authentication Indicator | 9F 56 | 1 byte | 80<br>(Issuer Authentication Mandatory)                                                                                                                                                                                                            | 0E 01 |
|                                 |       |        |                                                                                                                                                                                                                                                    |       |
| UDK A (for ARQC, MAC, ENC)      | N/A   |        | 31 E6 FB A5 03 1E 57C7<br>Because the PAN is different from the baseline card, the UDK is also different (as the UDK is based on the PAN and PAN Sequence Number). The MDK associated with the baseline card, however, is also used for this BIN.  | 80 00 |
| UDK B (for ARQC, MAC, ENC)      | N/A   |        | 0B 19 BF 5F 43 16 4E 44<br>Because the PAN is different from the baseline card, the UDK is also different (as the UDK is based on the PAN and PAN Sequence Number). The MDK associated with the baseline card, however, is also used for this BIN. | 80 00 |

### 6.3. Test Card 2 (Previously Test Card 21)

This section outlines the profile for Test Card 2 - Card containing a 19-digit PAN.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

#### 6.3.1. Magnetic Stripe - Track Data

Track 1:  
B4427808001112223337^VISA ACQUIRER TEST CARD  
02^15122011822200646000000

Track 2:  
4427808001112223337=151220118222646

#### 6.3.2. Chip Data:

| Data Element                                | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                   | DGI   |
|---------------------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                             | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45<br>53 54 20 43 41 52 44 20 30 32<br><br>(VISA ACQUIRER TEST CARD 02)                                                                                                                                                                                                                                                                                                    | 01 01 |
| Track 2 Equivalent Data                     | 57    | 0x 12  | 44 27 80 80 01 11 22 23 33 7D 15 12 20 11 82 22<br>98 3F                                                                                                                                                                                                                                                                                                                                                                | 01 01 |
| Track 1 Discretionary Data                  | 9F 1F | 0x 10  | 31 38 32 32 32 30 30 39 38 33 30 30 30 30 30                                                                                                                                                                                                                                                                                                                                                                            | 01 01 |
|                                             |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| Application Primary Account Number (Signed) | 5A    | 0x 0A  | 44 27 80 80 01 11 22 23 33 7F<br><br>(This is a 19 digit account number)                                                                                                                                                                                                                                                                                                                                                | 03 01 |
|                                             |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| Issuer Public Key Certificate               | 90    | 0x 80  | 01 30 67 3B 92 8A B3 9C 7D EA D7 08 F1 72 41<br>85 29 88 2C DD 63 6F B1 CC A2 08 06 CB 5B 89<br>16 1D 8F 99 64 5E 45 D0 EB 3A 41 87 44 0F 3B<br>61 3D B8 5A 5E C6 DE A6 0E 68 BB 07 B4 9D 9B<br>35 F0 69 04 14 C6 B9 85 1C 0F 50 3B 19 9C 2B<br>08 3E 9E 8D 8B 25 57 EE 41 03 EF 96 46 EC CC<br>F6 C9 AF 90 9F 66 E4 C9 2D 45 D1 16 24 4D E0<br>CC 0B F3 D9 BE 86 78 3E 35 C4 7E 39 E9 7D B6<br>C5 94 C6 AA D6 F1 63 6A | 02 01 |
| Certification Authority Public Key Index    | 8F    | 0x 01  | 99<br><br>(Visa CA Test Key of 1024 bits)                                                                                                                                                                                                                                                                                                                                                                               | 02 02 |
| Signed Application Data                     | 93    | 0x 80  | 29 CE CE A1 9A 43 6F EE A1 72 87 4C 8D D5 D9<br>13 E1 4A 05 79 1F A6 4B C1 E5 97 BD A3 A6 B9<br>82 2E C5 8C FF 5D A2 AC 2D 10 26 D9 BF 4B 82<br>FC CF 4D C1 5A 99 A9 C5 56 06 20 8A B9 5D 71<br>6E 3C FA EA BB 39 27 00 FA F2 8A 20 D2 06 DD<br>0A 63 5B 95 95 E8 4B 0D A8 26 00 1B 5D 44 0E<br>19 53 84 F7 EB AF 70 59 B3 48 8E 1F 44 3C F3<br>4A DA 07 C6 91 6D 59 2B 2C 72 C1 E0 E8 BC F2<br>A1 A7 DC 6C 46 80 76 81 | 02 03 |

| Data Element                                              | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                           | DGI   |
|-----------------------------------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Issuer Public Key Modulus                                 |       | 0x 80  | B0 21 8A A1 21 57 02 EA E1 CD 8D 8C F4 1C B4 62 60 8F F2 33 B1 8E 1D 48 59 7D 20 FB FE F9 C9 CF 9A 63 4E 0C 22 26 E4 F1 A6 CE 0D E2 5D 15 0F 4A F2 EA D3 D8 C9 98 44 F9 49 0B 05 66 85 2B AC AA 44 AA 6B 8A 6A 82 48 9C B8 61 E2 31 0C 2B E6 A2 55 7D 6A 3A 3F 62 14 B2 FE A9 DC BE 27 98 B6 40 96 A5 02 89 98 20 61 23 E4 21 9C C2 75 AC 09 1D A6 5A 17 74 C4 22 BB EF 6D DE 18 0F E3 F0 C7 3F |       |
| Issuer Public Key Exponent                                | 9F 32 | 0X 01  | 03                                                                                                                                                                                                                                                                                                                                                                                              | 02 02 |
| Issuer Private Key Exponent                               | N/A   | 0x 80  | 75 6B B1 C0 C0 E4 AC 9C 96 89 09 08 A2 BD CD 96 EB 0A A1 77 CB B4 13 85 90 FE 15 FD 54 A6 86 8A 66 EC DE B2 C1 6F 43 4B C4 89 5E 96 E8 B8 B4 DC A1 F1 E2 90 86 65 83 50 DB 5C AE 44 58 C7 C8 70 66 D6 06 43 DF BB BA A4 64 6D C8 3B 62 C3 3F DD 59 93 53 CA 63 5C 12 CD D0 9C 3C 5C 48 72 CF 17 09 BA 07 9F 79 80 D7 B2 2D 10 17 64 59 C7 16 01 D9 23 C7 D1 FC 57 32 4E C1 D8 4C 09 39 E0 CF 9B |       |
| Issuer Public Key Remainder                               | 92    | 0x 24  | 27 98 B6 40 96 A5 02 89 98 20 61 23 E4 21 9C C2 75 AC 09 1D A6 5A 17 74 C4 22 BB EF 6D DE 18 0F E3 F0 C7 3F                                                                                                                                                                                                                                                                                     | 02 02 |
| Certificate Expiration Date<br><br>(for information only) | N/A   |        | December 2030                                                                                                                                                                                                                                                                                                                                                                                   |       |
|                                                           |       |        |                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| UDK A (for ARQC, MAC, ENC)                                | N/A   |        | 07 1A 1A 08 20 EF 23 A4<br><br>(Because the PAN is different from the baseline card, the UDK is also different (as the UDK is based on the PAN and PAN Sequence Number). The MDK associated with the baseline card, however, is also used for this BIN.)                                                                                                                                        | 80 00 |
| UDK B (for ARQC, MAC, ENC)                                | N/A   |        | F8 75 29 B5 DA 52 10 85<br><br>(Because the PAN is different from the baseline card, the UDK is also different (as the UDK is based on the PAN and PAN Sequence Number). The MDK associated with the baseline card, however, is also used for this BIN.)                                                                                                                                        | 80 00 |

## 6.4. Test Card 3

This section outlines the profile for Test Card 3 – T=1 card with mandatory Issuer Authentication supported and DDA.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.4.1. Magnetic Stripe – Track Data

Magnetic Stripe Track 1:  
B4761739001010036^VISA ACQUIRER TEST CARD 03^151220111844007990000000

Magnetic Stripe Track 2:  
4761739001010036=15122011184479989

### 6.4.2. Chip Data:

- o Card must support the T=1 (rather than T=0) protocol.

| Data Element                                      | Tag   | Length | Value                                                                                                                                                                                                                                        | DGI   |
|---------------------------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45<br>53 54 20 43 41 52 44 20 30 33<br><br>(VISA ACQUIRER TEST CARD 03)                                                                                                                         | 01 01 |
| Track 2 Equivalent Data                           | 57    | 0x 11  | 47 61 73 90 01 01 00 36 D1 51 22 01 11 84 40 48<br>89                                                                                                                                                                                        | 01 01 |
| Track 1 Discretionary Data                        | 9F 1F | 0x 10  | 31 31 38 34 34 30 30 30 34 38 30 30 30 30 30 30                                                                                                                                                                                              | 01 01 |
| Application Interchange Profile                   | --    | 0x 02  | 7C 00<br>(DDA, SDA, Cardholder Verification, Terminal Risk Management & Issuer Authentication performed and supported)                                                                                                                       | 07 02 |
| Application Primary Account Number (PAN) (Signed) | 5A    | 0x 08  | 47 61 73 90 01 01 00 36                                                                                                                                                                                                                      | 03 01 |
| IAC Denial                                        | 9F 0E | 0x 05  | 08 00 00 00 00<br><br>• Decline Offline for Dynamic Data Authentication failure                                                                                                                                                              | 03 02 |
| Cardholder Verification Method                    | 8E    | 0x 12  | 0000 0000 0000 0000 0403 0103 1E03 0203<br>1F00<br><br>Amount X = 00000000<br>Amount Y = 00000000<br><br>• CVM Code 1 '0403'<br>o Offline Enciphered PIN, if Terminal supports<br>o Fail cardholder Verification if this CVM is unsuccessful | 03 02 |

| Data Element                                        | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | DGI   |
|-----------------------------------------------------|-------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|                                                     |       |        | <ul style="list-style-type: none"> <li>CVM Code 2 '0103'               <ul style="list-style-type: none"> <li>Offline (Plaintext) PIN, if terminal supports CVM</li> </ul> </li> <li>Fail cardholder Verification if this CVM is unsuccessful CVM Code 3 '1E03'               <ul style="list-style-type: none"> <li>Signature, if terminal supports CVM</li> <li>Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>CVM Code 4 '0203'               <ul style="list-style-type: none"> <li>Online PIN, if terminal supports CVM</li> <li>Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>CVM Code 5 '1F00'               <ul style="list-style-type: none"> <li>No CVM Required, Always</li> <li>Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> </ul> |       |
| ICC Public Key Certificate                          | 9F 46 | 0x 90  | 86 8A 4E BE 29 CC 89 06 81 0F 90 F4 5B 7C 2D CA 73 D8 C6 3C 8A B5 8E 2D 44 9F 2D AB F6 21 DF 21 BA 99 7C 38 3D FC AA 75 E1 64 A7 0F 65 45 03 94 3B 2D E5 CB F0 90 B9 1A 0B 30 93 03 6D FA 74 FA 0C 2B B9 68 92 8F 65 EC EB 01 D8 BF 38 FA DC 34 2A E9 94 C3 A5 67 7F C5 AD 3A 79 41 DC 5F 71 59 22 E3 57 12 E6 6C 58 10 BF 2F 98 69 4A 70 BB 9A 4C 20 CA B5 12 CF E8 D1 FF 84 74 F2 88 63 C7 9C 19 AE E1 4D 4E 10 4C 46 26 B9 62 BB 07 D1 EE 15                                                                                                                                                                                                                                                                                                                                                                                            | 02 04 |
| ICC Public Key Exponent                             | 9F 47 | 0x 01  | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 02 02 |
| ICC Public Key Remainder                            | 9F 48 | 0x 2A  | FB DA DA 20 08 2F D6 D0 43 9B C9 08 5D 12 F4 F9 06 AF 8D A6 60 DC 8A 9A A5 A6 B4 B5 92 29 92 D7 65 06 16 0E CB 3F 9B 53 27 C5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 02 02 |
| ICC Private (Secret) Key Exponent                   | --    | 0x 90  | 82 79 9D A1 F1 B9 E2 AA 81 0D 0C 2F 8B 0D 31 E6 7D 5E E4 2E DA 60 15 E2 EA 7D 26 93 58 B6 3C B7 F0 D5 4D 29 C6 B7 3C F5 C1 3F AF 3C 04 94 B2 00 A2 BC C8 CB 49 23 9C 3E 5D 36 17 6E 16 E0 D6 9A 06 EB B4 27 45 F2 A8 CC 31 F2 A2 F4 90 CD 46 BF 18 E3 00 F0 54 D0 D4 81 E3 CF AE 10 0F 22 93 8D 08 42 E8 9A AB 34 76 BB CC 1B D4 3E 18 5C DF DC 80 48 8D EE 33 E2 93 43 7E 54 57 89 30 CD B2 96 F2 69 50 C1 40 07 33 69 28 80 E1 F0 D4 55 07 33                                                                                                                                                                                                                                                                                                                                                                                            | 81 01 |
| Dynamic Data Authentication Data Object List (DDOL) | 9F 49 | 0x 03  | 9F 37 04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 02 02 |
| ICC Public Key Modulus                              |       | 0x 90  | C3 B6 6C 72 EA 96 D3 FF C1 93 92 47 50 93 CA D9 BC 0E 56 46 47 90 20 D4 5F BB B9 DD 05 11 5B 13<br>E9 3F F3 BE AA 12 DB 70 A1 DF 86 DA 06 DF 0B 00 F4 1B 2D 30 ED B5 6A 5D 8B D1 23 25 22 51 41 E7 0A 61 8E 3A E8 EB FD 34 0A DD 68 9B 27 E5 FF 1F 64 AD 29 41 A6 31 D5 91 B7 03 45 5C A0 A0 86 F2 C4 E7 42 DD ED D2 FB DA DA 20 08 2F D6 D0 43 9B C9 08 5D 12 F4 F9 06 AF 8D A6 60 DC 8A 9A A5 A6 B4 B5 92 29 92 D7 65                                                                                                                                                                                                                                                                                                                                                                                                                    | 81 03 |

| Data Element                                                                                                                                                                                                                                           | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | DGI        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|                                                                                                                                                                                                                                                        |       |        | 06 16 0E CB 3F 9B 53 27 C5                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |
| Issuer Public Key Certificate<br>(Issuer Public Key of 1152 bits signed by the Visa CA Test Key of 1152 bits)<br><br>(for CA index 95)                                                                                                                 | 90    | 0x 90  | 8B 39 01 F6 25 30 48 A8 B2 CB 08 97 4A 42 45<br>D9 0E 1F 0C 4A 2A 69 BC A4 69 61 5A 71 DB 21<br>EE 7B 3A A9 42 00 CF AE DC D6 F0 A7 D9 AD<br>0B F7 92 13 B6 A4 18 D7 A4 9D 23 4E 5C 97 15<br>C9 14 0D 87 94 0F 2E 04 D6 97 1F 4A 20 4C 92<br>7A 45 5D 4F 8F C0 D6 40 2A 79 A1 CE 05 AA 3A<br>52 68 67 32 98 53 F5 AC 2F EB 3C 6F 59 FF 6C<br>45 3A 72 45 E3 9D 73 45 14 61 72 57 95 ED 73<br>09 70 99 96 3B 82 EB F7 20 3C 1F 78 A5 29 14<br>0C 18 2D BB E6 B4 2A E0 0C 02 | 02 01      |
| Issuer Public Key Modulus (length of 1152 bits)<br><br>(This is provided for information only; it is not personalized on the card)                                                                                                                     | N/A   | 0x 90  | A6 87 AF 61 9B 88 CB AD 37 19 03 C8 95 79 B5<br>89 0D 60 5F 90 5B 09 3C 1F 85 68 01 AE 33 C1<br>2E 65 D0 2B 64 45 4D 99 21 46 82 83 ED 39 78<br>35 90 9B CB B2 F6 59 46 08 33 BA AC 1C 75 34<br>3F F6 71 EB 93 F0 49 53 C6 AE F4 28 F0 7E E2<br>8F C9 AB FB 65 CF 6A 96 1B 4A 08 5A F2 97 CD<br>14 53 CF 47 19 86 88 83 D2 0A 8F 62 4E 45 92<br>0B A3 C9 33 F5 E4 44 7D 4A 32 E5 93 6E 5A 13<br>39 32 9B B4 E8 DD 8B F0 04 4C E4 42 8E 24 D0<br>86 6F AE FD 23 48 80 9D 71 |            |
| Issuer Public Key Exponent                                                                                                                                                                                                                             | 9F 32 | 0x 01  | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 02 02      |
| Issuer Private Key Exponent<br><br>(This is provided for information only; it is not personalized on the card)                                                                                                                                         | N/A   |        | 6F 05 1F 96 67 B0 87 C8 CF 66 02 85 B8 FB CE<br>5B 5E 40 3F B5 92 06 28 15 03 9A AB C9 77 D6<br>1E EE 8A C7 98 2E 33 BB 6B 84 57 02 9E 26 50<br>23 B5 BD 32 77 4E E6 2E B0 22 7C 72 BD A3 78<br>2A A4 4B F2 62 A0 30 E2 84 74 A2 C4 E1 C2 B9<br>50 76 44 D0 79 DD 12 6E 89 F9 F5 67 4E BC 47<br>0D B5 57 53 DE 45 1F 2D 09 54 42 3A 47 00 81<br>4F AE 3F 0D 99 84 45 6D 7C B0 62 35 73 45 7E<br>0B 7E 85 CC 97 AA D0 AD 4A 54 D2 52 35 5C<br>4B A2 43 51 43 CD EF BB DC 2B |            |
| Issuer Public Key Remainder                                                                                                                                                                                                                            | 92    | 0x 24  | 33 F5 E4 44 7D 4A 32 E5 93 6E 5A 13 39 32 9B<br>B4 E8 DD 8B F0 04 4C E4 42 8E 24 D0 86 6F AE<br>FD 23 48 80 9D 71                                                                                                                                                                                                                                                                                                                                                          | 02 02      |
| Certification Authority Public Key Index                                                                                                                                                                                                               | 8F    | 0x 01  | 95<br><br>(Visa CA Test Key of 1152 bits)                                                                                                                                                                                                                                                                                                                                                                                                                                  | 02 02      |
| Certificate Expiration Date<br><br>(for information only)                                                                                                                                                                                              | N/A   |        | 12 15<br>December 2015                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |
| Signed Static Application Data<br>Note: The Signed Application Data is created using the PAN and PAN Sequence Number only. This allows the same Signed Application Data to be used on cards with different data elements (e.g., different IACs, etc.). | 93    | 0x 90  | 15 ED 7F AE BD 5A 2B 0C B4 C2 AC DB F3 EB<br>C1 29 8B B8 06 6A E8 4A 6B FE B5 EC 0D F2<br>C2 B3 C7 7A 39 EA A4 38 03 E9 FC AB 2F 6D 69<br>CE 4D 9D C1 71 6F 9E 2F 2C A1 12 9C 0F 4D<br>FF 25 DD 8C 90 AF 9E 73 82 C5 8F ED A2 06<br>FC 00 60 71 24 3B 7C 27 36 84 84 A2 14 F8 1C<br>34 23 34 2D A5 60 4C 07 49 17 21 D6 0A 68 CF<br>D1 0A 56 CB DE 20 DA 43 FF E2 A1 81 11 6B 07<br>46 D7 1D 43 AF 8A 32 6F CD A3 30 0D 72 A3<br>CD 95 58 6A 5C A7 A4 88 52 30 11 AE 75 99 | 02 03      |
|                                                                                                                                                                                                                                                        |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |
| PIN Try Limit                                                                                                                                                                                                                                          | --    | 0x 01  | 0F                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 80 10/9010 |
| PIN Try Counter                                                                                                                                                                                                                                        | 9F 17 | 0x 01  | Initialized to PIN Try Limit.                                                                                                                                                                                                                                                                                                                                                                                                                                              | 90 10      |
| Reference PIN                                                                                                                                                                                                                                          | --    | 0x 08  | 24 12 34 FF FF FF<br>(Shows the Reference PIN block.<br>The PIN is = 1234)                                                                                                                                                                                                                                                                                                                                                                                                 | 80 10      |

| Data Element                    | Tag   | Length | Value                                                                                                                                                                                                                    | DGI   |
|---------------------------------|-------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Issuer Authentication Indicator | 9F 56 | 0x 01  | 80 <ul style="list-style-type: none"><li>Issuer Authentication Mandatory</li></ul>                                                                                                                                       | 0E 01 |
| Application Default Action Code | 9F 52 | 0x 04  | 60 00 00 00 <ul style="list-style-type: none"><li>If Issuer Authentication performed and failed, decline transaction.</li><li>If Issuer Authentication is mandatory and no ARPC received, decline transaction.</li></ul> | 0E 01 |
|                                 |       |        |                                                                                                                                                                                                                          |       |
| UDK A (for ARQC, MAC & ENC)     |       |        | 22 52 91 39 B5 DC 93 F9                                                                                                                                                                                                  | 80 00 |
| UDK B (for ARQC, MAC & ENC)     |       |        | EB 9A 52 A6 FE 41 51 50                                                                                                                                                                                                  | 80 00 |

## 6.5. Test Card 4

This section outlines the profile for Test Card 4 - Card without Terminal Risk Management set in the Application Interchange Profile and with 'Floor Limit Exceeded' set in the IAC - Denial.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.5.1. Chip Data:

| Data Element                    | Tag   | Length | Value                                                                                                                | DGI   |
|---------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                 | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53<br>54 20 43 41 52 44 20 30 34<br><br>(VISA ACQUIRER TEST CARD 04) | 01 01 |
| Application Interchange Profile | --    | 0x 02  | 54 00                                                                                                                | 07 02 |
| Issuer Action Code – Denial     | 9F 0E | 0x 05  | 00 00 00 80 00<br><br>• Transaction Exceeds Floor Limit                                                              | 03 02 |

## 6.6. Test Card 5 (Previously Test Card 22)

This section outlines the profile for Test Card 22 - Card containing five applications (3 x Credit and 2 x Debit), each with the same AID but with a unique suffix and other unique characteristics:

- o Application #1 – Visa Credit (first priority): Expired application set to decline offline due to IAC – Denial setting. Application has a non-ASCII Application Preferred Name.
- o Application #2 – Visa Debit (second priority): Expired application set to decline offline due to IAC – Denial setting. Application has a non-ASCII Application Preferred Name.
- o Application #3 – Visa Credit (third priority): Expired application set to decline offline due to IAC – Denial setting. Application has a non-ASCII Application Preferred Name.
- o Application #4 – Visa Debit (fourth priority): Valid application with unique PAN. Application has a non-ASCII Application Preferred Name.
- o Application #5 – Visa Credit (fifth priority): Valid application with a unique PAN. Application has a non-ASCII Application Preferred Name.

VSDC Applet Version: 2.5.1

Changes to make from baseline card:

### 6.6.1. Magnetic Stripe – Track Data

Magnetic Stripe Track 1:  
 B4761739001010010^VISA ACQUIRER TEST CARD  
 05^15122011143800575000000

Magnetic Stripe Track 2:  
 4761739001010010=15122011143857589

### 6.6.2. Chip Data:

- o Card contains five applications but each AID has a unique suffix along with other unique characteristics.

#### Application 01

| Data Element                                      | Tag   | Length | Value                                                                                                                | DGI                 |
|---------------------------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------|---------------------|
| Cardholder Name                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54<br>20 43 41 52 44 20 30 35<br><br>(VISA ACQUIRER TEST CARD 05) | 01 01               |
|                                                   |       |        |                                                                                                                      |                     |
| Application Identifier (AID) – for application 01 | 4F    | 0x 08  | A0 00 00 00 03 10 10 01<br><br>(The suffix is 01).                                                                   | Set at install time |
| Application Priority Indicator                    | 87    | 0x 01  | 01<br>(Application is 1st priority and does not require cardholder confirmation)                                     | 91 02               |

|                              |       |       |                                                                                                                                                                                                                                                                                                                                                                                              |       |
|------------------------------|-------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Application Label            | 50    | 0x 0D | 56 49 53 41 20 43 52 45 44 49 54 20 31<br>(VISA CREDIT 1)                                                                                                                                                                                                                                                                                                                                    | 91 02 |
| Application Preferred Name   | 9F 12 | 0x 0D | B2 D8 E1 D0 20 BA E0 D5 D4 D8 E2 20 31<br>(Виса Кредит 1)                                                                                                                                                                                                                                                                                                                                    | 91 02 |
| Issuer Code Table Index      | 9F 11 | 0x 01 | 05                                                                                                                                                                                                                                                                                                                                                                                           | 91 02 |
| Language Preference          | 5F 2D | 0x 08 | 72 75 65 73 64 65 65 6E<br>(ruesdeen)                                                                                                                                                                                                                                                                                                                                                        | 91 02 |
|                              |       |       |                                                                                                                                                                                                                                                                                                                                                                                              |       |
| Application Expiration Date  | 5F 24 | 0x 03 | 09 12 31                                                                                                                                                                                                                                                                                                                                                                                     | 03 02 |
| Issuer Action Code – Online  | 9F 0F | 0x 05 | F0 00 00 98 00<br><br><ul style="list-style-type: none"> <li>Offline data authentication not performed</li> <li>Offline Static Data Authentication failure</li> <li>Chip data missing</li> <li>PAN on terminal exception file</li> <li>Transaction exceeds floor limit</li> <li>Transaction selected randomly for online transmission</li> <li>Merchant forced transaction online</li> </ul> | 03 02 |
| IAC – Denial                 | 9F 0E | 0x 05 | 00 40 00 00 00<br><ul style="list-style-type: none"> <li>If application expired, decline offline</li> </ul>                                                                                                                                                                                                                                                                                  | 03 02 |
| Issuer Action Code – Default | 9F 0D | 0x 05 | F0 00 00 88 00<br><br><ul style="list-style-type: none"> <li>Offline data authentication not performed</li> <li>Offline Static Data Authentication failure</li> <li>Chip data missing</li> <li>PAN on terminal exception file</li> <li>Transaction exceed floor limit</li> <li>Merchant forced transaction online</li> </ul>                                                                 | 03 02 |

## Application 02

| Data Element                                      | Tag   | Length | Value                                                                                                         | DGI                 |
|---------------------------------------------------|-------|--------|---------------------------------------------------------------------------------------------------------------|---------------------|
| Cardholder Name                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 20 43 41 52 44 20 30 35<br>(VISA ACQUIRER TEST CARD 05) | 01 01               |
|                                                   |       |        |                                                                                                               |                     |
| Application Identifier (AID) – for application 02 | 4F    | 0x 08  | A0 00 00 00 03 10 10 02                                                                                       | Set at install time |
| Application Priority Indicator                    | 87    | 0x 01  | 02<br>(Application is 2nd priority and does not require cardholder confirmation)                              | 91 02               |
| Application Label                                 | 50    | 0x 0C  | 56 49 53 41 20 44 45 42 49 54 20 31<br>(VISA DEBIT 1)                                                         | 91 02               |
| Application Preferred Name                        | 9F 12 | 0x 0C  | B2 D8 E1 D0 20 B4 D5 D1 D5 E2 20 31<br>(Виса Дебет 1)                                                         | 91 02               |
| Issuer Code Table Index                           | 9F 11 | 0x 01  | 05                                                                                                            | 91 02               |

|                              |       |       |                                                                                                                                                                                                                                                                                                                                                                                          |       |
|------------------------------|-------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Language Preference          | 5F 2D | 0x 08 | 72 75 65 73 64 65 65 6E<br>(ruesdeen)                                                                                                                                                                                                                                                                                                                                                    | 91 02 |
| Application Expiration Date  | 5F 24 | 0x 03 | 09 12 31                                                                                                                                                                                                                                                                                                                                                                                 | 03 02 |
| Issuer Action Code – Online  | 9F 0F | 0x 05 | F0 00 00 98 00<br><ul style="list-style-type: none"> <li>Offline data authentication not performed</li> <li>Offline Static Data Authentication failure</li> <li>Chip data missing</li> <li>PAN on terminal exception file</li> <li>Transaction exceeds floor limit</li> <li>Transaction selected randomly for online transmission</li> <li>Merchant forced transaction online</li> </ul> | 03 02 |
| IAC – Denial                 | 9F 0E | 0x 05 | 00 40 00 00 00<br><ul style="list-style-type: none"> <li>If application expired, decline offline</li> </ul>                                                                                                                                                                                                                                                                              | 03 02 |
| Issuer Action Code – Default | 9F 0D | 0x 05 | F0 00 00 88 00<br><ul style="list-style-type: none"> <li>Offline data authentication not performed</li> <li>Offline Static Data Authentication failure</li> <li>Chip data missing</li> <li>PAN on terminal exception file</li> <li>Transaction exceed floor limit</li> <li>Merchant forced transaction online</li> </ul>                                                                 | 03 02 |

### Application 3

| Data Element                                      | Tag   | Length | Value                                                                                                            | DGI                 |
|---------------------------------------------------|-------|--------|------------------------------------------------------------------------------------------------------------------|---------------------|
| Cardholder Name                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54<br>20 43 41 52 44 20 30 35<br>(VISA ACQUIRER TEST CARD 05) | 01 01               |
| Application Identifier (AID) – for application 03 | 4F    | 0x 08  | A0 00 00 00 03 10 10 03                                                                                          | Set at install time |
| Application Priority Indicator                    | 87    | 0x 01  | 03<br>(Application is 3 <sup>rd</sup> priority and does not require cardholder confirmation)                     | 91 02               |
| Application Label                                 | 50    | 0x 0D  | 56 49 53 41 20 43 52 45 44 49 54 20 32<br>(VISA CREDIT 2)                                                        | 91 02               |
| Application Preferred Name                        | 9F 12 | 0x 0D  | B2 D8 E1 D0 20 BA E0 D5 D4 D8 E2 20 32<br>(Виса Кредит 2)                                                        | 91 02               |
| Issuer Code Table Index                           | 9F 11 | 0x 01  | 05                                                                                                               | 91 02               |
| Language Preference                               | 5F 2D | 0x 08  | 72 75 65 73 64 65 65 6E<br>(ruesdeen)                                                                            | 91 02               |
| Application Expiration Date                       | 5F 24 | 0x 03  | 09 12 31                                                                                                         | 03 02               |

|                              |       |       |                                                                                                                                                                                                                                                                                                                                                                                  |       |
|------------------------------|-------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Issuer Action Code – Online  | 9F 0F | 0x 05 | F0 00 00 98 00<br><ul style="list-style-type: none"><li>Offline data authentication not performed</li><li>Offline Static Data Authentication failure</li><li>Chip data missing</li><li>PAN on terminal exception file</li><li>Transaction exceeds floor limit</li><li>Transaction selected randomly for online transmission</li><li>Merchant forced transaction online</li></ul> | 03 02 |
| IAC – Denial                 | 9F 0E | 0x 05 | 00 40 00 00 00<br><ul style="list-style-type: none"><li>If application expired, decline offline</li></ul>                                                                                                                                                                                                                                                                        | 03 02 |
| Issuer Action Code – Default | 9F 0D | 0x 05 | F0 00 00 88 00<br><ul style="list-style-type: none"><li>Offline data authentication not performed</li><li>Offline Static Data Authentication failure</li><li>Chip data missing</li><li>PAN on terminal exception file</li><li>Transaction exceed floor limit</li><li>Merchant forced transaction online</li></ul>                                                                | 03 02 |

### Application 04

| Data Element                                      | Tag   | Length | Value                                                                                                                | DGI                 |
|---------------------------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------|---------------------|
| Cardholder Name                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54<br>20 43 41 52 44 20 30 35<br><br>(VISA ACQUIRER TEST CARD 05) | 01 01               |
| Track 2 Equivalent Data                           | 57    | 0x 11  | 47 61 73 90 01 01 02 26 D1 51 22 01 14 83 57 48 89                                                                   | 01 01               |
| Track 1 Discretionary Data                        | 9F 1F | 0x 10  | 31 39 34 39 39 30 30 37 34 38 30 30 30 30 30 30                                                                      | 01 01               |
|                                                   |       |        |                                                                                                                      |                     |
| Application Identifier (AID) – for application 04 | 4F    | 0x 08  | A0 00 00 00 03 10 10 04                                                                                              | Set at install time |
| Application Priority Indicator                    | 87    | 0x 01  | 04<br>(Application is 4th priority and does not require cardholder confirmation)                                     | 91 02               |
| Application Label                                 | 50    | 0x 0C  | 56 49 53 41 20 44 45 42 49 54 30 32<br>(VISA DEBIT 2)                                                                | 91 02               |
| Application Preferred Name                        | 9F 12 | 0x 0C  | B2 D8 E1 D0 20 B4 D5 D1 D5 E2 20 32<br>(Виса Дебет 2)                                                                | 91 02               |
| Issuer Code Table Index                           | 9F 11 | 0x 01  | 05                                                                                                                   | 91 02               |
| Language Preference                               | 5F 2D | 0x 08  | 72 75 65 73 64 65 65 6E<br>(ruesdeen)                                                                                | 91 02               |
|                                                   |       |        |                                                                                                                      |                     |
| Application Primary Account Number (PAN) (Signed) | 5A    | 0x 08  | 47 61 73 90 01 01 02 26                                                                                              | 03 01               |
| Application PAN Sequence Number (Signed)          | 5F 34 | 0x 01  | 05                                                                                                                   | 03 01               |
|                                                   |       |        |                                                                                                                      |                     |

|                                                                                                                                                                                                                                                        |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Signed Static Application Data<br>Note: The Signed Application Data is created using the PAN and PAN Sequence Number only. This allows the same Signed Application Data to be used on cards with different data elements (e.g., different IACs, etc.). | 93  | 0x 90 | 48 E6 98 4B AB 8E D1 4F 61 66 09 37 4D 19 5C 0B BF 7C B8 89 9B 9C 30 1A 75 D2 FF 70 87 98 5C 21 76 3F 41 A9 5F A0 73 4D 22 3D CE DA B7 D9 60 67 1F 1C D3 1A 56 34 70 98 69 ED FA 7C 3C 67 CE D0 88 8A 29 F0 86 D9 1A 42 6F B0 07 97 02 46 91 99 68 83 91 69 7D 2D F4 F6 51 66 A7 46 B4 65 12 4B B7 32 F7 E1 28 3B 58 3E 50 7E E1 5E 58 9C 48 DA C2 00 13 37 3A 2A B0 D1 37 D8 53 5F 39 9A 93 05 6B C1 34 12 3A 5F A4 F1 5C 0C B3 33 8A 68 0E 5B | 02 03 |
| UDK A (for ARQC, MAC & ENC)                                                                                                                                                                                                                            | N/A |       | 02 E9 49 F2 16 15 C2 A4                                                                                                                                                                                                                                                                                                                                                                                                                         | 80 00 |
| UDK B (for ARQC, MAC & ENC)                                                                                                                                                                                                                            | N/A |       | 61 9B 40 E9 DC 25 2C F8                                                                                                                                                                                                                                                                                                                                                                                                                         | 80 00 |

### Application 05

| Data Element                                      | Tag   | Length | Value                                                                                                         | DGI                 |
|---------------------------------------------------|-------|--------|---------------------------------------------------------------------------------------------------------------|---------------------|
| Cardholder Name                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 20 43 41 52 44 20 30 35<br>(VISA ACQUIRER TEST CARD 05) | 01 01               |
| Track 2 Equivalent Data                           | 57    | 0x 11  | 47 61 73 90 01 01 22 22 D1 51 22 01140038 76 89                                                               | 01 01               |
| Track 1 Discretionary Data                        | 9F 1F | 0x 10  | 31 34 30 30 33 30 30 38 37 36 30 30 30 30 30 30                                                               | 01 01               |
| Application Identifier (AID) – for application 05 | 4F    | 0x 08  | A0 00 00 00 03 10 10 05                                                                                       | Set at install time |
| Application Priority Indicator                    | 87    | 0x 01  | 05<br>(Application is 5th priority and does not require cardholder confirmation)                              | 91 02               |
| Application Label                                 | 50    | 0x 0D  | 56 49 53 41 20 43 52 45 44 49 54 20 33<br>(VISA CREDIT 3)                                                     | 91 02               |
| Application Preferred Name                        | 9F 12 | 0x 0B  | B2 D8 E1 D0 20 BA E0 D5 D4 D8 E2 20 33<br>(Виса Кредит 3)                                                     | 91 02               |
| Issuer Code Table Index                           | 9F 11 | 0x 01  | 05                                                                                                            | 91 02               |
| Language Preference                               | 5F 2D | 0x 08  | 72 75 65 73 64 65 65 6E<br>(ruesdeen)                                                                         | 91 02               |
| Application Primary Account Number (PAN) (Signed) | 5A    | 0x 08  | 47 61 73 90 01 01 22 22                                                                                       | 03 01               |
| Application PAN Sequence Number (Signed)          | 5F 34 | 0x 01  | 09                                                                                                            | 03 01               |

|                                                                                                                                                                                                                                                        |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Signed Static Application Data<br>Note: The Signed Application Data is created using the PAN and PAN Sequence Number only. This allows the same Signed Application Data to be used on cards with different data elements (e.g., different IACs, etc.). | 93 | 0x 90 | 51 8D 8C 7C 77 CA 01 D3 F5 34 7E A3 0A 34 92 83 9F 5B 1 <sup>a</sup> 3E B3 65 40 08 63 CA 3C C9 C2 98 C0 9E B7 85 C0 6F 8E 6E 65 83 AC B0 0A 8C 3 <sup>a</sup> 49 F6 1 <sup>a</sup> 60 F 1 3 <sup>a</sup> CC DF 73 D6 DA AF 7B E0 31 00 A8 BF AF E6 D9 CD 3E D9 A0 BD 58 21 23 29 00 47 6B EE 71 96 87 75 A1 27 88 28 25 8 <sup>a</sup> 46 13 E0 52 0A EF 6E 9 <sup>a</sup> 7F B3 58 9E 2D F6 8F 59 EB 2E 59 C4 72 CC B2 BA 6D E7 DA 71 97 37 CA 3B 39 2E 56 8B B2 0F BC EA 09 14 9C CB C5 2E 4B 59 12 B5 D8 F5 BE DD AB | 02 03 |
|                                                                                                                                                                                                                                                        |    |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |
| UDK A (for ARQC, MAC & ENC)                                                                                                                                                                                                                            |    |       | 25 B0 7 <sup>a</sup> 73 40 29 B9 6B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 80 00 |
| UDK B (for ARQC, MAC & ENC)                                                                                                                                                                                                                            |    |       | 46 37 D9 89 79 73 F2 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 80 00 |

## 6.7. Test Card 6

This section outlines the profile for Test Card 6 – Dual Interface (DI) card supporting MSD and qVSDC with Cryptogram Version Number 10 on the contactless interface, and a Long PDOL, DDA and Language Preference on the VSDC contact interface.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.7.1. Contact VSDC Application Data

The following table defines the data to be used in personalizing the contact VSDC application.

| Data Element                                  | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | DGI   |
|-----------------------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                               | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53<br>54 20 43 41 52 44 20 30 36<br><br>(VISA ACQUIRER TEST CARD 06)                                                                                                                                                                                                                                                                                                                                                                                                                            | 01 01 |
| Language Preference                           | 5F 2D | 0x 06  | 6A 61 6B 6F 7A 68<br><br>Japanese, Korean, Chinese<br>(ja ko zh)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 91 02 |
| Processing Options<br>Data Object List (PDOL) | 9F 38 | 2D     | 9F 02 06 9F 03 06 9F 1A 02 95 05 5F 2A 02 9A 03 9C<br>01 9F 37 04 9F 35 01 9F 33 03 9F 40 05 5F 36 01 9F<br>7A 01 9F 09 02 9F 15 02 9F 66 10                                                                                                                                                                                                                                                                                                                                                                                                    | 91 02 |
| Application Interchange<br>Profile            | --    | 0x 02  | 7C 00<br>(DDA, SDA, Cardholder Verification, Terminal Risk<br>Management & Issuer Authentication performed and<br>supported)                                                                                                                                                                                                                                                                                                                                                                                                                    | 07 02 |
| ICC Public Key<br>Certificate                 | 9F 46 | 0x 90  | 95 CA 64 D5 3D A9 78 60 30 89 98 59 20 E7 B7 AA<br>55 E8 A3 24 D7 A1 96 9B 3B 61 E8 A5 7B 8E E5 8B<br>F6 8B 00 B7 D1 AC 33 05 EC 64 FD 6F EC 58 14 F3<br>F6 11 5 <sup>a</sup> 55 B9 1E 6D AC FE 5D B4 3D 84 19 9C 8D<br>15 3D E6 0C 9F 7C 1 <sup>a</sup> AF A0 0C FE D9 B9 01 5C 7D<br>37 A6 17 42 49 DC FB 9E 12 71 8B 62 3C 77 83 C2<br>6B 01 D4 7 <sup>a</sup> 9D 5B A1 01 4B 2C DB 08 A7 59 DE F4<br>58 50 3F FF 3 <sup>a</sup> 2E 9D 8B 1F FD 8C 99 CA 43 B1 B9<br>D6 3 <sup>a</sup> 6C F5 77 8F 7E 54 3B AD 4B D1 8 <sup>a</sup> 9C 7D 4B | 02 02 |
| ICC Public Key<br>Exponent                    | 9F 47 | 0x 01  | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 02 02 |
| ICC Public Key<br>Remainder                   | 9F 48 |        | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 02 02 |
| ICC Key Coefficient                           |       |        | 4A 1D 3B 2C 01 BA BF 70 9B 31 E8 89 8E 92 5E 2E<br>D6 1B 8F 89 2D B8 D9 FE AD 58 68 AF 89 E4 7B 92<br>1A 24 6F 53 AD F2 9A D7                                                                                                                                                                                                                                                                                                                                                                                                                   | 82 01 |
| ICC Key Exponent 1                            |       |        | 95 69 80 1D C7 E3 57 86 0D 11 38 4D 0E 75 62 D4<br>4C E7 A3 41 E6 E7 B7 66 0F D4 71 5E 9B 13 BC 69<br>23 94 E9 71 F0 3F 5B 43                                                                                                                                                                                                                                                                                                                                                                                                                   | 82 03 |
| ICC Key Exponent 2                            |       |        | 8E 56 7B 6F D9 49 F8 C1 B9 46 F1 16 8A 64 C2 54<br>4E D5 E4 42 92 D2 BA 5E 41 67 A6 AE E7 29 1C 0B                                                                                                                                                                                                                                                                                                                                                                                                                                              | 82 02 |

|                                                                                                                                                    |       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|                                                                                                                                                    |       |       | B6 8C 88 1B 34 F2 20 3B                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| ICC Key Prime 1                                                                                                                                    |       |       | E0 1E 40 2C AB D5 03 49 13 99 D4 73 95 B0 14 3E<br>73 5B 74 E2 DA 5B 93 19 17 BE AA 0D E8 9D 9A 9D<br>B5 5F 5E 2A E8 5F 08 E5                                                                                                                                                                                                                                                                                                                                           | 82 05 |
| ICC Key Prime 2                                                                                                                                    |       |       | D5 81 B9 27 C5 EE F5 22 95 EA 69 A1 CF 97 23 7E<br>76 40 D6 63 DC 3C 17 8D 62 1B 7A 06 5A BD AA 11<br>91 D2 CC 28 CF 6B 30 59                                                                                                                                                                                                                                                                                                                                           | 82 04 |
| Dynamic Data<br>Authentication Data<br>Object List (DDOL)                                                                                          | 9F 49 | 0x 03 | 9F 37 04                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 02 02 |
| Issuer Public Key<br>Certificate<br>(Issuer Public Key of<br>1152 bits signed by the<br>Visa CA Test Key of<br>1152 bits)<br><br>(for CA index 95) | 90    | 0x 90 | 8B 39 01 F6 25 30 48 A8 B2 CB 08 97 4° 42 45 D9 0E<br>1F 0C 4° 2° 69 BC A4 69 61 5° 71 DB 21 EE 7B 3° A9<br>42 00 CF AE DC D6 F0 A7 D9 AD 0B F7 92 13 B6 A4<br>18 D7 A4 9D 23 4E 5C 97 15 C9 14 0D 87 94 0F 2E<br>04 D6 97 1F 4° 20 4C 92 7° 45 5D 4F 8F C0 D6 40 2°<br>79 A1 CE 05 AA 3° 52 68 67 32 98 53 F5 AC 2F EB<br>3C 6F 59 FF 6C 45 3° 72 45 E3 9D 73 45 14 61 72 57<br>95 ED 73 09 70 99 96 3B 82 EB F7 20 3C 1F 78 A5 29<br>14 0C 18 2D BB E6 B4 2° E0 0C 02 | 02 01 |
| Issuer Public Key<br>Modulus (length of 1152<br>bits)<br><br>(This is provided for<br>information only; it is not<br>personalized on the<br>card)  |       | 0x 90 | A6 87 AF 61 9B 88 CB AD 37 19 03 C8 95 79 B5 89<br>0D 60 5F 90 5B 09 3C 1F 85 68 01 AE 33 C1 2E 65<br>D0 2B 64 45 4D 99 21 46 82 83 ED 39 78 35 90 9B<br>CB B2 F6 59 46 08 33 BA AC 1C 75 34 3F F6 71 EB<br>93 F0 49 53 C6 AE F4 28 F0 7E E2 8F C9 AB FB 65<br>CF 6A 96 1B 4A 08 5A F2 97 CD 14 53 CF 47 19 86<br>88 83 D2 0A 8F 62 4E 45 92 0B A3 C9 33 F5 E4 44<br>7D 4° 32 E5 93 6E 5° 13 39 32 9B B4 E8 DD 8B F0<br>04 4C E4 42 8E 24 D0 86 6F AE FD 23 48 80 9D 71 |       |
| Issuer Public Key<br>Exponent                                                                                                                      | 9F 32 | 0x 01 | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 02 02 |
| Issuer Private Key<br>Exponent<br><br>(This is provided for<br>information only; it is not<br>personalized on the<br>card)                         | N/A   |       | 6F 05 1F 96 67 B0 87 C8 CF 66 02 85 B8 FB CE 5B<br>5E 40 3F B5 92 06 28 15 03 9° AB C9 77 D6 1E EE<br>8° C7 98 2E 33 BB 6B 84 57 02 9E 26 50 23 B5 BD<br>32 77 4E E6 2E B0 22 7C 72 BD A3 78 2° A4 4B F2<br>62 A0 30 E2 84 74 A2 C4 E1 C2 B9 50 76 44 D0 79<br>DD 12 6E 89 F9 F5 67 4E BC 47 0D B5 57 53 DE 45<br>1F 2D 09 54 42 3° 47 00 81 4F AE 3F 0D 99 84 45<br>6D 7C B0 62 35 73 45 7E 0B 7E 85 CC 97 AA D0 AD<br>4A 54 D2 52 35 5C 4B A2 43 51 43 CD EF BB DC 2B |       |
| Issuer Public Key<br>Remainder                                                                                                                     | 92    | 0x 24 | 33 F5 E4 44 7D 4A 32 E5 93 6E 5A 13 39 32 9B B4<br>E8 DD 8B F0 04 4C E4 42 8E 24 D0 86 6F AE FD 23<br>48 80 9D 71                                                                                                                                                                                                                                                                                                                                                       | 02 02 |
| Certification Authority<br>Public Key Index                                                                                                        | 8F    | 0x 01 | 95<br><br>(Visa CA Test Key of 1152 bits)                                                                                                                                                                                                                                                                                                                                                                                                                               | 02 02 |
| Certificate Expiration<br>Date<br><br>(for information only)                                                                                       | N/A   |       | 12 15<br>December 2015                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |
| Card Authentication<br>Related Data                                                                                                                | 9F 69 | 0x 05 | 01 00 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 02 02 |
| Signed Static<br>Application Data<br>Note: The Signed<br>Application Data is<br>created using the PAN<br>and PAN Sequence<br>Number only. This     | 93    | 0x 90 | 91 9D 6C 21 0B 39 81 D1 C9 9B 3A D5 5E DF 36 A1<br>38 FF AD 54 D8 38 FA 40 62 2A B9 70 46 E0 5E A6<br>E6 23 0A B8 9D 5B E8 71 11 4E B5 43 1B 97 40 3B<br>8C 3D 2D 4C A9 BB 62 5A C1 3F D8 C6 B8 25 43 36<br>56 CB 56 55 7A AC 39 6D 94 5F 6D 40 14 FB 6E 71<br>E8 DB EA 74 B2 85 E9 CF 3F CE AB DF A6 1D 5A 4B                                                                                                                                                          | 02 03 |

|                                                                                                                        |  |  |                                                                                                                                                       |  |
|------------------------------------------------------------------------------------------------------------------------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| allows the same Signed Application Data to be used on cards with different data elements (e.g., different IACs, etc.). |  |  | E1 6D AA A4 33 F7 F2 64 4B 17 8A 7D D9 3D A9 8B<br>B9 D1 0E 84 29 8B DB 6B 6A E0 2D 04 E6 E5 55 8C<br>77 E7 9F 82 C9 E0 46 DF 82 1D D0 27 7A B9 D0 0C |  |
|------------------------------------------------------------------------------------------------------------------------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------|--|

### 6.7.2. Contactless PPSE, MSD and qVSDC Data

The following table defines the PPSE, MSD and qVSDC data to be personalized to support the Visa Contactless Payment Specifications (VCPS) feature.

| Proximity Payment System Environment (PPSE)    |      |                                                   |
|------------------------------------------------|------|---------------------------------------------------|
| Data Element                                   | Tag  | Value                                             |
| <b>DGI 9102 : SELECT Command Response Data</b> |      |                                                   |
| FCI Proprietary Template                       | A5   | Value N/A for template tag                        |
| FCI Issuer Discretionary Data                  | BF0C | Value N/A for template tag                        |
| Directory Entry Template                       | 61   | Value N/A for template tag                        |
| Application Identifier                         | 4F   | A0 00 00 00 03 10 10                              |
| Application Label                              | 50   | 56 49 53 41 20 43 52 45 44 49 54<br>"VISA CREDIT" |

| Magnetic Stripe Data (MSD)                          |      |                               |
|-----------------------------------------------------|------|-------------------------------|
| <b>DGI 9206 : GPO Command Response Data for MSD</b> |      |                               |
| Application Interchange Profile                     | 82   | 00 80 - MSD is supported      |
| Application File Locator (AFL)                      | 94   | 08 01 01 00 - SFI 1 Record 1  |
| <b>DGI 0101 : Record Data (MSD)</b>                 |      |                               |
| Track 2 Equivalent Data                             | 57   | See DGI 0101 in Baseline card |
| Cardholder Name                                     | 5F20 | See DGI 0101 in Baseline card |
| Track 1 Discretionary Data                          | 9F1F | See DGI 0101 in Baseline card |

| Quick Visa Smart Debit and Credit (qVSDC)                                   |      |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DGI 9103 : SELECT Command Response Data for Contactless Transactions</b> |      |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Application Label                                                           | 50   | See DGI 9102 in Baseline card                                                                                                                                                                                                                                                                                                                                                                                   |
| Processing Options Data Object List (PDOL)                                  | 9F38 | 9F 66 04 - Terminal Transaction Qualifiers<br>9F 02 06 - Amount, Authorized<br>9F 03 06 - Amount, Other<br>9F 1A 02 - Terminal Country Code<br>95 05 - Terminal Verification Result<br>5F 2A 02 - Transaction Currency Code<br>9A 03 - Transaction Date<br>9C 01 - Transaction Type<br>9F 37 04 - Unpredictable Number<br>This PDOL is needed on applications where qVSDC uses cryptogram version number CVN 10 |
| <b>DGI 9207 : GPO Command Response Data for qVSDC</b>                       |      |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Application Interchange Profile                                             | 82   | 20 00 - DDA is supported                                                                                                                                                                                                                                                                                                                                                                                        |

|                                                          |       |                                                                                                                                                                 |
|----------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Application File Locator (AFL)                           | 94    | 18 01 01 01 - SFI 3 Record 1<br>10 01 02 00 - SFI 2 Records 1-2<br>Note that SFI 3 proceeds SFI 2                                                               |
| Issuer Application Data                                  | 9F10  | 06 01 0A 03 00 00 00 - CVN 10                                                                                                                                   |
| <b>DGI 0E01 – Internal Data</b>                          |       |                                                                                                                                                                 |
| Application Currency Code                                | 9F51  | 08 40                                                                                                                                                           |
| Application Default Action (ADA)                         | 9F 52 | 80 40 00 00<br>- If Issuer Authentication failure, transmit next transaction online<br>- If PIN Try Limit exceeded on previous transaction, decline transaction |
| Consecutive Transaction Limit (International)            | 9F53  | 7F (127 consecutive offline international transactions)                                                                                                         |
| Cumulative Total Transaction Amount Limit (CTTAL)        | 9F54  | This tag is not used in the baseline image                                                                                                                      |
| Issuer Authentication Indicator                          | 9F56  | 00                                                                                                                                                              |
| Issuer Country Code                                      | 9F57  | 08 40                                                                                                                                                           |
| Lower Consecutive Offline Limit                          | 9F58  | 7F                                                                                                                                                              |
| Upper Consecutive Offline Limit                          | 9F59  | 7F                                                                                                                                                              |
| Cumulative Total Transaction Amount Upper Limit (CTTAUL) | 9F5C  | 99 99 99 99 99 99                                                                                                                                               |
| Available Offline Spending Amount (access permission)    | 9F5D  | 01 – Allow retrieval of AOSA                                                                                                                                    |
| Card Additional Processes                                | 9F68  | 84 00 00 00<br>- Low Value (LV Only) check supported<br>- Offline transactions in non-matching currencies are allowed                                           |
| Card Transaction Qualifiers                              | 9F6C  | 10 00<br>- Terminate if Offline Data Authentication fails and reader supports contact VSDC                                                                      |
| VLP Reset Threshold                                      | 9F6D  | 00 00 00 00 10 00                                                                                                                                               |
| VLP Funds Limit                                          | 9F77  | 99 99 99 99 99 99                                                                                                                                               |
| VLP Single Transaction Limit                             | 9F78  | 00 00 00 00 10 00                                                                                                                                               |
| VLP Available Funds                                      | 9F79  | 99 99 99 99 99 99 (initial value)                                                                                                                               |

## 6.8. Test Card 7 (Previously Test Card 23)

This section outlines the profile for Test Card 7 – Card that ensures correct Terminal Action Codes— Service Not Allowed feature set in the terminal.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.8.1. Chip Data

| Data Element                 | Tag   | Length | Value                                                                                                                | DGI   |
|------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name              | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54<br>20 43 41 52 44 20 30 37<br><br>(VISA ACQUIRER TEST CARD 7)  | 01 01 |
|                              |       |        |                                                                                                                      |       |
| Application Usage Control    | 9F 07 | 0x 02  | AB 80<br><br>• Valid for domestic transactions only                                                                  | 03 02 |
| Issuer Action Codes— Default | 9F 0D | 0x 05  | 00 00 00 00 00                                                                                                       | 03 02 |
| Issuer Action Codes— Denial  | 9F 0E | 0x 05  | 00 00 00 00 00                                                                                                       | 03 02 |
| Issuer Action Codes— Online  | 9F 0F | 0x 05  | 00 00 00 00 00                                                                                                       | 03 02 |
| Issuer Country Code          | 5F 28 | 0x 02  | 08 11<br><br>(This is a fake country code to ensure terminal will treat transaction as an international transaction) | 03 02 |

## 6.9. Test Card 8 (Previously Test Card 26)

This section outlines the profile for Test Card 8 – Card created to allow fallback testing.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

- o PSE is installed and personalized.
- o VSDC is installed only; VSDC is not personalized.

### 6.9.1. Magnetic Stripe – Track Data

Magnetic Stripe Track 1

B4761739001010267^VISA ACQUIRER TEST CARD  
08^151220113506004600000000

Magnetic Stripe Track 2

4761739001010267=15122011350646089

## 6.10. Test Card 9 (Previously Test Card 30)

This section outlines the profile for Test Card 9 - Card containing a CVM that the terminal does not support and the CVM is not on the list of CVMs that must be supported by the terminal. For this test, the condition for the RFU CVM is “apply next CVM if CVM is unsuccessful”.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.10.1. Chip Data

| Data Element                   | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | DGI   |
|--------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53<br>54 20 43 41 52 44 20 30 39<br><br>(VISA ACQUIRER TEST CARD 9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 01 01 |
|                                |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |
| Cardholder Verification Method | 8E    | 0x 0E  | 0000 0000 0000 0000 5D00 1E03 1F00<br><br>Amount X = 00000000<br>Amount Y = 00000000<br><br><ul style="list-style-type: none"> <li>o CVM Code 1 '5D00' <ul style="list-style-type: none"> <li>o CVM which is RFU (Unrecognized CVM), Always</li> <li>o Apply next CVM if CVM is unsuccessful</li> </ul> </li> <li>o CVM Code 2 '1E03' <ul style="list-style-type: none"> <li>o Signature, if terminal supports CVM</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>o CVM Code 3 '1F00' <ul style="list-style-type: none"> <li>o No CVM Required, Always (Cannot fail CVM)</li> </ul> </li> </ul> | 03 02 |
| Issuer Action Code – Default   | 9F 0D | 0x 05  | 00 00 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 03 02 |
| Issuer Action Code – Denial    | 9F 0E | 0x 05  | 00 00 80 00 00<br><ul style="list-style-type: none"> <li>• If Cardholder Verification is unsuccessful, decline offline.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 03 02 |
| Issuer Action Code – Online    | 9F 0F | 0x 05  | 00 00 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 03 02 |

## 6.11. Test Card 10 (Previously Test Card 31)

This section outlines the profile for Test Card 10 (Card contains a CVM that the terminal does not support and the CVM is not on the list of CVMs that must be supported by the terminal. For this test, the condition for the RFU CVM is “fail CVM processing if this CVM is unsuccessful”).

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.11.1. Chip Data

| Data Element                   | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DGI   |
|--------------------------------|-------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53<br>54 20 43 41 52 44 20 31 30<br><br>(VISA ACQUIRER TEST CARD 10)                                                                                                                                                                                                                                                                                                                                                                                                  | 01 01 |
|                                |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |
| Cardholder Verification Method | 8E    | 0x 0E  | 0000 0000 0000 0000 1D00 1E03 1F00<br><br>Amount X = 00000000<br>Amount Y = 00000000<br><br><ul style="list-style-type: none"> <li>o CVM Code 1 '1D00'</li> <li>o CVM which is RFU (Unrecognized CVM), Always</li> <li>o Fail CVM processing if this CVM is unsuccessful</li> <li>o CVM Code 2 '1E03'</li> <li>o Signature, if terminal supports CVM</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> <li>o CVM Code 3 '1F00'</li> <li>o No CVM Required, Always (Cannot fail CVM)</li> </ul> | 03 02 |
| Issuer Action Code – Default   | 9F 0D | 0x 05  | 00 00 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 03 02 |
| Issuer Action Code – Denial    | 9F 0E | 0x 05  | 00 00 80 00 00<br><br><ul style="list-style-type: none"> <li>• If Cardholder Verification is unsuccessful, decline offline.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                | 03 02 |
| Issuer Action Code – Online    | 9F 0F | 0x 05  | 00 00 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 03 02 |

## 6.12. Test Card 11

This section outlines the profile for Test Card 11 - Dual Interface (DI) card that supports MSD and qVSDC with Cryptogram Version Number 17 on the contactless interface, and DDA and multiple applications on the contact interface.

On the contact interface, the card contains 3 applications:

- o Unknown application
- o Blocked application
- o Valid application

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.12.1. Magnetic Stripe – Track Data

Magnetic Stripe Track 1:  
B4761739001010119^VISA ACQUIRER TEST CARD  
11^15122011143800281000000

Magnetic Stripe Track 2:  
4761739001010119=15122011143828189

### 6.12.2. Chip Data

#### Application 01

Configure the card with an Unknown application.

| Data Element                   | Tag   | Length | Value                                                                                                                | DGI   |
|--------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45<br>53 54 20 43 41 52 44 20 31 31<br><br>(VISA ACQUIRER TEST CARD 11) | 01 01 |
|                                |       |        |                                                                                                                      |       |
| Application Identifier         | 4F    | 0x 07  | A0 00 00 00 01 11 11<br><br>(AID is added during install time not perso time)                                        | NA    |
| Application Priority Indicator | 87    | 0x 01  | 01                                                                                                                   | 91 02 |

#### Application 02

IMPORTANT: The vendor developing the test card must block this application after personalizing it. This can be accomplished by sending an APPLICATION BLOCK Issuer Script command to the card (Refer to EMV 4.2 – Book 3 – Section 6.5.1)

| Data Element                   | Tag   | Length | Value                                                                                                                | DGI   |
|--------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45<br>53 54 20 43 41 52 44 20 31 31<br><br>(VISA ACQUIRER TEST CARD 11) | 01 01 |
| Application Identifier         | 4F    | 0x 07  | A0 00 00 00 03 10 10 01<br><br>(AID is added during install time not perso time)                                     | NA    |
| Application Priority Indicator | 87    | 0x 01  | 02                                                                                                                   | 91 02 |

### Application 3

The following table defines the data to be used in personalizing the contact VSDC application.

| Data Element                       | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | DGI   |
|------------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                    | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45<br>53 54 20 43 41 52 44 20 31 31<br><br>(VISA ACQUIRER TEST CARD 11)                                                                                                                                                                                                                                                                                                                                                       | 01 01 |
| Track 2 Equivalent Data            | 57    | 0x 11  | 47 61 73 90 01 01 01 19 D1 51 22 01 11 43 83 03<br>89                                                                                                                                                                                                                                                                                                                                                                                                                      | 01 01 |
| Track 1 Discretionary Data         | 9F 1F | 0x 10  | 31 31 34 33 38 30 30 33 30 33 30 30 30 30 30                                                                                                                                                                                                                                                                                                                                                                                                                               | 01 01 |
|                                    |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| Application Identifier             | 4F    | 0x 07  | A0 00 00 00 03 10 10 02<br><br>(AID is added during install time not perso time)                                                                                                                                                                                                                                                                                                                                                                                           |       |
| Application Primary Account Number | 5A    | 0x 08  | 47 61 73 90 01 01 01 19                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 03 01 |
| Application Priority Indicator     | 87    | 0x 01  | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 91 02 |
|                                    |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| Application Interchange Profile    | --    | 0x 02  | 7C 00<br>(DDA, SDA, Cardholder Verification, Terminal Risk Management & Issuer Authentication performed and supported)                                                                                                                                                                                                                                                                                                                                                     | 07 02 |
|                                    |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| Service Code                       | 5F30  | 0x 02  | 02 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| Application Expiration Date        | 5F 24 | 0x 03  | 15 12 31                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 03 02 |
|                                    |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| ICC Public Key Certificate         | 9F 46 | 0x 90  | 24 56 9D 09 15 91 E9 F0 59 1F 04 ED 1D 12 11<br>2C B9 B7 6B 04 C6 CC 95 01 4E 98 1E 93 4F 40<br>80 C7 6D 59 01 DF E9 42 31 59 0A 9D C7 D1 1E<br>D3 0B A2 80 C1 42 19 81 B5 3B 1C B8 96 FE 5D<br>73 1E FA 4E 48 0F 90 F7 5D C7 E1 F7 0C 2B FE<br>63 21 1E 93 6E 52 2C CF 52 FD 0F E8 F1 33 95<br>FB 9F FF FB 5A 33 8C C3 4A E4 77 9E A5 2E 85<br>EC CD 81 61 EB CC EE 5F 43 D7 12 F7 40 D3<br>96 6E 0E 71 95 80 68 25 64 BE 0C 34 EC B8 4B<br>12 1C D4 66 D2 FE 30 27 3B 99 | 02 02 |
| ICC Public Key Exponent            | 9F 47 | 0x 01  | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 02 02 |
| ICC Public Key Remainder           | 9F 48 |        | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 02 02 |
| ICC Key Coefficient                |       |        | 4A 1D 3B 2C 01 BA BF 70 9B 31 E8 89 8E 92 5E                                                                                                                                                                                                                                                                                                                                                                                                                               | 82 01 |

|                                                                                                                                                    |       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|                                                                                                                                                    |       |       | 2E D6 1B 8F 89 2D B8 D9 FE AD 58 68 AF 89 E4<br>7B 92 1A 24 6F 53 AD F2 9A D7                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| ICC Key Exponent 1                                                                                                                                 |       |       | 95 69 80 1D C7 E3 57 86 0D 11 38 4D 0E 75 62<br>D4<br>4C E7 A3 41 E6 E7 B7 66 0F D4 71 5E 9B 13 BC<br>69 23 94 E9 71 F0 3F 5B 43                                                                                                                                                                                                                                                                                                                                                          | 82 03 |
| ICC Key Exponent 2                                                                                                                                 |       |       | 8E 56 7B 6F D9 49 F8 C1 B9 46 F1 16 8A 64 C2<br>54<br>4E D5 E4 42 92 D2 BA 5E 41 67 A6 AE E7 29 1C<br>0B B6 8C 88 1B 34 F2 20 3B                                                                                                                                                                                                                                                                                                                                                          | 82 02 |
| ICC Key Prime 1                                                                                                                                    |       |       | E0 1E 40 2C AB D5 03 49 13 99 D4 73 95 B0 14<br>3E<br>73 5B 74 E2 DA 5B 93 19 17 BE AA 0D E8 9D 9A<br>9D B5 5F 5E 2A E8 5F 08 E5                                                                                                                                                                                                                                                                                                                                                          | 82 05 |
| ICC Key Prime 2                                                                                                                                    |       |       | D5 81 B9 27 C5 EE F5 22 95 EA 69 A1 CF 97 23<br>7E<br>76 40 D6 63 DC 3C 17 8D 62 1B 7A 06 5A BD AA<br>11 91 D2 CC 28 CF 6B 30 59                                                                                                                                                                                                                                                                                                                                                          | 82 04 |
| Dynamic Data<br>Authentication Data<br>Object List (DDOL)                                                                                          | 9F 49 | 0x 03 | 9F 37 04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 02 02 |
| Issuer Public Key<br>Certificate<br>(Issuer Public Key of<br>1152 bits signed by the<br>Visa CA Test Key of<br>1152 bits)<br><br>(for CA index 95) | 90    | 0x 90 | 8B 39 01 F6 25 30 48 A8 B2 CB 08 97 4A 42 45<br>D9 0E 1F 0C 4A 2A 69 BC A4 69 61 5A 71 DB 21<br>EE 7B 3A A9 42 00 CF AE DC D6 F0 A7 D9 AD<br>0B F7 92 13 B6 A4 18 D7 A4 9D 23 4E 5C 97 15<br>C9 14 0D 87 94 0F 2E 04 D6 97 1F 4A 20 4C 92<br>7A 45 5D 4F 8F C0 D6 40 2A 79 A1 CE 05 AA 3A<br>52 68 67 32 98 53 F5 AC 2F EB 3C 6F 59 FF 6C<br>45 3A 72 45 E3 9D 73 45 14 61 72 57 95 ED 73<br>09 70 99 96 3B 82 EB F7 20 3C 1F 78 A5 29 14<br>0C 18 2D BB E6 B4 2A E0 0C 02                | 02 01 |
| Issuer Public Key<br>Modulus (length of 1152<br>bits)<br><br>(This is provided for<br>information only; it is not<br>personalized on the<br>card)  | N/A   | 0x 90 | A6 87 AF 61 9B 88 CB AD 37 19 03 C8 95 79 B5<br>89<br>0D 60 5F 90 5B 09 3C 1F 85 68 01 AE 33 C1 2E<br>65<br>D0 2B 64 45 4D 99 21 46 82 83 ED 39 78 35 90<br>9B<br>CB B2 F6 59 46 08 33 BA AC 1C 75 34 3F F6 71<br>EB 93 F0 49 53 C6 AE F4 28 F0 7E E2 8F C9 AB<br>FB 65 CF 6A 96 1B 4A 08 5A F2 97 CD 14 53 CF<br>47 19 86 88 83 D2 0A 8F 62 4E 45 92 0B A3 C9<br>33 F5 E4 44 7D 4A 32 E5 93 6E 5A 13 39 32 9B<br>B4 E8 DD 8B F0 04 4C E4 42 8E 24 D0 86 6F AE<br>FD 23 48 80 9D 71       |       |
| Issuer Public Key<br>Exponent                                                                                                                      | 9F 32 | 0x 01 | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 02 02 |
| Issuer Private Key<br>Exponent<br><br>(This is provided for<br>information only; it is not<br>personalized on the<br>card)                         | N/A   |       | 6F 05 1F 96 67 B0 87 C8 CF 66 02 85 B8 FB CE<br>5B<br>5E 40 3F B5 92 06 28 15 03 9A AB C9 77 D6 1E<br>EE<br>8A C7 98 2E 33 BB 6B 84 57 02 9E 26 50 23 B5<br>BD<br>32 77 4E E6 2E B0 22 7C 72 BD A3 78 2A A4 4B<br>F2<br>62 A0 30 E2 84 74 A2 C4 E1 C2 B9 50 76 44 D0<br>79<br>DD 12 6E 89 F9 F5 67 4E BC 47 0D B5 57 53 DE<br>45 1F 2D 09 54 42 3A 47 00 81 4F AE 3F 0D 99<br>84 45 6D 7C B0 62 35 73 45 7E 0B 7E 85 CC 97<br>AA D0 AD 4A 54 D2 52 35 5C 4B A2 43 51 43 CD<br>EF BB DC 2B |       |
| Issuer Public Key                                                                                                                                  | 92    | 0x 24 | 33 F5 E4 44 7D 4A 32 E5 93 6E 5A 13 39 32 9B                                                                                                                                                                                                                                                                                                                                                                                                                                              | 02 02 |

|                                                                                                                                                                                                                                                                                         |       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Remainder                                                                                                                                                                                                                                                                               |       |       | B4<br>E8 DD 8B F0 04 4C E4 42 8E 24 D0 86 6F AE FD<br>23 48 80 9D 71                                                                                                                                                                                                                                                                                                                                                                                                       |       |
| Certification Authority<br>Public Key Index                                                                                                                                                                                                                                             | 8F    | 0x 01 | 95<br><br>(Visa CA Test Key of 1152 bits)                                                                                                                                                                                                                                                                                                                                                                                                                                  | 02 02 |
| Certificate Expiration<br>Date<br><br>(for information only)                                                                                                                                                                                                                            | N/A   |       | 12 15<br>December 2015                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |
| Card Authentication<br>Related Data                                                                                                                                                                                                                                                     | 9F 69 | 0x 05 | 01 00 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 02 02 |
| Signed Static<br>Application Data<br>Note: The Signed<br>Application Data is<br>created using the PAN<br>and PAN Sequence<br>Number only. This<br>allows the same Signed<br>Application Data to be<br>used on cards with<br>different data elements<br>(e.g., different IACs,<br>etc.). | 93    | 0x 90 | 8F 48 E6 91 40 34 94 05 76 88 B2 2B 23 7E F0<br>EE 40 23 85 39 BB 9D E9 9A 97 DC 2C 47 B3<br>42 7F 29 26 51 BF 53 8B B8 9C 04 6F 86 CE 05<br>C5 57 8C C1 20 07 F3 D4 F8 43 68 47 66 2D F7<br>8C B3 85 AF B8 15 B7 E2 80 97 C0 A5 20 F6 7D<br>42 67 A3 53 1E 6C 7C EB 76 10 B1 13 A3 69 C0<br>D5 89 25 15 FE 06 2B F7 BA 16 DA 57 C0 40 95<br>24 50 07 E1 B3 8B 78 23 B9 AB 4A 51 77 A1 83<br>48 AD 4C E7 A8 E9 9F 44 04 C2 56 B4 06 12 86<br>79 4D 8B 41 B2 CF 42 E4 02 2B | 02 03 |
| UDK A (for ARQC, MAC<br>& ENC)                                                                                                                                                                                                                                                          |       |       | 31 E6 FB A4 02 1F 57 C7                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 80 00 |
| UDK B (for ARQC, MAC<br>& ENC)                                                                                                                                                                                                                                                          |       |       | 0B 19 BF 5E 43 16 4F 45                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 80 00 |

### 6.12.3. Contactless PPSE, MSD and qVSDC Data

The following table defines the PPSE, MSD and qVSDC data to be personalized to support the Visa Contactless Payment Specifications (VCPS) feature.

| Proximity Payment System Environment (PPSE)         |      |                                                    |
|-----------------------------------------------------|------|----------------------------------------------------|
| Data Element                                        | Tag  | Value                                              |
| <b>DGI 9102 : SELECT Command Response Data</b>      |      |                                                    |
| FCI Proprietary Template                            | A5   | Value N/A for template tag                         |
| FCI Issuer Discretionary Data                       | BF0C | Value N/A for template tag                         |
| Directory Entry Template                            | 61   | Value N/A for template tag                         |
| Application Identifier                              | 4F   | A0 00 00 00 03 10 10                               |
| Application Label                                   | 50   | 56 49 53 41 20 43 52 45 44 49 54<br>"VISA CREDIT"  |
| <b>Magnetic Stripe Data (MSD)</b>                   |      |                                                    |
| <b>DGI 9206 : GPO Command Response Data for MSD</b> |      |                                                    |
| Application Interchange Profile                     | 82   | 00 80 - MSD is supported                           |
| Application File Locator (AFL)                      | 94   | 08 01 01 00 - SFI 1 Record 1                       |
| <b>DGI 0101 : Record Data (MSD)</b>                 |      |                                                    |
| Track 2 Equivalent Data                             | 57   | 47 61 73 90 01 01 01 19 D1 51 22 01 11 43 83 03 89 |

|                 |      |                                                                                  |
|-----------------|------|----------------------------------------------------------------------------------|
| Cardholder Name | 5F20 | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 20 43<br>41 52 44 20 31 31 |
|                 |      | (VISA ACQUIRER TEST CARD 11)                                                     |

|                            |      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----------------------------|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Track 1 Discretionary Data | 9F1F | 31 | 31 | 34 | 33 | 38 | 30 | 30 | 38 | 39 | 33 | 30 | 30 | 30 | 30 | 30 | 30 |
|----------------------------|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|

### Quick Visa Smart Debit and Credit (qVSDC)

## DGI 9103 : SELECT Command Response Data for Contactless Transactions

|                                            |      |                                                                                                                                                                                                                                                    |
|--------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Application Label                          | 50   | See DGI 9102 in Baseline card                                                                                                                                                                                                                      |
| Processing Options Data Object List (PDOL) | 9F38 | 9F 66 04 - Terminal Transaction Qualifiers<br>9F 02 06 - Amount, Authorized<br>9F 37 04 - Unpredictable Number<br>5F 2A 02 – Transaction Currency Code<br>This PDOL is needed on applications where qVSDC uses<br>cryptogram version number CVN 17 |

## DGI 9207 : GPO Command Response Data for qVSDC

|                                 |      |                                                                                                   |
|---------------------------------|------|---------------------------------------------------------------------------------------------------|
| Application Interchange Profile | 82   | 20 00 - DDA is supported                                                                          |
| Application File Locator (AFL)  | 94   | 18 01 01 01 - SFI 3 Record 1<br>10 01 02 00 - SFI 2 Records 1-2<br>Note that SFI 3 proceeds SFI 2 |
| Issuer Application Data         | 9F10 | 06 01 11 03 00 00 00 - CVN 17                                                                     |

## DGI 0E01 – Internal Data

|                                                          |       |                                                                                                                       |
|----------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------|
| Application Currency Code                                | 9F51  | 08 40                                                                                                                 |
| Application Default Action (ADA)                         | 9F 52 | 00 00 00 00                                                                                                           |
| Consecutive Transaction Limit (International)            | 9F53  | 7F (127 consecutive offline international transactions)                                                               |
| Cumulative Total Transaction Amount Limit (CTTAL)        | 9F54  | This tag is not used in the baseline image                                                                            |
| Issuer Authentication Indicator                          | 9F56  | 00                                                                                                                    |
| Issuer Country Code                                      | 9F57  | 08 40                                                                                                                 |
| Lower Consecutive Offline Limit                          | 9F58  | 7F                                                                                                                    |
| Upper Consecutive Offline Limit                          | 9F59  | 7F                                                                                                                    |
| Cumulative Total Transaction Amount Upper Limit (CTTAUL) | 9F5C  | 99 99 99 99 99 99                                                                                                     |
| Available Offline Spending Amount (access permission)    | 9F5D  | 01 – Allow retrieval of AOSA                                                                                          |
| MSD Offset                                               | 9F67  | 1E                                                                                                                    |
| Card Additional Processes                                | 9F68  | 84 00 00 00<br>- Low Value (LV Only) check supported<br>- Offline transactions in non-matching currencies are allowed |
| Card Transaction Qualifiers                              | 9F6C  | 10 00<br>- Terminate if Offline Data Authentication fails and reader supports contact VSDC                            |
| VLP Reset Threshold                                      | 9F6D  | 00 00 00 00 10 00                                                                                                     |
| VLP Funds Limit                                          | 9F77  | 99 99 99 99 99 99                                                                                                     |
| VLP Single Transaction Limit                             | 9F78  | 00 00 00 00 10 00                                                                                                     |
| VLP Available Funds                                      | 9F79  | 99 99 99 99 99 99 (initial value)                                                                                     |

## 6.13. Test Card 12

This section outlines the profile for Test Card 12 - Card supporting Geographic Restrictions check and is restricted to domestic transactions only.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.13.1. Chip Data

| Data Element                           | Tag   | Length | Value                                                                                                                | DGI   |
|----------------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                        | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54<br>20 43 41 52 44 20 31 32<br><br>(VISA ACQUIRER TEST CARD 12) | 01 01 |
| Processing Options<br>Data Object List | 9F 38 | 0x 03  | 9F 1A 02                                                                                                             | 91 02 |
| Geographic Indicator                   | 9F 55 | 0x 01  | 80                                                                                                                   | 0E 01 |
| Issuer Country Code                    | 9F 57 | 0x 02  | 08 11                                                                                                                | 0E 01 |

## 6.14. Test Card 13

This section outlines the profile for Test Card 13 - Card containing proprietary data in PSE and application. The card also includes a 6-digit Offline plaintext PIN.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.14.1. Magnetic Stripe – Track Data

Magnetic Stripe Track 1:  
B4761739001011133^VISA ACQUIRER TEST CARD 13^1512  
2201045900454000000

Magnetic Track 2:  
4761739001011133=15122201045945489

### 6.14.2. Chip Data

PSE SELECT Response:

Note: This card must include the PSE and personalize Tag C2 in the PSE with a value of SAMPLE.

| Data Element                     | Tag   | Length | Value                                                               | DGI   |
|----------------------------------|-------|--------|---------------------------------------------------------------------|-------|
| DF Name                          | 84    | 0x 0E  | 31 50 41 59 2E 53 59 53 2E 44 44 46 30 31                           |       |
| FCI Proprietary Template         | A5    | 0x 0E  | 88 01 01 BF 0C 08 C2 06 53 41 4D 50 4C 45<br>(see below of parsing) | 91 02 |
| SFI of Directory Elementary File | 88    | 0x 01  | 01                                                                  | 91 02 |
| FCI Issuer Discretionary Data    | BF 0C | 0x 08  | C2 06 53 41 4D 50 4C 45<br>(see below for parsing)                  | 91 02 |
| Proprietary Tag                  | C2    | 0x 06  | 53 41 4D 50 4C 45 (Sample)                                          | 91 02 |

PSE Directory Record:

Within the PSE directory record is Tag 73 which contains Tags 5F56 (Issuer Country Code) & Tag DF99

| Data Element                     | Tag   | Length | Value                                                       | DGI   |
|----------------------------------|-------|--------|-------------------------------------------------------------|-------|
| Application Identifier           | 4F    | 0x 07  | A0 00 00 00 03 10 10                                        | 91 02 |
| Application Label                | 50    | 0x 0A  | 56 69 73 61 20 44 45 42 49 54<br>(VISA DEBIT)               | 91 02 |
| Application Priority Indicator   | 87    | 0x 01  | 01                                                          | 91 02 |
| Directory Discretionary Template | 73    | 0x 0B  | 5F 56 03 78 79 7A DF 99 02 80 80<br>(see below for parsing) | 91 02 |
| Issuer Country Code (Alpha)      | 5F 56 |        | 78 79 7A (xyz)                                              | 01 01 |
| Proprietary Tag                  | DF 99 | 0x 02  | 80 80                                                       | 01 01 |

## Application Data

The following table defines the data to be used in personalizing the VSDC application.

| Data Element                                                     | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | DGI   |
|------------------------------------------------------------------|-------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                                                  | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53<br>54 20 43 41 52 44 20 31 33<br><br>(VISA ACQUIRER TEST CARD 13)                                                                                                                                                                                                                                                                                                                                                           | 01 01 |
| Track 2 Equivalent Data                                          | 57    | 0x 11  | 4761739001011133D15122201045951989                                                                                                                                                                                                                                                                                                                                                                                                                                             | 01 01 |
| Track 1 Discretionary Data                                       | 9F 1F | 0x 10  | 31 30 34 35 39 30 30 35 31 39 30 30 30 30 30 30                                                                                                                                                                                                                                                                                                                                                                                                                                | 01 01 |
|                                                                  |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |
| Application Label                                                | 50    | 0x 0A  | 56 69 73 61 20 44 45 42 49 54<br>(VISA DEBIT)                                                                                                                                                                                                                                                                                                                                                                                                                                  | 91 02 |
| Application Preferred Name                                       | 9F 12 | 0x 0E  | 44 45 42 49 54 4F 20 44 45 20 56 49 53 41<br><br>DEBITO DE VISA<br><br>(Preferred Name contains spaces)                                                                                                                                                                                                                                                                                                                                                                        | 91 02 |
|                                                                  |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |
| Application Primary Account Number (PAN) (Signed)                | 5A    | 0x 08  | 47 61 73 90 01 01 11 33                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0301  |
| Application PAN Sequence Number (Signed)                         | 5F 34 | 0x 01  | 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0301  |
|                                                                  |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |
| Proprietary Tag<br>(Must be personalized at the end of DGI 0302) | C3    | 0x 06  | 53 41 4D 50 4C 45                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 03 02 |
| Service Code                                                     | 5F 30 | 0x 02  | 02 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 03 02 |
| Cardholder Verification Method List (CVM)                        | 8E    | 0x 10  | 0000 0000 0000 0000 4103 5E03 4203 1F00<br><br>Amount X = 00000000<br>Amount Y = 00000000<br><br>CVM Code 1 '4103'<br>Offline (Plaintext) PIN, if terminal supports CVM<br>Apply next CVM<br><br>CVM Code 2 '5E03'<br>Signature, if terminal supports CVM<br>Apply Next CVM<br><br>CVM Code 3 '4203'<br>Online PIN, if terminal supports CVM<br>Apply Next CVM<br><br>CVM Code 4 '1F00'<br>No CVM Required, Always<br>Fail cardholder verification if this CVM is unsuccessful | 03 02 |
|                                                                  |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |

|                                                                                                                                                                                                                                                        |       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Issuer Public Key Certificate<br>(Issuer Public Key of 1152 bits signed by the Visa CA Test Key of 1152 bits)<br><br>(for CA index 95)                                                                                                                 | 90    | 0x 90 | 8B 39 01 F6 25 30 48 A8 B2 CB 08 97 4A 42 45 D9<br>0E 1F 0C 4A 2A 69 BC A4 69 61 5A 71 DB 21 EE 7B<br>3A A9 42 00 CF AE DC D6 F0 A7 D9 AD 0B F7 92 13<br>B6 A4 18 D7 A4 9D 23 4E 5C 97 15 C9 14 0D 87 94<br>0F 2E 04 D6 97 1F 4A 20 4C 92 7A 45 5D 4F 8F C0<br>D6 40 2A 79 A1 CE 05 AA 3A 52 68 67 32 98 53 F5<br>AC 2F EB 3C 6F 59 FF 6C 45 3A 72 45 E3 9D 73 45<br>14 61 72 57 95 ED 73 09 70 99 96 3B 82 EB F7 20<br>3C 1F 78 A5 29 14 0C 18 2D BB E6 B4 2A E0 0C 02    | 02 01 |
| Issuer Public Key Modulus (length of 1152 bits)<br><br>(This is provided for information only; it is not personalized on the card)                                                                                                                     |       | 0x 90 | A6 87 AF 61 9B 88 CB AD 37 19 03 C8 95 79 B5 89<br>0D 60 5F 90 5B 09 3C 1F 85 68 01 AE 33 C1 2E 65<br>D0 2B 64 45 4D 99 21 46 82 83 ED 39 78 35 90 9B<br>CB B2 F6 59 46 08 33 BA AC 1C 75 34 3F F6 71 EB<br>93 F0 49 53 C6 AE F4 28 F0 7E E2 8F C9 AB FB 65<br>CF 6A 96 1B 4A 08 5A F2 97 CD 14 53 CF 47 19 86<br>88 83 D2 0A 8F 62 4E 45 92 0B A3 C9 33 F5 E4 44<br>7D 4A 32 E5 93 6E 5A 13 39 32 9B B4 E8 DD 8B F0<br>04 4C E4 42 8E 24 D0 86 6F AE FD 23 48 80 9D 71    |       |
| Issuer Public Key Exponent                                                                                                                                                                                                                             | 9F 32 | 0x 01 | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 02 02 |
| Issuer Private Key Exponent<br><br>(This is provided for information only; it is not personalized on the card)                                                                                                                                         |       |       | 6F 05 1F 96 67 B0 87 C8 CF 66 02 85 B8 FB CE 5B<br>5E 40 3F B5 92 06 28 15 03 9A AB C9 77 D6 1E EE<br>8A C7 98 2E 33 BB 6B 84 57 02 9E 26 50 23 B5 BD<br>32 77 4E E6 2E B0 22 7C 72 BD A3 78 2A A4 4B F2<br>62 A0 30 E2 84 74 A2 C4 E1 C2 B9 50 76 44 D0 79<br>DD 12 6E 89 F9 F5 67 4E BC 47 0D B5 57 53 DE 45<br>1F 2D 09 54 42 3A 47 00 81 4F AE 3F 0D 99 84 45<br>6D 7C B0 62 35 73 45 7E 0B 7E 85 CC 97 AA D0 AD<br>4A 54 D2 52 35 5C 4B A2 43 51 43 CD EF BB DC 2B    |       |
| Issuer Public Key Remainder                                                                                                                                                                                                                            | 92    | 0x 24 | 33 F5 E4 44 7D 4A 32 E5 93 6E 5A 13 39 32 9B B4<br>E8 DD 8B F0 04 4C E4 42 8E 24 D0 86 6F AE FD 23<br>48 80 9D 71                                                                                                                                                                                                                                                                                                                                                          | 02 02 |
| Certification Authority Public Key Index                                                                                                                                                                                                               | 8F    | 0x 01 | 95<br><br>(Visa CA Test Key of 1152 bits)                                                                                                                                                                                                                                                                                                                                                                                                                                  | 02 02 |
| Certificate Expiration Date<br><br>(for information only)                                                                                                                                                                                              |       |       | 12 15<br>December 2015                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |
| Signed Static Application Data<br>Note: The Signed Application Data is created using the PAN and PAN Sequence Number only. This allows the same Signed Application Data to be used on cards with different data elements (e.g., different IACs, etc.). | 93    | 0x 90 | 13 6C 1B 97 5D F2 F0 E4 CB 97 F3 D0 16 61 92 CE<br>7B F9 9F 39 6A 63 41 0E E7 36 49 95 0E 12 DA EE<br>EA 51 FB D6 20 AD 48 73 1C 98 0F E3 BB DE 7C<br>C9 DF E5 38 10 6F 86 7A 84 28 CE 7A DB A1 95 AD<br>7F 25 BD 7A A6 C5 8C F8 D9 80 A6 79 74 61 59 0B<br>5F FE C9 DA 2B DF AC 6E 9D 5B EA 9A A3 A0 9F<br>51 DF AF 4C A7 D7 C2 1B D7 F0 2F DC D1 8A 90<br>8C A7 C4 91 01 FE 2B 1B 86 C6 3A 35 C1 FB 76 21<br>64 32 C2 59 BE 83 37 CF 5E 13 CF 2B FA AE 4C 8C<br>30 60 A6 | 02 03 |
| Application Default Action Code                                                                                                                                                                                                                        | 9F 52 | 0X 04 | 60 00 00 00<br><ul style="list-style-type: none"> <li>If Issuer Authentication performed and failed, decline transaction</li> <li>If Issuer Authentication mandatory and no ARPC received, decline transaction</li> </ul>                                                                                                                                                                                                                                                  |       |

|                             |       |       |                                                                                        |               |
|-----------------------------|-------|-------|----------------------------------------------------------------------------------------|---------------|
| PIN Try Limit               | --    | 0x 01 | 7F                                                                                     | 80<br>10/9010 |
| PIN Try Counter             | 9F 17 | 0x 01 | 03                                                                                     | 90 10         |
| Reference PIN               | --    | 0x 08 | 26 12 34 12 FF FF FF FF<br><br>(Shows the Reference PIN block.<br>The Pin is = 123412) | 80 10         |
|                             |       |       |                                                                                        |               |
| UDK A (for ARQC, MAC & ENC) |       |       | 6D 4A 10 A4 6E C4 54 58                                                                | 80 00         |
| UDK B (for ARQC, MAC & ENC) |       |       | 6E 92 15 86 3E 64 2F A1                                                                | 80 00         |

## 6.15. Test Card 14

This section outlines the profile for Test Card 14- A card with a PDOL requesting a long string of data – the terminal must return 97 zero-value bytes followed by the Transaction Date).

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.15.1. Chip Data

| Data Element                           | Tag   | Length | Value                                                                                                                | DGI   |
|----------------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                        | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53<br>54 20 43 41 52 44 20 31 34<br><br>(VISA ACQUIRER TEST CARD 14) | 01 01 |
| Processing Objects Data<br>Object List | 9F 38 | 0x 0E  | 9F 1A 02 9F 7A 01 9F 02 06 5F 2A 02 9A 64                                                                            | 91 02 |

## 6.16. Test Card 15

This section outlines the profile for Test Card 15 - Card containing both a record with a length of two bytes and one with zero bytes.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.16.1. Chip Data

| Data Element                              | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                               | DGI   |
|-------------------------------------------|-------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                           | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53<br>54 20 43 41 52 44 20 31 35<br><br>(VISA ACQUIRER TEST CARD 15)                                                                                                                                                                                                                                                                                                | 01 01 |
| Application Currency code                 | 9F 42 | 0x 02  | 08 40                                                                                                                                                                                                                                                                                                                                                                                                               | 03 02 |
| Application Effective Date                | 5F 25 | 0x 03  | 95 07 01                                                                                                                                                                                                                                                                                                                                                                                                            | 03 02 |
| Application Expiration Date               | 5F 24 | 0x 03  | 15 12 31                                                                                                                                                                                                                                                                                                                                                                                                            | 03 02 |
| Application Version Number                | 9F 08 | 0x 02  | 00 8C                                                                                                                                                                                                                                                                                                                                                                                                               | 03 02 |
| Issuer Country Code                       | 5F 28 | 0x 02  | 08 40                                                                                                                                                                                                                                                                                                                                                                                                               | 03 02 |
| Service Code                              | 5F 30 | 0x 02  | 02 01                                                                                                                                                                                                                                                                                                                                                                                                               | 03 02 |
| Application Usage Control                 | 9F 07 | 0x 02  | FF 00                                                                                                                                                                                                                                                                                                                                                                                                               | 03 02 |
| Cardholder Verification Method List (CVM) | 8E    | 0x 0E  | 00 00 00 00 00 00 00 00 1E 03 02 03 1F 00<br>Amount X = '00000000'<br>Amount Y = '00000000'<br><br>o CVM Code 1 '1E03'<br>o Signature, if supported<br>o Fail CVM<br>o CVM Code 2 '0203'<br>o Online PIN , if terminal supports CVM<br>o Fail cardholder verification if this CVM is unsuccessful<br>o CVM Code 3 '1F00'<br>o No CVM Required, Always<br>o Fail cardholder verification if this CVM is unsuccessful | 03 02 |
| IAC – Default                             | 9F 0D | 0x 05  | F0 40 00 88 00                                                                                                                                                                                                                                                                                                                                                                                                      | 03 02 |
| IAC – Denial                              | 9F 0E | 0x 05  | 00 10 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                      | 03 02 |
| IAC – Online                              | 9F 0F | 0x 05  | F0 40 00 98 00                                                                                                                                                                                                                                                                                                                                                                                                      | 03 02 |
| Issuer Public Key Certificate             | 90    | 81 80  | 6F C4 63 DD D0 2A 73 B3 5C 84 DA A7 26 EE<br>4D 3F 25 32 66 22 F1 D8 2A 07 48 11 AE 2B<br>1B 9A 67 CB 58 D9 55 73 5E E6 35 D5 71 F3<br>9B 5C E0 F6 4D 71 AF 73 2D 83 F3 7E 2B D5<br>6D 67 22 13 76 C9 9B 14 3B 05 30 F2 FC EA<br>B2 FE 63 50 C6 2F CE A0 C1 63 E4 BD 84 EC                                                                                                                                          | 02 01 |

|                                                          |       |       |                                                                                                                                              |       |
|----------------------------------------------------------|-------|-------|----------------------------------------------------------------------------------------------------------------------------------------------|-------|
|                                                          |       |       | B8 43 42 D0 5E BF B6 8F 6A 9E 49 96 D2 CA<br>B9 63 96 2E 54 8A 5B EE F5 EF FF D0 19 55<br>B9 2A B5 06 4B AC B0 C8 BC 3E 1C 40 28 6D<br>FE FC |       |
| Issuer Public Key<br>Exponent                            | 9F 32 | 0x 01 | 03                                                                                                                                           | 02 02 |
| Issuer Public Key<br>Remainder                           | 92    | 0x 14 | D8 0D 54 FB EC B3 E7 6B 0B 57 1A 70 1D FF 35 D3<br>61 D9 F9 B3                                                                               | 02 02 |
| Certificate Authority<br>Public Key Index                | 8F    | 0x 01 | 99                                                                                                                                           | 02 02 |
| Certificate Expiration<br>Date<br>(for information only) | N/A   |       | 12 30<br>December 2030                                                                                                                       |       |
| ICC Public Key<br>Remainder                              | 9F 48 | 0x 00 | N/A                                                                                                                                          | 02 02 |

## 6.17. Test Card 16

This section outlines the profile for Test Card 16 – A card containing two applications with the same AIDs, but each with a unique suffix.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.17.1. Chip Data

- o Visa Credit: This is the first priority application. It is expired, requires cardholder confirmation, and the IAC is set to decline offline if application is expired
- o Visa Debit: This is the second priority application. It is not expired and does not require cardholder confirmation.

### Application 01

| Data Element                                      | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                      | DGI                 |
|---------------------------------------------------|-------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Cardholder Name                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53<br>54 20 43 41 52 44 20 31 36<br><br>(VISA ACQUIRER TEST CARD 16)                                                                                                                                                                                                                                                                                       | 01 01               |
| Application Identifier (AID) – for application 08 | 4F    | 0x 08  | A0 00 00 00 03 10 10 08<br><br>(The suffix is 08).                                                                                                                                                                                                                                                                                                                                                         | Set at install time |
| Application Priority Indicator                    | 87    | 0x 01  | 81<br>(Application is 1st priority and requires cardholder confirmation)                                                                                                                                                                                                                                                                                                                                   | 91 02               |
| Application Label                                 | 50    | 0x 0B  | 56 49 53 41 20 43 52 45 44 49 54<br>(VISA CREDIT)                                                                                                                                                                                                                                                                                                                                                          | 91 02               |
| Application Expiration Date                       | 5F 24 | 0x 03  | 05 12 31                                                                                                                                                                                                                                                                                                                                                                                                   | 03 02               |
| Issuer Action Code – Online                       | 9F 0F | 0x 05  | F0 00 00 98 00<br><br><ul style="list-style-type: none"> <li>• Offline data authentication not performed</li> <li>• Offline Static Data Authentication failure</li> <li>• Chip data missing</li> <li>• PAN on terminal exception file</li> <li>• Transaction exceeds floor limit</li> <li>• Transaction selected randomly for online transmission</li> <li>• Merchant forced transaction online</li> </ul> | 03 02               |
| IAC – Denial                                      | 9F 0E | 0x 05  | 00 40 00 00 00<br><br><ul style="list-style-type: none"> <li>• If application expired, decline offline.</li> </ul>                                                                                                                                                                                                                                                                                         | 03 02               |

|                              |       |       |                                                                                                                                                                                                                                                                                                                                   |       |
|------------------------------|-------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Issuer Action Code – Default | 9F 0D | 0x 05 | F0 00 00 88 00 <ul style="list-style-type: none"> <li>• Offline data authentication not performed</li> <li>• Offline Static Data Authentication failure</li> <li>• Chip data missing</li> <li>• PAN on terminal exception file</li> <li>• Transaction exceed floor limit</li> <li>• Merchant forced transaction online</li> </ul> | 03 02 |
|------------------------------|-------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|

## Application 02

| Data Element                                      | Tag   | Length | Value                                                                                                            | DGI                 |
|---------------------------------------------------|-------|--------|------------------------------------------------------------------------------------------------------------------|---------------------|
| Cardholder Name                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53<br>54 20 43 41 52 44 20 31 36<br>(VISA ACQUIRER TEST CARD 16) | 01 01               |
|                                                   |       |        |                                                                                                                  |                     |
| Application Identifier (AID) – for application 09 | 4F    | 0x 08  | A0 00 00 00 03 10 10 09                                                                                          | Set at install time |
| Application Priority Indicator                    | 87    | 0x 01  | 02<br>(Application is 2nd priority and does not require cardholder confirmation)                                 | 91 02               |
| Application Label                                 | 50    | 0x 0A  | 56 49 53 41 20 44 45 42 49 54<br>(VISA DEBIT)                                                                    | 91 02               |
| Application Preferred Name                        | 9F 12 | 0x 0B  | 56 49 53 41 20 44 45 42 49 54 4F<br>(VISA DEBITO)                                                                | 91 02               |

## 6.18. Test Card 17

This section outlines the profile for Test Card 17 - Card configured to support minimum requirements.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.18.1. Magnetic Stripe – Track Data

Magnetic Stripe Track 1:  
B4761739001010176^VISA ACQUIRER TEST CARD  
17^15122011483500949000000

Magnetic Stripe Track 2:  
4761739001010176=15122011483594989

### 6.18.2. Chip Data:

- o Magnetic Stripe Image card where the CDOLs contain the minimum data elements

| Data Element                                      | Tag   | Length | Value                                                                                                                                                        | DGI   |
|---------------------------------------------------|-------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53<br>54 20 43 41 52 44 20 31 37<br><br>(VISA ACQUIRER TEST CARD 17)                                         | 01 01 |
| Track 2 Equivalent Data                           | 57    | 0x 11  | 47 61 73 90 01 01 01 76 D1 51 22 01 14 83 55 57 89                                                                                                           | 01 01 |
| Track 1 Discretionary Data                        | 9F 1F | 0x 10  | 31 34 38 33 35 30 30 35 35 37 30 30 30 30 30 30                                                                                                              | 01 01 |
|                                                   |       |        |                                                                                                                                                              |       |
| Application Interchange Profile                   | 82    | 0x 02  | 08 00<br>• Terminal Risk Management to be performed                                                                                                          | 91 04 |
| AFL List                                          | 94    | 0x 08  | 08 01 01 00 18 01 02 00                                                                                                                                      | 91 04 |
|                                                   |       |        |                                                                                                                                                              |       |
| Application Primary Account Number (PAN) (Signed) | 5A    | 0x 08  | 47 61 73 90 01 01 01 76                                                                                                                                      | 03 01 |
| Application PAN Sequence Number                   | 5F 34 | NA     | (remove this tag)                                                                                                                                            | 03 01 |
|                                                   |       |        |                                                                                                                                                              |       |
| IAC—Default                                       | 9F 0D | 0x 05  | 10 40 00 88 00<br><br>• PAN on terminal exception file<br>• Expired application<br>• Transaction exceeds floor limit<br>• Merchant forced transaction online | 03 02 |
| IAC—Online                                        | 9F 0F | 0x 05  | 10 40 00 98 00<br><br>• PAN on terminal exception file<br>• Expired application<br>• Transaction exceeds floor limit<br>• Merchant forced transaction online | 03 02 |

| Data Element                             | Tag   | Length | Value                                                                           | DGI   |
|------------------------------------------|-------|--------|---------------------------------------------------------------------------------|-------|
| CDOL 1                                   | 8C    | 0x 02  | 95 05<br>• Terminal Verification Results                                        | 03 02 |
| CDOL 2                                   | 8D    | 0x 04  | 8A 02 95 05<br>• Authorization Response Code<br>• Terminal Verification Results | 03 02 |
| Application Currency Code                | 9F 42 | NA     | (remove this tag)                                                               | 03 02 |
| Application Effective Date               | 5F 25 | NA     | (remove this tag)                                                               | 03 02 |
|                                          |       |        |                                                                                 |       |
| Issuer Public Key Certificate            | 90    | NA     | (remove this tag)                                                               | 02 01 |
| Certification Authority Public Key Index | 8F    | NA     | (remove this tag)                                                               | 02 02 |
| Issuer PK Exponent                       | 9F 32 | NA     | (remove this tag)                                                               | 02 02 |
| Issuer PK Remainder                      | 92    | NA     | (remove this tag)                                                               | 02 02 |
| Signed Static Application Data           | 93    | NA     | (remove this tag)                                                               | 02 03 |
|                                          |       |        |                                                                                 |       |
| Cryptogram Version Number                | C6    | 0x 01  | 0C                                                                              | 07 01 |
| Derivation Key Index                     | --    | 0x 01  | 00                                                                              | 07 01 |
|                                          |       |        |                                                                                 |       |
| UDK A (for ARQC, MAC & ENC)              |       |        | D9 98 A2 C7 C7 8B 44 E8                                                         | 80 00 |
| UDK B (for ARQC, MAC & ENC)              |       |        | BF 55 4D 70 0F AC 25 6F                                                         | 80 00 |

## 6.19. Test Card 18 (Previously Test Card 49)

This section outlines the profile for Test Card 18, a T=1 card containing an Issuer Public Key Certificate signed by the Visa CA Test Key of 1984 bits.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.19.1. Chip Data

- Card must support the T=1 (rather than T=0) protocol.
- Card must support a specific CVM list.

| Data Element                    | Tag   | Length | Value                                                                                                                                                                     | DGI   |
|---------------------------------|-------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                 | 5F 20 | 0x1A   | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45<br>53 54 20 43 41 52 44 20 31 38<br><br>(VISA ACQUIRER TEST CARD 18)<br><br>Note: This value changes with each test card. | 01 01 |
| Application Interchange Profile | 82    | 0x 02  | 7C 00<br>o DDA<br>o SDA<br>o Cardholder verification<br>o Terminal Risk Management is performed<br>O Issuer Authentication is supported)                                  | 91 04 |
| AFL List                        | 94    | 0x 0C  | 08 01 01 00 10 01 05 00 18 01 02 01                                                                                                                                       | 91 04 |

| Data Element                      | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | DGI   |
|-----------------------------------|-------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Verification Method    | 8E    | 0x 12  | 0000 0000 0000 0000 0201 4103 1E03 0203 1F00<br><br>Amount X = 00000000<br>Amount Y = 00000000<br><br><ul style="list-style-type: none"> <li>• CVM Code 1 '0201'               <ul style="list-style-type: none"> <li>o Online PIN, if cash/cashback</li> <li>o Fail cardholder Verification if this CVM is unsuccessful</li> </ul> </li> <li>• CVM Code 2 '4103'               <ul style="list-style-type: none"> <li>o Offline (Plaintext) PIN, if terminal supports CVM</li> <li>o Apply succeeding cardholder Verification if this CVM is unsuccessful</li> </ul> </li> <li>• CVM Code 3 '1E03'               <ul style="list-style-type: none"> <li>o Signature, if terminal supports CVM</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>• CVM Code 4 '0203'               <ul style="list-style-type: none"> <li>o Online PIN, if terminal supports CVM</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>• CVM Code 5 '1F00'               <ul style="list-style-type: none"> <li>o No CVM Required, Always</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> </ul> | 03 02 |
|                                   |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |
| ICC Public Key Certificate        | 9F 46 | 0x 90  | AB F8 F4 B5 61 A0 C4 4F D6 A2 4A 92 57 39 51 6D<br>B1 97 8F 3D F2 87 4C AF 26 50 92 9D 9C AE E6 EF<br>A7 1A AB CC 7B A3 07 22 F5 07 F3 8F 28 50 71 D8<br>31 D8 EA 92 38 AA AE B0 17 0D 41 8D 59 1B C6 D6<br>21 78 64 A9 7E DB 14 D9 D8 2A E1 16 34 03 4E 84<br>AF 66 A5 6E 62 DF C7 0D ED FF 1F FF F7 F4 84 2D<br>5D 7F E9 DB 3C 82 10 26 3B EF CF 9E 1F FA 69 88<br>1A 2A C6 91 7A FB 11 0A D6 C6 A0 70 9C 98 92 4C<br>C3 8D C8 17 10 33 5C 9B B8 6A CB DD BF 9A 68 B7<br>40 14 EC 6A C3 D0 DD 52 DA B7 32 ED 71 5D F2 44<br>87 62 68 0E F5 FE D1 5B D3 9A EC 28 CA C1 BA 12<br>6B 60 B2 9B 7A 81 74 A4 2A 07 B6 F0 0F E3 CD 02<br>77 9A FA 3B 26 C4 27 AD F9 91 47 83 F9 C8 FC 76<br>68 7C C9 56 08 22 1D 9E 94 99 1D 03 47 21 EE 90<br>A6 73 A5 F9 11 23 30 11 72 D3 FB AA 24 A4 3F 87<br>9E 75 15 26 4B E1 BB                                                                                                                                                                                                                                                                                                                                                                              | 02 04 |
| ICC Public Key Exponent           | 9F 47 | 0x 01  | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 02 05 |
| ICC Public Key Remainder          | 9F 48 | 0x 23  | F8 8B 39 7E C7 62 52 9C F6 94 C7 AF 7E 26 99 3E<br>FD F4 A9 C0 77 B2 B6 D6 25 BF 2B EC 16 0F 4D 65<br>CF 7C FF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 02 05 |
| ICC Private (Secret) Key Exponent |       | 0x F0  | 8C CF F1 D8 8B AF 16 90 82 28 6E 95 15 00 96 81<br>EC 2A 67 14 52 AB 90 45 66 CF 43 A3 56 57 3B 8A<br>D3 4D AD 36 F2 0A C7 6E 2B 88 79 95 D2 36 DE 84<br>72 9B 05 D1 17 9D 7C CC 51 06 9B 8A A8 E1 61 72<br>AA 92 56 22 B9 2C 6E EB 22 6F 57 63 B4 60 89 B7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 81 01 |

| Data Element                                                                                                                           | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | DGI   |
|----------------------------------------------------------------------------------------------------------------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|                                                                                                                                        |       |        | FB 9F 17 E6 F4 04 5C D9 4E 6D C4 34 53 A7 60 1C<br>5B B8 FE AD A0 9D 12 9C ED C8 EA A4 C8 40 ED A7<br>4C 4A 35 E7 7D 8D 22 13 93 14 B5 45 00 75 D2 BA<br>83 CE D3 C3 E3 8E 57 5C 7A EC EF BF 2D DC CE 02<br>F5 67 FA 4C 40 65 82 81 9D 18 21 A6 E5 0B 0B FE<br>56 4B E3 2B 5B C0 D1 85 F9 D0 44 98 A5 E6 59 A4<br>08 BA E2 B4 82 0F F0 2D 6F 91 3F 9F 0F 60 33 03<br>9B D5 6B AE 49 BA 92 A7 E7 D8 6A F5 36 FF D0 5E<br>4F 68 A6 11 5F 0C 23 A1 B7 3D 1E 2F 2E 6A D3 D1<br>D4 83 BA 21 E3 17 81 22 F5 47 FF 48 5C 91 2B AB                                                                                                                                                                                                                                                                                           |       |
| Dynamic Data Authentication Data Object List (DDOL)                                                                                    | 9F 49 | 0x 03  | 9F 37 04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 02 05 |
| ICC Public Key Modulus                                                                                                                 |       | 0x F0  | D3 37 EA C4 D1 86 A1 D8 C3 3C A5 DF 9F 80 E1 C2<br>E2 3F 9A 9E 7C 01 58 68 1A 36 E5 75 01 82 D9 50<br>3C F4 83 D2 6B 10 2B 25 41 4C B6 60 BB 52 4D C6<br>AB E8 88 B9 A3 6C 3B 32 79 89 E9 4F FD 52 12 2B<br>FF DB 81 34 15 C2 A6 60 B3 A7 03 15 8E 90 CE 93<br>F9 6E A3 DA 6E 06 8B 45 F5 A4 A6 4E 7D 7B 10 2A<br>89 95 7E 04 70 EB 9B EB 64 AD 5F F7 2C 61 64 7A<br>F2 6F 50 DB 3C 53 B3 1F 2D B3 9B 5D C2 B5 97 32<br>D0 8A B4 F1 F1 E1 5E 82 6C 38 3A 01 E0 03 AF F4<br>9F 71 9A 99 F9 FD BD 8A F2 62 95 68 ED 98 65 16<br>FF 01 33 11 3B D0 19 74 24 AC CE 45 5F EF 86 86<br>E9 23 A5 EC 14 40 99 11 D0 B2 99 E9 F3 A3 7C 31<br>E8 D3 0C D9 15 30 AE A4 2C 8D 24 49 E9 F8 8B 39<br>7E C7 62 52 9C F6 94 C7 AF 7E 26 99 3E FD F4 A9<br>C0 77 B2 B6 D6 25 BF 2B EC 16 0F 4D 65 CF 7C FF                            | 81 03 |
| Issuer Public Key Certificate<br>(Issuer Public Key of 1976 bits signed by the Visa CA Test Key of 1984 bits)<br><br>(for CA index 94) | 90    | 0x F8  | 41 41 11 96 60 EF 1B AD B7 69 34 32 64 7F 42 1A<br>8B C5 E2 C5 AF 6F 27 D1 EF A7 10 A0 D4 EE 14 82<br>CE 3A D0 1F 3C 53 76 51 E1 34 BB AE D4 1E 23 A8<br>52 09 50 DE 5E F6 C7 B7 CA BC C9 87 8B 46 10 B8<br>FC 95 02 44 40 52 17 EC 48 35 12 B3 35 7E 52 89<br>F9 AD 6C 5B 32 FC 36 61 9F 74 B6 1D 54 6B 73 E4<br>1B 69 51 04 B5 C2 46 E2 00 A0 BD 42 05 94 44 08<br>C2 C8 52 87 05 80 97 65 F3 3B 50 7C 41 1E 3A A3<br>D6 FA 9F BA 97 1B 91 14 B9 35 50 B9 8A 11 3C 93<br>06 7B 0D 6F A6 9D EB 92 8A 0D 8D 5D 95 E4 9B 64<br>8D CA 20 61 AB 24 D1 77 37 DF 5A 9A 66 F5 3E F3<br>DA F4 E0 EE C5 A7 0D D5 A2 5D 4A 70 45 C9 0A 55<br>32 84 69 4F 78 CF EC 1D A8 11 1E B9 D9 6F C9 BB<br>DE CF AD 40 B6 18 D9 9D 73 D3 4E C1 91 0D D1 F5<br>E2 98 EB 44 F4 25 82 07 E6 9E CE A7 FA 54 8A 68<br>56 74 12 B0 F2 27 8F 3A | 02 01 |
| Issuer Public Key Modulus (length of 1976 bits)<br><br>(This is provided for information only; it is not personalized on the card)     |       | 0x F7  | CD 7F 2E EC 56 37 5D 6B F8 8F 5F E8 D9 3C DE F2<br>F0 0D 93 35 4F 6A EE CF 83 AE EC 43 E1 A2 AB 9E<br>A8 9D 1B F5 68 9B 55 DC 65 67 6D D7 BD 65 5A 08<br>58 A6 DD A0 1A A3 41 13 B4 8B 57 7C 0E 1C CF 16<br>E3 F9 02 A3 FC B9 FF 84 49 74 11 80 F7 21 07 51<br>91 8B F4 06 C7 CF A0 0C 8F 3E 7C 5F 03 46 CC 6A<br>AD 31 41 31 17 37 98 66 70 45 11 5F BF 39 A7 62<br>1C C9 87 B2 64 9A 24 12 54 06 CD 62 95 4F 77 C9<br>3C D4 89 11 A0 8F 70 79 6B 25 97 8C 95 1F DA 57<br>CF 4C A5 F2 F8 F9 5D 49 2F 03 CB 5D 55 D3 E6 22<br>70 2F 6E FA 02 1E 7E 14 1E 49 48 85 82 AE 8D B2<br>91 94 6D 04 6C 77 CE BB 6C 09 DF 65 65 4F 53 30<br>0B 77 58 A0 4A ED 59 6C 3E 7C 6E 6A C5 9B 7F 59<br>66 C3 90 06 DE CC 3C 96 C2 B3 E4 36 10 32 E1 31                                                                               |       |

| Data Element                                                                                                                                                                                                                                           | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | DGI               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|                                                                                                                                                                                                                                                        |       |        | 72 BB 49 87 53 F1 E9 4E 2F 86 4D BA 1E 09 23 F0<br>15 9F 6E 6E E5 F1 57                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |
| Issuer Public Key Exponent                                                                                                                                                                                                                             | 9F 32 | 0x 01  | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 02 02             |
| Issuer Private Key Exponent<br><br>(This is provided for information only; it is not personalized on the card)                                                                                                                                         |       | 0x F7  | 88 FF 74 9D 8E CF 93 9D 50 5F 95 45 E6 28 94 A1<br>F5 5E 62 23 8A 47 49 DF AD 1F 48 2D 41 17 1D 14<br>70 68 BD 4E 45 BC E3 E8 43 9A 49 3A 7E 43 91 5A<br>E5 C4 93 C0 11 C2 2B 62 78 5C E4 FD 5E BD DF 64<br>97 FB 57 17 FD D1 55 02 DB A2 B6 55 FA 16 04 E1<br>0B B2 A2 AF 2F DF C0 08 5F 7E FD 94 AC D9 DD 9C<br>73 76 2B 76 0F 7A 65 99 A0 2E 0B 95 2A 26 6F 96<br>BD DB AF CC 43 11 6D 61 8D 59 DE 2E 7E B3 77 D8<br>F7 ED 70 B6 FE 6B B4 BD 0F E1 75 4E 71 FE 4E F9<br>BA 31 FA 3C EF E9 37 81 69 FA A2 38 58 16 14 A1<br>DF BC 65 A1 22 9C 98 68 51 EC 49 74 DF 02 AB 2C<br>35 D8 B3 30 31 BB 7A 71 4F 16 B4 66 4E 5B 62 0E<br>FA 3A A7 0D CE 64 F6 C7 34 E7 D4 D0 F5 00 B2 0E<br>C4 94 60 9E 44 1D 66 96 DE C7 69 00 7A B0 4F 3E<br>76 A6 97 A3 76 7A B7 77 A9 51 6B 9C 8C FD 81 77<br>E8 7C 1C 21 FF 53 BB | 02 02             |
| Issuer Public Key Remainder                                                                                                                                                                                                                            | 92    | 0x 23  | DE CC 3C 96 C2 B3 E4 36 10 32 E1 31 72 BB 49 87<br>53 F1 E9 4E 2F 86 4D BA 1E 09 23 F0 15 9F 6E 6E<br>E5 F1 57                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 02 02             |
| Certification Authority Public Key Index                                                                                                                                                                                                               | 8F    | 0x 01  | 94<br><br>(Visa CA Test Key of 1984 bits)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 02 02             |
| Certificate Expiration Date<br><br>(for information only)                                                                                                                                                                                              |       |        | 12 15<br>(December 2015)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |
| Signed Static Application Data<br>Note: The Signed Application Data is created using the PAN and PAN Sequence Number only. This allows the same Signed Application Data to be used on cards with different data elements (e.g., different IACs, etc.). | 93    | 0x F7  | 81 7A EC 9C D6 FF DF AE 19 C1 EC F1 BF DB FC 90 2E AC<br>71 D2 E9 34 23 37 71 E0 B7 2B 57 D6 F9 96 83 E8 27 BB 7C 7D<br>5F CB 7F ED E0 19 B6 D1 58 1F 6E DD 02 D0 BB CD C9 07 10<br>78 C2 20 68 93 9C 6D 67 60 48 A7 CC BA 37 2E 8E DF 5A F4<br>FF 2F 6E 17 B0 37 5C E1 3C 57 D6 37 83 F4 48 80 9E 35 79 85<br>C4 71 5F FA FD 21 86 F6 A8 18 7C 43 10 2E 72 4C 08 32 A1<br>4F 2C C9 72 1F 30 E1 AF 5F 8E A0 EB 81 7F BC 7E 33 8E EB<br>C9 82 CA 65 BF 3F 24 62 5F 5F 84 A5 C4 93 B2 B8 97 AF 92 87<br>65 CD 4A E2 A1 0B 6F 31 8B 8B CC B2 F2 8D 6C B9 83 99 E6<br>14 0E 07 1A 29 55 5C 57 17 E1 3C 58 04 F7 AD 45 08 93 E4 8A<br>CE 1A 7B 78 6A 36 13 EE 1A 6D 26 1E AD 7C 38 6F 24 84 6D<br>E9 55 80 7A ED F3 A4 4F A5 45 89 EA 1F 7F AD A8 D0 30 44<br>1E 5E C1 E2 D5 F0 C6 69 92 B3 87 C4 C8 2F BF 8A          | 02 03             |
|                                                                                                                                                                                                                                                        |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |
| PIN Try Limit                                                                                                                                                                                                                                          | --    | 0x 01  | 0F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 80<br>10/901<br>0 |
| PIN Try Counter                                                                                                                                                                                                                                        | 9F 17 | 0x 01  | Initialized to PIN Try Limit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 90 10             |
| Reference PIN                                                                                                                                                                                                                                          | --    | 0x 08  | 24 12 34 FF FF FF FF FF<br><br>(Shows the Reference PIN block.<br>The PIN is = 1234)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 80 10             |

## 6.20. Test Card 19 (Previously Test Card 50)

This section outlines the profile for Test Card 19, a card containing the Visa RID with the Plus PIX and a suffix of '01'.

Plus is a deposit access product that offers worldwide cash access and other around-the-clock financial services through the Visa Global ATM Network. The PLUS Program can be added to any banking card and complements the utility of other Visa products:

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.20.1. Magnetic Stripe – Track Data

Magnetic Stripe Track 1:

B4761739001010671^VISA ACQUIRER TEST CARD  
19^15122201350600076000000

Magnetic Stripe Track 2:

4761739001010671=15122201350607689

### 6.20.2. Chip Data:

| Data Element                    | Tag   | Length | Value                                                                                                                                                        | DGI                 |
|---------------------------------|-------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Cardholder Name                 | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 20 43 41<br>52 44 20 31 39<br><br>(VISA ACQUIRER TEST CARD 19)                                         | 01 01               |
| Track 2 Equivalent Data         | 57    | 0x 11  | 47 61 73 90 01 01 06 71 D1 51 22 20 13 50 65 81 89                                                                                                           | 01 01               |
| Track 1 Discretionary Data      | 9F 1F | 0x 10  | 31 33 35 30 36 30 30 35 38 31 30 30 30 30 30 30                                                                                                              | 01 01               |
|                                 |       |        |                                                                                                                                                              |                     |
| Application Identifier (AID)    | 4F    | 0x 08  | A0 00 00 00 03 80 10 01<br>(This is the Visa RID with the PLUS PIX and a suffix of '01').                                                                    | Set at install time |
| Application Label               | 50    | 0x 04  | 50 4C 55 53<br><br>(PLUS)                                                                                                                                    | 91 02               |
| Application Preferred Name      | 9F 12 |        | Remove this tag                                                                                                                                              | 91 02               |
| Issuer Code Table Index         | 9F 11 |        | Remove this tag                                                                                                                                              | 91 02               |
| Application Priority Indicator  | 87    |        | Remove this tag                                                                                                                                              | 91 02               |
|                                 |       |        |                                                                                                                                                              |                     |
| Application Interchange Profile | 82    | 0x 02  | 1C 00<br><ul style="list-style-type: none"> <li>Offline Static Data Authentication is NOT supported</li> <li>Cardholder Verification is supported</li> </ul> | 91 04               |

| Data Element                                                                                                                          | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | DGI   |
|---------------------------------------------------------------------------------------------------------------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|                                                                                                                                       |       |        | <ul style="list-style-type: none"> <li>Terminal Risk Management to be performed</li> <li>Issuer Authentication is supported</li> </ul>                                                                                                                                                                                                                                                                                                                                                       |       |
| AFL List                                                                                                                              | 94    | 0x 08  | 08 01 01 00 18 01 02 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 91 04 |
|                                                                                                                                       |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| Application Primary Account Number (PAN) (Signed)                                                                                     | 5A    | 0x 08  | 47 61 73 90 01 01 06 71                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 03 01 |
|                                                                                                                                       |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| Service Code                                                                                                                          | 5F 30 | 0x 02  | 02 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 03 02 |
| Application Usage Control                                                                                                             | 9F 07 | 0x 02  | C2 00<br><br>Byte 1 <ul style="list-style-type: none"> <li>BIT 8 = 1 Valid for domestic cash transactions</li> <li>BIT 7 = 1 Valid for international cash transactions</li> <li>BIT 6 = 0 Not valid for domestic goods</li> <li>BIT 5 = 0 Not valid for international goods</li> <li>BIT 4 = 0 Not valid for domestic services</li> <li>BIT 3 = 0 Not valid for international services</li> <li>BIT 2 = 1 Valid at ATMs</li> <li>BIT 1 = 0 Not valid at terminals other than ATMs</li> </ul> | 03 02 |
| IAC – Denial                                                                                                                          | 9F 0E | 0x 05  | 00 00 80 00 00<br><br><ul style="list-style-type: none"> <li>If cardholder verification unsuccessful, decline offline</li> </ul>                                                                                                                                                                                                                                                                                                                                                             | 03 02 |
| IAC – Online                                                                                                                          | 9F 0F | 0x 05  | 00 00 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 03 02 |
| Cardholder Verification Method List (CVM)                                                                                             | 8E    | 0x 0A  | 0000 0000 0000 0000 0203<br>Amount X = 00000000<br>Amount Y = 00000000<br><br><ul style="list-style-type: none"> <li>CVM Code 1 '0203'               <ul style="list-style-type: none"> <li>Online PIN, if terminal supports CVM</li> <li>Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> </ul>                                                                                                                                                                    | 03 02 |
|                                                                                                                                       |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| Signed Static Application Data                                                                                                        | 93    |        | Remove from card.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 02 03 |
| Issuer Public Key Certificate<br>(Issuer Public Key of 896 bits signed by the Visa CA Test Key of 1024 bits)<br><br>(for CA index 99) | 90    |        | Remove from card.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 02 01 |

| Data Element                                                                                                                      | Tag   | Length | Value             | DGI   |
|-----------------------------------------------------------------------------------------------------------------------------------|-------|--------|-------------------|-------|
| Issuer Public Key Modulus (length of 896 bits)<br><br>(This is provided for information only; it is not personalized on the card) |       |        | Remove from card. |       |
| Issuer Public Key Exponent                                                                                                        | 9F 32 |        | Remove from card. | 02 02 |
| Issuer Private Key Exponent<br><br>(This is provided for information only; it is not personalized on the card)                    |       |        | Remove from card. |       |
| Issuer Public Key Remainder                                                                                                       | 92    |        | Remove from card. | 02 02 |
| Certification Authority Public Key Index                                                                                          | 8F    |        | Remove from card. | 02 02 |
| Certificate Expiration Date<br><br>(for information only)                                                                         |       |        | Remove from card. |       |
|                                                                                                                                   |       |        |                   |       |
| UDK A (for ARQC, MAC & ENC)                                                                                                       |       |        | D120BA08 813CEA69 | 80 00 |
| UDK B (for ARQC, MAC & ENC)                                                                                                       |       |        | 3C1C5E7E F31481E4 | 80 00 |

## 6.21. Test Card 20

This section outlines the profile for Test Card 20, a Visa Electron card with a non-usable magnetic stripe.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.21.1. Magnetic Stripe – Track Data

Magnetic Stripe Track 1:  
 B0000000000000000^VISA ACQUIRER TEST CARD  
 20^151222100000000000000000

Magnetic Stripe Track 2:  
 0000000000000000=15122211143888489

### 6.21.2. Chip Data

| Data Element               | Tag   | Length | Value                                                                                                                | DGI   |
|----------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name            | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 20<br>43 41 52 44 20 32 30<br><br>(VISA ACQUIRER TEST CARD 20) | 01 01 |
| Track 2 Equivalent Data    | 57    | 0x 11  | 47 61 73 90 01 01 00 10 D1 51 22 21 11 43 80 44 89                                                                   | 01 01 |
| Track 1 Discretionary Data | 9F 1F | 0x 10  | 31 31 34 33 38 30 30 30 34 34 30 30 30 30 30 30                                                                      | 01 01 |
|                            |       |        |                                                                                                                      |       |
| Application Label          | 50    | 0x 0D  | 56 49 53 41 20 45 4C 45 43 54 52 4F 4E<br><br>(VISA ELECTRON)<br><br>The label contains a space.                     | 91 02 |
|                            |       |        |                                                                                                                      |       |
| Application Preferred Name | 9F 12 | 0x 10  | 45 4C 45 43 54 52 4F 4E 20 44 45 20 56 49 53 41<br><br>(ELECTRON DE VISA)                                            | 91 02 |
|                            |       |        |                                                                                                                      |       |
| Service Code               | 5F 30 | 0x 02  | 02 21                                                                                                                | 03 02 |

## 6.22. Test Card 21 (Previously Test Card 32)

This section outlines the profile for Test Card 21 (Card configured with PIN Try Limit Exceeded and fallback to signature).

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

**IMPORTANT:** The vendor developing the test card must set the PIN Try Limit to 00 (to block the PIN) when personalizing this test card.

### 6.22.1. Chip Data

| Data Element                   | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                     | DGI               |
|--------------------------------|-------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Cardholder Name                | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 20<br>43 41 52 44 20 32 31<br><br>(VISA ACQUIRER TEST CARD 21)                                                                                                                                                                                                      | 01 01             |
| Cardholder Verification Method | 8E    | 0x 0E  | 0000 0000 0000 0000 4103 4203 1E03<br>Amount X = '00000000'<br>Amount Y = '00000000'<br><br>o CVM Code 1 '4103'<br>o Offline PIN, if supported<br>o Apply succeeding CVM<br>o CVM Code 2 '4203'<br>o Online PIN, if supported<br>o Apply succeeding CVM<br>o CVM Code 3 '1E03'<br>o Signature, if supported<br>o Fail CVM | 03 02             |
| Issuer Action Code – Default   | 9F 0D | 0x 05  | 00 00 00 00 00                                                                                                                                                                                                                                                                                                            | 03 02             |
| Issuer Action Code – Denial    | 9F 0E | 0x 05  | 00 00 80 00 00<br><br>(If cardholder verification unsuccessful, decline offline)                                                                                                                                                                                                                                          | 03 02             |
| Issuer Action Code – Online    | 9F 0F | 0x 05  | 00 00 00 00 00                                                                                                                                                                                                                                                                                                            | 03 02             |
| PIN Try Limit                  | --    | 0x 01  | 00<br><br>See important note above.                                                                                                                                                                                                                                                                                       | 80<br>10/901<br>0 |
| PIN Try Counter                | 9F 17 | 0x 01  | Initialized to PIN Try Limit.                                                                                                                                                                                                                                                                                             | 90 10             |
| Reference PIN                  | --    | 0x 08  | 24 12 34 FF FF FF FF FF<br><br>(Shows the Reference PIN block.<br>The Pin is = 1234)                                                                                                                                                                                                                                      | 80 10             |

## 6.23. Test Card 22 (Previously Test Card 33)

This section outlines the profile for Test Card 22 - Card with the PIN Try Limit exceeded.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

**IMPORTANT:** The vendor developing the test card must set the PIN Try Limit to 00 (to block the PIN) when personalizing this test card.

### 6.23.1. Chip Data

| Data Element                   | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DGI   |
|--------------------------------|-------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 20<br>43 41 52 44 20 32 32<br><br>(VISA ACQUIRER TEST CARD 22)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 01 01 |
| Cardholder Verification Method | 8E    | 0x 10  | 0000 0000 0000 0000 0103 1E03 0203 1F00<br><br>Amount X = 00000000<br>Amount Y = 00000000<br><br><ul style="list-style-type: none"> <li>o CVM Code 1 '0103' <ul style="list-style-type: none"> <li>o Offline (Plaintext) PIN, if terminal supports CVM</li> <li>o Fail cardholder Verification if this CVM is unsuccessful</li> </ul> </li> <li>o CVM Code 2 '1E03' <ul style="list-style-type: none"> <li>o Signature, if terminal supports CVM</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>o CVM Code 3 '0203' <ul style="list-style-type: none"> <li>o Online PIN , if terminal supports CVM</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>o CVM Code 4 '1F00' <ul style="list-style-type: none"> <li>o No CVM Required, Always</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> </ul> | 03 02 |
| Issuer Action Code – Default   | 9F 0D | 0x 05  | 00 00 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 03 02 |
| Issuer Action Code – Denial    | 9F 0E | 0x 05  | 00 00 20 00 00<br><br>(If PIN Try Limit Exceeded, decline offline)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 03 02 |
| Issuer Action Code – Online    | 9F 0F | 0x 05  | 00 00 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 03 02 |

| Data Element               | Tag   | Length | Value                                                                                   | DGI               |
|----------------------------|-------|--------|-----------------------------------------------------------------------------------------|-------------------|
| Application Default Action | 9F 52 | 0x 04  | 00 40 00 00<br><br>(If PIN Try Limit Exceeded on previous transaction, decline offline) | 0E 01             |
| PIN Try Limit              | --    | 0x 01  | 00<br><br>See important note above.                                                     | 80<br>10/901<br>0 |
| PIN Try Counter            | 9F 17 | 0x 01  | Initialized to PIN Try Limit.                                                           | 90 10             |
| Reference PIN              | --    | 0x 08  | 24 12 34 FF FF FF FF FF<br><br>(Shows the Reference PIN block.<br>The Pin is = 1234)    | 80 10             |

## 6.24. Test Card 23 (Previously Test Card 39)

This section outlines the profile for Test Card 23 - Card containing a CVM List where the first CVM is Offline PIN/signature.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.24.1. Chip Data

| Data Element                   | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DGI   |
|--------------------------------|-------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 20 43 41 52 44 20 32 33<br><br>(VISA ACQUIRER TEST CARD 23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 01 01 |
| Cardholder Verification Method | 8E    | 0x 14  | 0000 0000 0000 0000 0303 0201 0103 0203 1E03 1F00<br><br>Amount X = 0000 0000<br>Amount Y = 0000 0000<br><br><ul style="list-style-type: none"> <li>o CVM Code 1 '0303'               <ul style="list-style-type: none"> <li>o Offline PIN/signature, if terminal supports</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>o CVM Code 2 '0201'               <ul style="list-style-type: none"> <li>o Online PIN, if cash/cashback</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>o CVM Code 3 '0103'               <ul style="list-style-type: none"> <li>o Offline PIN, if terminal supports</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>o CVM Code 4 '0203'               <ul style="list-style-type: none"> <li>o Online PIN, if terminal supports</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>o CVM Code 5 '1E03'               <ul style="list-style-type: none"> <li>o Signature, if terminal supports</li> <li>o Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>o CVM Code 6 '1F00'               <ul style="list-style-type: none"> <li>o No CVM required, always</li> </ul> </li> </ul> | 03 02 |
| IAC – Default                  | 9F 0D | 0x 05  | 00 00 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 03 02 |

| Data Element    | Tag   | Length | Value                                                                             | DGI               |
|-----------------|-------|--------|-----------------------------------------------------------------------------------|-------------------|
| IAC – Denial    | 9F 0E | 0x 05  | 00 00 80 00 00<br>o If cardholder verification is unsuccessful, decline offline   | 03 02             |
| IAC – Online    | 9F 0F | 0x 05  | 00 00 00 00 00                                                                    | 03 02             |
|                 |       |        |                                                                                   |                   |
| PIN Try Limit   | --    | 0x 01  | 0F                                                                                | 80<br>10/90<br>10 |
| PIN Try Counter | 9F 17 | 0x 01  | Initialized to PIN Try Limit.                                                     | 90 10             |
| Reference PIN   | --    | 0x 08  | 24 12 34 FF FF FF FF FF<br><br>(Shows the Reference PIN block. The PIN is = 1234) | 80 10             |

## 6.25. Test Card 24 (Previously Test Card 41)

This section outlines the profile for Test Card 24 - Card is personalized with a 16-digit account number and the unused fields are padded with Fs to the maximum account length.

VSDC Applet Version: 2.71

Changes to make from baseline card:

### 6.25.1. Magnetic Stripe – Track Data

Magnetic Stripe Track 1:  
B4761739001010416^VISA ACQUIRER TEST CARD  
24^15122011161600215000000

Magnetic Stripe Track 2:  
4761739001010416=15122011161621589

### 6.25.2. Chip Data:

NOTE: Because the PAN contains padded Fs, the card requires new Signed Application Data. The vendor creating the test cards may either use the Signed Application Data provided below or generate their own. For simplicity, it is recommended (but not required) that only the PAN and PAN Sequence Number be included in the SDA data.

| Data Element                                      | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                | DGI   |
|---------------------------------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 20<br>43 41 52 44 20 32 34<br><br>(VISA ACQUIRER TEST CARD 24)                                                                                                                                                                                                                                                                                                 | 01 01 |
| Track 2 Equivalent Data                           | 57    | 0x 11  | 47 61 73 90 01 01 04 16 D1 51 22 01 11 61 69 13 89                                                                                                                                                                                                                                                                                                                                                                   | 01 01 |
| Track 1 Discretionary Data                        | 9F 1F | 0x 10  | 31 31 36 31 36 30 30 39 31 33 30 30 30 30 30 30                                                                                                                                                                                                                                                                                                                                                                      | 01 01 |
|                                                   |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| Application Primary Account Number (PAN) (Signed) | 5A    | 0x 0A  | 47 61 73 90 01 01 04 16 FF FF                                                                                                                                                                                                                                                                                                                                                                                        | 03 01 |
|                                                   |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| Issuer Public Key Certificate                     | 90    | 0x 80  | 6F C4 63 DD D0 2A 73 B3 5C 84 DA A7 26 EE 4D 3F<br>25 32 66 22 F1 D8 2A 07 48 11 AE 2B 1B 9A 67 CB<br>58 D9 55 73 5E E6 35 D5 71 F3 9B 5C E0 F6 4D 71<br>AF 73 2D 83 F3 7E 2B D5 6D 67 22 13 76 C9 9B 14<br>3B 05 30 F2 FC EA B2 FE 63 50 C6 2F CE A0 C1 63<br>E4 BD 84 EC B8 43 42 D0 5E BF B6 8F 6A 9E 49 96<br>D2 CA B9 63 96 2E 54 8A 5B EE F5 EF FF D0 19 55<br>B9 2A B5 06 4B AC B0 C8 BC 3E 1C 40 28 6D FE FC | 02 01 |

| Data Element                                                                                               | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                             | DGI   |
|------------------------------------------------------------------------------------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Issuer Public Key Modulus                                                                                  |       | 0x 70  | BD BA DB 8E C4 F4 89 C0 D6 0E 14 63 2C CE AA 41<br>C8 DF D1 2E CF 36 51 DB 4C 84 7D BA 8C 75 5D 6E<br>2F 46 2C FD 99 E1 75 61 EE 6E 6A C6 0F 31 58 57<br>90 C6 F9 5F 06 5E 7D 2A 2C 73 19 07 0B FC B9 44<br>8B 51 27 B6 C9 09 63 DE 7F 62 11 FD 34 EB AA 00<br>47 50 62 81 47 A8 D4 DB 9A A9 0D A8 D8 0D 54 FB<br>EC B3 E7 6B 0B 57 1A 70 1D FF 35 D3 61 D9 F9 B3 |       |
| Issuer Public Key Exponent                                                                                 | 9F 32 | 0x 01  | 03                                                                                                                                                                                                                                                                                                                                                                | 02 02 |
| Issuer Private Key Exponent<br>(This is provided for information only; it is not personalized on the card) |       |        | 7E 7C 92 5F 2D F8 5B D5 E4 09 62 EC C8 89 C6 D6<br>85 EA 8B 74 8A 24 36 92 33 02 FE 7C 5D A3 93 9E<br>CA 2E C8 A9 11 40 F8 EB F4 49 9C 84 0A 20 E5 8F<br>B5 D9 FB 94 AE E9 A8 C5 A1 FD A6 B6 8A 4F 7C DF<br>CE 54 7A 5F 99 E4 6E 98 9A 5F 2F DF 9C 63 96 56<br>C0 95 A2 47 2F 6F B5 81 F7 67 02 E5 D3 40 45 18<br>24 2B 58 E0 36 6E 5E 90 D8 78 18 89 B8 9E A0 3B |       |
| Issuer Public Key Remainder                                                                                | 92    | 0x 14  | D8 0D 54 FB EC B3 E7 6B 0B 57 1A 70 1D FF 35 D3<br>61 D9 F9 B3                                                                                                                                                                                                                                                                                                    | 02 02 |
| Certificate Authority Public Key Index                                                                     | 8F    | 0x 01  | 99                                                                                                                                                                                                                                                                                                                                                                | 02 02 |
| Certificate Expiration Date<br>(for information only)                                                      | N/A   |        | 12 30<br>December 2030                                                                                                                                                                                                                                                                                                                                            |       |
| Signed Static Application Data                                                                             | 93    | 0x 70  | 94 94 BE 05 33 CC B6 65 7A 82 BB 86 21 7D EC B9 7A<br>4B AA 92 19 5C 64 88 62 40 F1 5F F6 99 36 0D 4F FD A3<br>2B 6D 3C 02 85 BB EC F8 45 E2 DD BA 3B E1 D6 95 BB<br>C5 A8 42 FD 8B 5F 50 7F 5C A2 65 46 92 A5 18 17 C8 77<br>B2 19 92 11 C8 97 0E 87 7F F8 0A 05 B4 30 10 0A 3C 31<br>7D BC 6A 67 3F C1 E0 6D CF ED 91 5A 21 BB D3 E4 97<br>F0 98 52 D2 94 BB 17 | 02 03 |
|                                                                                                            |       |        |                                                                                                                                                                                                                                                                                                                                                                   |       |
| UDK A (for ARQC, MAC & ENC)                                                                                |       |        | 2C A7 1C 7B F7 77 9F A0                                                                                                                                                                                                                                                                                                                                           | 80 00 |
| UDK B (for ARQC, MAC & ENC)                                                                                |       |        | 2E 30 13 3B CF 1B 9B 03                                                                                                                                                                                                                                                                                                                                           | 80 00 |

## 6.26. Test Card 25 (Previously Test Card 43)

This section outlines the profile for Test Card 25 - Card is personalized without a PAN Sequence Number.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.26.1. Magnetic Stripe – Track Data

Magnetic Track 1:  
 B4761739001010432^VISA ACQUIRER TEST CARD  
 25^15122011631100696000000  
 Magnetic Stripe Track 2:  
 4761739001010432=15122011631169689

### 6.26.2. Chip Data:

| Data Element                                      | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | DGI   |
|---------------------------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 20<br>43 41 52 44 20 32 35<br><br>(VISA ACQUIRER TEST CARD 25)                                                                                                                                                                                                                                                                                                                                                    | 01 01 |
| Track 2 Equivalent Data                           | 57    | 0x 11  | 47 61 73 90 01 01 04 32 D1 51 22 01 16 31 14 20 89                                                                                                                                                                                                                                                                                                                                                                                                                      | 01 01 |
| Track 1 Discretionary Data                        | 9F 1F | 0x 10  | 31 36 33 31 31 30 30 34 32 30 30 30 30 30 30                                                                                                                                                                                                                                                                                                                                                                                                                            | 01 01 |
|                                                   |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| Application Primary Account Number (PAN) (Signed) | 5A    | 0x 08  | 47 61 73 90 01 01 04 32                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 03 01 |
| Application PAN Sequence Number (Signed)          | 5F 34 | 0x 01  | Not personalized                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 03 01 |
|                                                   |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| Signed Static Application Data                    | 93    | 0x 90  | 07 A6 C0 42 CA 44 C6 AD 59 10 FF E5 1D 49 9A 8F C9<br>B7 4F 92 C0 C2 BF 5C 76 1A EF 95 5C 1F AD C7 93 31<br>83 E6 FF 32 5F F8 99 23 CC 1D 0D FD 50 A9 5A 5A EB<br>A3 45 18 1B DC 2E 12 E2 15 B7 33 B6 40 BC 12 FE 85<br>0F 2F E6 C4 7F C0 9D E7 C8 19 7B FE C2 DC 77 EA 86<br>DE F4 E9 3C FD 9A 7A B0 74 73 72 E5 40 F3 9F CD 03<br>EC 95 73 5B FC 40 15 CD 7A BD 59 01 C7 82 4F 1A D1<br>82 15 CD B9 64 9A 06 B1 E9 AB CC AD 34 B5 7E 60 D2<br>A9 DB 16 67 06 31 04 CC | 02 03 |
|                                                   |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| UDK A (for ARQC, MAC & ENC)                       | N/A   |        | 08 7E B5 67 36 53 58 73                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 80 00 |
| UDK B (for ARQC, MAC & ENC)                       | N/A   |        | 97 8B 7F F8 30 F6 B8 E2                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 80 00 |

## 6.27. Test Card 26 (Previously Test Card 44)

This section outlines the profile for Test Card 26 - Card is personalized with a PAN Sequence Number of "11".

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.27.1. Magnetic Stripe – Track Data

Magnetic Stripe Track 1:  
B4761739001010440^VISA ACQUIRER TEST CARD  
26^15122011966100231000000

Magnetic Stripe Track 2:  
4761739001010440=15122011966123189

### 6.27.2. Chip Data

| Data Element                                      | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | DGI   |
|---------------------------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54<br>20 43 41 52 44 20 32 36<br><br>(VISA ACQUIRER TEST CARD 26)                                                                                                                                                                                                                                                                                                                                                    | 01 01 |
| Track 2 Equivalent Data                           | 57    | 0x 11  | 47 61 73 90 01 01 04 40 D1 51 22 01 19 66 18 94 89                                                                                                                                                                                                                                                                                                                                                                                                                      | 01 01 |
| Track 1 Discretionary Data                        | 9F 1F | 0x 10  | 31 39 36 36 31 30 30 38 39 34 30 30 30 30 30 30                                                                                                                                                                                                                                                                                                                                                                                                                         | 01 01 |
|                                                   |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| Application Primary Account Number (PAN) (Signed) | 5A    | 0x 08  | 47 61 73 90 01 01 04 40                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 03 01 |
| Application PAN Sequence Number (Signed)          | 5F 34 | 0x 01  | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 03 01 |
|                                                   |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| Signed Static Application Data                    | 93    | 0x 90  | 72 29 AD F1 16 4C 43 95 D2 11 53 DE 1A 7E 32 E0 E6<br>F9 5D 2A 51 5F C0 D2 36 2F 74 4D 25 8F E6 2A 17 1F<br>B4 1F 45 A6 EE 9E F3 80 CC 88 D2 97 EC 71 4A 30<br>CD 56 59 C6 C6 59 95 62 C1 11 4F 96 EB 17 43 D9 B6<br>05 21 0C 6C 68 9D A2 04 0C E0 FF 09 26 01 62 7F F4<br>A8 B0 92 45 D1 EE 5A C8 71 37 74 E6 62 4C CB 53 1F<br>7A 92 EB 8C 4C 94 5D D2 8F 4B A6 E0 07 74 29 4F 6E<br>7D 8C 42 BF A7 11 4B 25 58 57 13 5E 8F 92 F8 85 EC<br>A2 96 46 12 06 D4 AD 77 EC | 02 03 |
|                                                   |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| UDK A (for ARQC, MAC & ENC)                       | N/A   |        | E8 20 CC BC EC CD 1E E8                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 80 00 |
| UDK B (for ARQC, MAC & ENC)                       | N/A   |        | 7D 7E 66 41 34 71 E8 15                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 80 00 |

## 6.28. Test Card 27 (Previously Test Card 45)

This section outlines the profile for Test Card 27 - Card is personalized with an IPK Certificate based on an 1144-bit Issuer Public Key.

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.28.1. Chip Data

| Data Element                                                                                                   | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | DGI   |
|----------------------------------------------------------------------------------------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                                                                                                | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54<br>20 43 41 52 44 20 32 37<br><br>(VISA ACQUIRER TEST CARD 27)                                                                                                                                                                                                                                                                                                                                                    | 01 01 |
| Issuer Public Key Certificate                                                                                  | 90    | 0x 90  | 12 36 DE 4D 2C 9D 1A 4D AB 5B 44 0F 37 EC A5 1A<br>68 F3 87 F0 D0 2D 44 EC 77 A9 01 B1 67 75 1E EA 5E<br>0C CA 31 71 00 9F 79 36 F1 5D 61 67 83 82 CC 5A 79<br>DF 75 4B 8C 2E D1 A3 18 3C ED 26 E9 0D C5 A8 DC<br>5C E4 8B D5 AB D4 1F 8B F3 2A 5D 1C 7A D9 A1 86<br>AF 79 42 87 29 39 F6 69 5A 64 2E 04 E6 67<br>31 03 7F 93 94 CC 31 A1 4A 9F F5 E2 56 DF 8E 50 2A<br>17 7A 75 4D 25 6A B1 E9 2A 28 BE CA 05 21 8A DD A6<br>09 A9 D2 14 77 30 5A B0 E6 7F 61 EC E5 FF | 02 01 |
| Issuer Public Key Modulus<br><br>(This is provided for information only; it is not personalized on the card)   | N/A   | 0x 8F  | E7 C5 0B 2B 8C 90 F9 24 97 5B 3C 9F 73 E4 3A 84<br>54 16 8F A0 1D 08 84 1D AF 37 70 3C 11 56 33 12<br>E0 33 80 04 9B F5 63 7E 23 DA 0B D7 92 0C 9E 27<br>51 0D 80 2E 5A 0C 4D 93 DE B7 74 C9 3A AC A9 0F<br>3A 59 25 89 FB 88 DD 6C EA B1 90 BE C5 A7 06 22<br>3A 69 8D C1 DC 89 89 EE CA BA 2E 8A 58 1F 9E E5<br>7B BE E7 30 74 FA BD EF 75 8B E8 0A 98 F6 B7 0B<br>67 99 D2 E8 81 A5 70 25 F3 25 68 CD F7 46 3E 44<br>7C 60 28 08 1F 71 A5 F4 76 D7 5E 3C D1 FA E3    |       |
| Issuer Private Key Exponent<br><br>(This is provided for information only; it is not personalized on the card) | N/A   | 0x 8F  | 9A 83 5C C7 B3 0B 50 C3 0F 92 28 6A 4D 42 D1 AD<br>8D 64 5F C0 13 5B 02 BE 74 CF A0 28 0B 8E CC B7<br>40 22 55 58 67 F8 EC FE C2 91 5D 3A 61 5D BE C4<br>E0 B3 AA C9 91 5D 89 0D 3F 24 F8 86 27 1D C6 0A<br>26 E6 19 06 A7 B0 93 89 A1 B4 FB 54 85 10 20 F8<br>C3 E9 9C 8F 14 BF 18 A9 3C A4 67 30 C0 7F 99 56<br>50 B3 EF 54 42 41 CE 20 3F 54 9B 48 10 17 05 6C<br>E7 67 5B F3 C4 E2 43 51 66 AB 49 28 BE 6D FC 97<br>7E EF 36 62 74 62 8F 86 BA 33 F7 FC C8 9B BB    |       |
| Issuer Public Exponent                                                                                         | 9F 32 | 0x 01  | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 02 02 |
| Issuer Prime 1 (p).                                                                                            | N/A   | 0x 48  | 0F CD E0 1B B2 5D 18 64 1C 5C 24 EE A2 76 B2 43<br>03 C4 75 F8 B5 93 8D 34 29 20 BB FA 79 44 88 98<br>95 8D 48 DF 4A A8 04 30 F8 8C 97 A5 DD DE 13 BF<br>37 88 76 F2 89 26 49 AE 47 A6 87 57 01 9E CC 65<br>CC ED 87 3A 2D F1 72 45                                                                                                                                                                                                                                     |       |
| Issuer Prime 2 (p).                                                                                            | N/A   | 0x 48  | 0E AA 41 FC 0D A0 F6 70 90 B8 66 34 48 C6 B8 A1<br>ED 2B 4D 9B 0B A3 D3 04 BA E1 F5 05 B8 CD 0F 70                                                                                                                                                                                                                                                                                                                                                                      |       |

|                                |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
|--------------------------------|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|                                |     |       | 29 89 44 1F D3 D8 CF FD EF 7F E7 23 1C FC 3E 4B<br>F4 50 AD 88 87 B3 57 95 19 97 71 FF 72 D2 11 68<br>DD 72 02 30 13 B3 9F 07                                                                                                                                                                                                                                                                                                                                        |       |
| Issuer Exponent 1 (d mod p-1)  | N/A | 0x 48 | 0A 89 40 12 76 E8 BA ED 68 3D 6D F4 6C 4F 21 82<br>02 82 F9 50 79 0D 08 CD 70 C0 7D 51 A6 2D B0 65<br>B9 08 DB 3F 87 1A AD 75 FB 08 65 19 3E 94 0D 2A<br>25 05 A4 A1 B0 C4 31 1E DA 6F 04 E4 AB BF 32 EE<br>88 9E 5A 26 C9 4B A1 83                                                                                                                                                                                                                                  |       |
| Issuer Exponent 2 (d mod q-1). | N/A | 0x 48 | 09 C6 D6 A8 09 15 F9 A0 60 7A EE CD 85 D9 D0 6B<br>F3 72 33 BC B2 6D 37 58 7C 96 A3 59 25 DE 0A 4A<br>C6 5B 82 BF E2 90 8A A9 4A 55 44 C2 13 52 D4 32<br>A2 E0 73 B0 5A 77 8F B8 BB BA 4B FF A1 E1 60 F0<br>93 A1 56 CA B7 CD 14 AF                                                                                                                                                                                                                                  |       |
| Issuer Coefficient             | N/A | 0x 48 | 09 A5 1E 9E E6 63 E0 A0 00 CA 07 8F 82 2F D2 2D<br>F3 B2 63 D9 1C DF D5 EF C6 B4 ED 8E E5 57 C0 4E<br>0A 94 1E 19 5B 2A 00 A7 36 23 40 3D 95 30 0C 0B<br>3F EA 7C BB D1 0D 17 F8 BE 98 A6 2E 21 61 8B E0<br>2F 9F DD E6 22 8A F5 23                                                                                                                                                                                                                                  |       |
| Issuer PK Remainder            | 92  | 0x 23 | 98 F6 B7 0B 67 99 D2 E8 81 A5 70 25 F3 25 68 CD F7<br>46 3E 44 7C 60 28 08 1F 71 A5 F4 76 D7 5E 3C D1 FA<br>E3                                                                                                                                                                                                                                                                                                                                                       | 02 02 |
| Signed Static Application Data | 93  | 0x 8F | 15 73 2D BE 4D 0C 9B E9 17 38 47 20 EF AC 24 45 4A<br>28 95 7E 8C AA 6B DD 0C E0 DD AC 49 4F C2 BE 3E<br>BE AD 29 54 1F 5C F3 C9 F0 73 C7 99 11 5A 9D FA 65<br>E4 88 86 05 84 62 A4 A0 B0 BD FD B8 99 95 DA A6 43<br>E7 C4 01 53 07 53 DC 37 62 4C 35 C5 F8 90 08 41 56<br>C3 D7 0D 58 E4 10 E3 A3 D5 4C EE 25 C4 66 B7 EE<br>7C 4D 5B 63 FC 13 95 1E E5 A7 FA 80 A1 00 3E 18 3B<br>88 52 43 37 81 2D 12 3D EA 82 6A 48 D6 DA 72 CD 71<br>EA 74 D9 C4 77 70 3D 93 1C | 02 03 |

## 6.29. Test Card 28 (Previously Test Card 46)

This section outlines the profile for Test Card 28 - Card is personalized with an Issuer URL and Issuer Discretionary Data. It also contains an Application Expiration Date = December 31, 2025.

VSDC Applet Version: 2.5.1

Changes to make from baseline card:

### 6.29.1. Magnetic Stripe – Track Data

Track 1:  
B4761739001010465^VISA ACQUIRER TEST CARD  
28^25122011172700509000000

Track 2:  
4761739001010465=25122011172750989

### 6.29.2. Chip PSE Data

Card must be personalized with a PSE and both the PSE/FCI and ADF/FCI contain Language Preference, Issuer Code Table Index and FCI Issuer Discretionary Data.

Changes to make from baseline card:

- o Include PSE. For this test card, PSE must include all mandatory data elements as well as the optional data elements of Language Preference (value = en), Issuer Code Table Index (value = 01) and FCI Issuer Discretionary Data.
- o Include Language Preference, Issuer Code Table Index, and FCI Issuer Discretionary Data.

| Data Element                  | Tag   | Length | Value                                                                                                                                                                 | DGI   |
|-------------------------------|-------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Language Preference           | 5F 2D | 0x 02  | 65 6E<br>(en)                                                                                                                                                         | 91 02 |
| Issuer Code Table Index       | 9F 11 | 0x 01  | 01                                                                                                                                                                    | 91 02 |
| FCI Issuer Discretionary Data | BF 0C | 0x 28  | 5F 50 25 68 74 74 70 3A 2F 2F 77 77 77 2E 41 42 43 42<br>41 4E 4B 2E 63 6F 6D 2F 41 30 30 30 30 30 30 30 33<br>31 30 31 30<br>"http://www.ABCBANK.com/A0000000031010" | 91 02 |

### 6.29.3. VSDC Application Data:

| Data Element            | Tag   | Length | Value                                                                                                                | DGI   |
|-------------------------|-------|--------|----------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name         | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 20<br>43 41 52 44 20 32 38<br><br>(VISA ACQUIRER TEST CARD 28) | 01 01 |
| Track 2 Equivalent Data | 57    | 0x 11  | 47 61 73 90 01 01 04 65 D2 51 22 01 11 72 74 56 89                                                                   | 01 01 |

| Data Element                                      | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | DGI   |
|---------------------------------------------------|-------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Track 1 Discretionary Data                        | 9F 1F | 0x 10  | 31 31 37 32 37 30 30 34 35 36 30 30 30 30 30 30                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 01 01 |
| FCI Issuer Discretionary Data                     | BF 0C | 0x 28  | 5F 50 25 68 74 74 70 3A 2F 2F 77 77 77 2E 41 42 43 42 41 4E 4B 2E 63 6F 6D 2F 41 30 30 30 30 30 30 30 33 31 30 31 30<br>"http://www.ABCBANK.com/A0000000031010"                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 91 02 |
| Application Primary Account Number (PAN) (Signed) | 5A    | 0x 08  | 47 61 73 90 01 01 04 65                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 03 01 |
| Application Expiration Date                       | 5F 24 | 0x 03  | 25 12 31                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 03 02 |
| Cardholder Verification Method List (CVM)         | 8E    | 0x 0E  | 0000 0000 0000 0000 0103 0203 1F00 <ul style="list-style-type: none"> <li>CVM Code 1 '0103' <ul style="list-style-type: none"> <li>Offline (Plaintext) PIN, if terminal supports CVM</li> <li>Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>CVM Code 2 '0203' <ul style="list-style-type: none"> <li>Online PIN, if terminal supports CVM</li> <li>Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> <li>CVM Code 3 '1F00' <ul style="list-style-type: none"> <li>No CVM Required, Always</li> <li>Fail cardholder verification if this CVM is unsuccessful</li> </ul> </li> </ul> | 03 02 |
| Issuer Action Code – Denial                       | 9F 0E | 0x 05  | 00 10 18 00 00 <ul style="list-style-type: none"> <li>Requested service not allowed for card product</li> <li>PIN entry required and PIN pad not present or not working</li> <li>PIN entry required, PIN pad present but PIN was not entered</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                    | 03 02 |
| Signed Static Application Data                    | 93    | 0x 70  | 8C 2B CD F9 28 5D C9 09 A7 51 5B 3D 02 B7 1A 16 C1 5C 60 82 B9 49 43 F4 B5 56 7B 30 87 85 51 CE 69 8F 14 43 3B 7A 1A 29 D1 EB 65 62 E4 9A 62 39 B6 32 6E E4 BC C9 B2 6D 0A 84 48 79 DF 04 6E AE 71 4F 57 37 35 F0 3F 1D 06 9E 63 19 5D B0 3B 1F D1 B9 56 A2 32 15 3A FB 3B 5B F3 4F 26 DA E1 30 E2 33 60 4D 3C 36 5D 14 79 2E 60 CF 06 EC 98 F7 4D 60 38 2C 1B F3 A2 AD ED 5A 0F 82 E5 74 F0 AD 3D E3 31 EA 26 30 4B EC 97 CF 21 D3 0F 02 29 C4                                                                                                                                                                                            | 02 03 |
| Application Currency Code                         | 9F 51 | 0x 02  | 08 40                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0E 01 |
| Application Default Action                        | 9F 52 | 0x 04  | 00 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0E 01 |
| Geographic Indicator                              | 9F 55 | 0x 01  | C0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0E 01 |
| Issuer Authentication                             | 9F 56 | 0x 01  | 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0E 01 |

| Data Element                              | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | DGI             |
|-------------------------------------------|-------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Indicator                                 |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |
| Issuer Country Code                       | 9F 57 | 0x 02  | 08 40                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0E 01           |
| Cumulative Total Transaction Amount Limit | 9F 54 | 0x 06  | 00 00 00 00 10 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0D 01           |
| Issuer Application Data                   | 9F 10 | 0x 09  | 06 01 0A 03 00 00 00 0F 03<br><br>This field needs to be personalized as follows: <ul style="list-style-type: none"> <li>• Visa Discretionary Data (as per TADR) = 06 01 0A 03 00 00 00</li> <li>• Issuer Discretionary Data containing:               <ul style="list-style-type: none"> <li>• 0x 0F (length of Issuer Discretionary Data to be returned)</li> <li>• 0x 03 (a code for the information to be returned by the card in the Issuer Discretionary Data field: VLP Available Funds &amp; Cumulative Total Transaction Amount)</li> </ul> </li> </ul> Note: This feature is only supported on VSDC Applet versions 2.4.1 and above. | 07 01           |
| VLP Available Funds                       | 9F 79 | 0x 06  | 00 00 00 00 00 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0B 01           |
| VLP Issuer Authorization Code             | 9F 74 | 0x 06  | 56 4C 50 31 31 31                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0B 01           |
| VLP Funds Limit                           | 9F 77 | 0x 06  | 00 00 00 00 05 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0D 01           |
| Application Interchange Profile (for VLP) | 82    | 0x 02  | 08 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 07 03           |
| VLP Application File Locator (for VLP)    | 94    | 0x 04  | 58 01 01 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 07 03           |
|                                           |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |
| Reference PIN                             |       | 0x 08  | 24 12 34 FF FF FF FF FF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 80 10           |
| PIN Try Limit                             |       | 0x 01  | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 80 10/<br>90 10 |
| PIN Try Counter                           |       | 0x 01  | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 90 10           |
|                                           |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |
| UDK A (for ARQC, MAC & ENC)               |       |        | 2C 37 CC EE 9B 4B EB 74                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 80 00           |
| UDK B (for ARQC, MAC & ENC)               |       |        | B8 84 2C 31 8B 31 13 7D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 80 00           |

6.30. Test Card 29 (Previously Test Card 47)

This section outlines the profile for Test Card 29 – This is a Blocked Card.  
VSDC Applet Version: 2.7.1

Changes to make from baseline card:

IMPORTANT: The vendor developing the test card must block the card after personalizing it. This can be accomplished by sending the EMV CARD BLOCK Issuer Script command to the card.

| Data Element    | Tag   | Length | Value                                                                                                             | DGI   |
|-----------------|-------|--------|-------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45 53 54 20 43 41 52 44 20 32 39<br><br>(VISA ACQUIRER TEST CARD 29) | 01 01 |

## 6.31. Test Card 30 (Previously Test Card 48)

This section outlines the profile for Test Card 30 - A card containing an Issuer Public Key Certificate signed by the Visa CA Test Key of 1408 bits).

VSDC Applet Version: 2.7.1

Changes to make from baseline card:

### 6.31.1. Chip Data

| Data Element                                                                                                                                                                                                                                                                                                                                                      | Tag   | Length | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DGI   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Cardholder Name                                                                                                                                                                                                                                                                                                                                                   | 5F 20 | 0x 1A  | 56 49 53 41 20 41 43 51 55 49 52 45 52 20 54 45<br>53 54 20 43 41 52 44 20 33 30<br><br>(VISA ACQUIRER TEST CARD 30)<br><br>Note: This value changes with each test card.                                                                                                                                                                                                                                                                                                                                                                                                     | 01 01 |
| Static Data Authentication                                                                                                                                                                                                                                                                                                                                        |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |
| Note To Vendors Regarding SDA-Related Data: The SDA-related data elements outlined in this section do not have to be used on the card. These data elements are provided as sample data. If the vendor wants to generate their own data, they may do so as long as the SDA data is valid test data. The sample data was created using the Modulus Exponent method. |       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |
| Issuer Public Key Certificate<br><br>(for CA index 92 )                                                                                                                                                                                                                                                                                                           | 90    | 0x B0  | 62 BC 5F 60 D2 7E 2C 61 16 03 C9 99 E0 0E A0 B3<br>9F 01 DC B4 01 B0 AD A8 F5 45 4E 5B 43 3B 2F E2<br>D6 B7 B0 55 F0 3D 27 14 EB 21 4D 5B 11 EB 4D 11<br>72 2D 69 04 A4 22 46 7F 35 08 78 42 BA 67 E6 0A<br>FF 09 39 4E 4A 2C AC B2 F7 C3 F2 1B 20 92 D1 93<br>28 E7 5C 49 9F E0 5B D1 63 F9 40 EC 8C A8 AB A3<br>81 45 E1 CD 58 8A F2 84 27 6F 76 F4 0C 38 46 28<br>66 3D 4D AB B8 F5 47 EE 03 99 CD F8 F9 10 9F 05<br>C6 44 7D 3C 3B D1 01 67 3C D5 15 42 B9 AC 8E 23<br>89 8D A2 FE 6C F2 89 B9 3C 24 C4 DB D0 E7 AE D9<br>A1 63 5A AA 03 B5 C8 2C 9B 01 7B 91 80 73 03 66 | 02 01 |
| Issuer Public Key Remainder                                                                                                                                                                                                                                                                                                                                       | 92    |        | (remove this tag)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| Signed Static Application Data                                                                                                                                                                                                                                                                                                                                    | 93    | 0x 70  | AE 4C F9 D4 9C D3 86 31 67 B9 A2 47 90 37 2F E1<br>F2 D7 1D C8 C0 68 ED 19 36 49 07 CE 09 E1 66 BB<br>07 77 BE D0 0D B7 D2 C2 E4 00 00 79 DA 50 27 9E<br>1F 6B CF 1D 70 3D 24 90 C0 63 57 B1 F7 51 22 ED<br>CB 9E 96 57 E4 DF 9E CC 00 F8 DF C6 5D A7 91 AC<br>AD 20 7F 11 50 95 67 6E 98 0C 4A 08 BE C8 A5 57<br>34 38 D0 52 74 E2 3A 1C 3B 0B 8E 05 62 72 5E A0                                                                                                                                                                                                             | 02 03 |
| Certification Authority Public Key Index                                                                                                                                                                                                                                                                                                                          | 8F    | 0x 01  | 92<br><br>(Visa CA Test Key of 1408 bits)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 02 02 |
| Certificate Expiration Date<br><br>(for information only)                                                                                                                                                                                                                                                                                                         |       |        | December 2030                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |

# Appendix A: Visa CA Test Public Keys for VSDC

These test keys need to be loaded into the terminal to support the tests associated with Static and Dynamic Data Authentication.

NOTE: Prior to deployment, these keys must be removed from the terminal and replaced with the Visa CA production keys.

## A.1: 1152 Bit VSDC TEST Key

This key is the Visa CA Public 1152 bit TEST key:

| Component                                        | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Registered Application Provider Identifier (RID) | A0 00 00 00 03                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Index                                            | 95                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Modulus                                          | BE 9E 1F A5 E9 A8 03 85 29 99 C4 AB 43 2D B2 86<br>00 DC D9 DA B7 6D FA AA 47 35 5A 0F E3 7B 15 08<br>AC 6B F3 88 60 D3 C6 C2 E5 B1 2A 3C AA F2 A7 00<br>5A 72 41 EB AA 77 71 11 2C 74 CF 9A 06 34 65 2F<br>BC A0 E5 98 0C 54 A6 47 61 EA 10 1A 11 4E 0F 0B<br>55 72 AD D5 7D 01 0B 7C 9C 88 7E 10 4C A4 EE 12<br>72 DA 66 D9 97 B9 A9 0B 5A 6D 62 4A B6 C5 7E 73<br>C8 F9 19 00 0E B5 F6 84 89 8E F8 C3 DB EF B3 30<br>C6 26 60 BE D8 8E A7 8E 90 9A FF 05 F6 DA 62 7B |
| Exponent                                         | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Secure Hash Algorithm-1 Hash                     | EE 15 11 CE C7 10 20 A9 B9 04 43 B3 7B 1D 5F 6E<br>70 30 30 F6                                                                                                                                                                                                                                                                                                                                                                                                          |
| Comments:                                        | The production version of Visa's 1152-bit CA public key is currently set to expire on December 31, 2015.                                                                                                                                                                                                                                                                                                                                                                |

## A.2: 1408 Bit VSDC TEST Key

This key is the Visa CA Public 1408 bit TEST key:

| Component                                        | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Registered Application Provider Identifier (RID) | A0 00 00 00 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Index                                            | 92                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Modulus                                          | 99 6A F5 6F 56 91 87 D0 92 93 C1 48 10 45 0E D8<br>EE 33 57 39 7B 18 A2 45 8E FA A9 2D A3 B6 DF 65<br>14 EC 06 01 95 31 8F D4 3B E9 B8 F0 CC 66 9E 3F<br>84 40 57 CB DD F8 BD A1 91 BB 64 47 3B C8 DC 9A<br>73 0D B8 F6 B4 ED E3 92 41 86 FF D9 B8 C7 73 57<br>89 C2 3A 36 BA 0B 8A F6 53 72 EB 57 EA 5D 89 E7<br>D1 4E 9C 7B 6B 55 74 60 F1 08 85 DA 16 AC 92 3F<br>15 AF 37 58 F0 F0 3E BD 3C 5C 2C 94 9C BA 30 6D<br>B4 4E 6A 2C 07 6C 5F 67 E2 81 D7 EF 56 78 5D C4<br>D7 59 45 E4 91 F0 19 18 80 0A 9E 2D C6 6F 60 08<br>05 66 CE 0D AF 8D 17 EA D4 6A D8 E3 0A 24 7C 9F |
| Exponent                                         | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Secure Hash Algorithm-1 Hash                     | 42 9C 95 4A 38 59 CE F9 12 95 F6 63 C9 63 E5 82<br>ED 6E B2 53                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Comments:                                        | The maximum expiration date for certificates issued using Visa's 1408-bit CA public key is December 31, 2016. Considered to have an anticipated lifetime to at least December 31, 2018.                                                                                                                                                                                                                                                                                                                                                                                       |

## A.3: 1984 Bit VSDC TEST Key

This key is the Visa CA Public 1984 bit TEST key, exponent 3:

| Component                                        | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Registered Application Provider Identifier (RID) | A0 00 00 00 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Index                                            | 94                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Modulus                                          | AC D2 B1 23 02 EE 64 4F 3F 83 5A BD 1F C7 A6 F6<br>2C CE 48 FF EC 62 2A A8 EF 06 2B EF 6F B8 BA 8B<br>C6 8B BF 6° B5 87 0E ED 57 9B C3 97 3E 12 13 03<br>D3 48 41 A7 96 D6 DC BC 41 DB F9 E5 2C 46 09 79<br>5C 0C CF 7E E8 6F A1 D5 CB 04 10 71 ED 2C 51 D2<br>20 2F 63 F1 15 6C 58 A9 2D 38 BC 60 BD F4 24 E1<br>77 6E 2B C9 64 80 78 A0 3B 36 FB 55 43 75 FC 53<br>D5 7C 73 F5 16 0E A5 9F 3° FC 53 98 EC 7B 67 75<br>8D 65 C9 BF F7 82 8B 6B 82 D4 BE 12 4A 41 6A B7<br>30 19 14 31 1E A4 62 C1 9F 77 1F 31 B3 B5 73 36<br>00 0D FF 73 2D 3B 83 DE 07 05 2D 73 03 54 D2 97<br>BE C7 28 71 DC CF 0E 19 3F 17 1A BA 27 EE 46 4C<br>6° 97 69 09 43 D5 9B DA BB 2° 27 EB 71 CE EB DA<br>FA 11 76 04 64 78 FD 62 FE C4 52 D5 CA 39 32 96<br>53 0A A3 F4 19 27 AD FE 43 4A 2D F2 AE 30 54 F8<br>84 06 57 A2 6E 0F C6 17 |
| Exponent                                         | 03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Secure Hash Algorithm-1 Hash                     | C4 A3 C4 3C CF 87 32 7D 13 6B 80 41 60 E4 7D 43<br>B6 0E 6E 0F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Comments:                                        | This key length is currently considered to have an anticipated lifetime to at least December 31, 2018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

# Appendix B: Terminal Action Code (TAC) Settings

This chapter provides the Terminal Action Code settings for terminals.  
Please refer to Visa 1.4.0, Section 10.2: Terminal Data for additional details.

## B.1: Terminal Action Code (TAC) settings for Terminals

This section provides the Terminal Action Code settings for terminals.

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TAC—Denial  | 0010000000<br>The TAC value causes a decline for the following conditions: <ul style="list-style-type: none"><li>• Service not allowed for card product</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                            |
| TAC—Online  | DC4004F800<br>This TAC value generates an online authorization when: <ul style="list-style-type: none"><li>• Offline data authentication is not performed or failed</li><li>• The PAN is on the terminal exception file</li><li>• The application is expired</li><li>• An Online PIN is entered</li><li>• The transaction exceeds the floor limit</li><li>• The upper (9F23) or lower consecutive offline limit (9F14) is exceeded</li><li>• The transaction is randomly selected for online processing</li><li>• The terminal forced the transaction online</li><li>• CDA failure</li></ul> |
| TAC—Default | DC4000A800<br>This TAC value generates a decline if the transaction cannot be sent online for authorization when: <ul style="list-style-type: none"><li>• Offline data authentication is not performed or failed</li><li>• The PAN is on the terminal exception file</li><li>• The application is expired</li></ul>                                                                                                                                                                                                                                                                          |

|  |                                                                                                                                                                                                                                       |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"><li>• The transaction exceeds the floor limit</li><li>• The Upper Consecutive Offline Limit (9F23) is exceeded</li><li>• The merchant forced the transaction online</li><li>• CDA failure</li></ul> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

NOTE: Markets not supporting offline data authentication in cards may remove the TAC—Online and TAC—Default settings for offline data authentication not performed resulting in a TAC—Online value of 584004F800 and a TAC—Default 584000A800.

## Appendix C: VSDC Stand-in Processing Conditions

This section provides the VSDC Stand-in Processing Conditions. When the Acquirer is connected to VCMS/VMTS and the transaction associated with one of the ADV Toolkit cards is sent online, VCMS/VMTS will use these conditions to make the online authorization decision.

NOTE: The Route to Issuer Defaults are not used as the transaction is processed in Stand-in. Only the Stand-in Authorization Response Defaults are used.

This information is valuable in determining the reason that VCMS/VMTS either approved or declined the online-initiated transaction.

For example, the VSDC Stand-in Authorization Response Default for expired application is “decline offline.” If the application is expired and the transaction is sent online to VCMS/VMTS, VCMS/VMTS will decline the transaction. VCMS/VMTS will indicate the decline in the Response Code (field 39) in the response message.

### VSDC Stand-In Processing Conditions

|   | Stand-In Condition                                          | Source | Route-to-Issuer Default | Stand-in Authorization Response Default |
|---|-------------------------------------------------------------|--------|-------------------------|-----------------------------------------|
| 1 | Transaction exceeds floor limit                             | TVR    | No                      | Approve                                 |
| 2 | Transaction selected randomly for online processing         | TVR    | No                      | Approve                                 |
| 3 | Cardholder verification failed                              | TVR    | Yes                     | Decline                                 |
| 4 | Unrecognized cardholder verification method                 | TVR    | Yes                     | Approve                                 |
| 5 | Offline PIN verification failed                             | CVR    | Yes                     | Decline                                 |
| 6 | PIN entry required and PIN pad not present or not working   | TVR    | Yes                     | Decline                                 |
| 7 | PIN entry required, PIN pad is present, but PIN not entered | TVR    | Yes                     | Decline                                 |

|    | Stand-In Condition                                                                    | Source     | Route-to-Issuer Default            | Stand-in Authorization Response Default |
|----|---------------------------------------------------------------------------------------|------------|------------------------------------|-----------------------------------------|
| 8  | Offline PIN try limit exceeded                                                        | CVR or TVR | Yes                                | Decline                                 |
| 9  | Exceeded total, domestic, or international counters                                   | CVR        | Yes                                | Approve                                 |
| 10 | Lower consecutive offline limit exceeded                                              | TVR        | Yes                                | Approve                                 |
| 11 | Upper consecutive offline limit exceeded                                              | TVR        | Yes                                | Approve                                 |
| 12 | Expired application                                                                   | TVR        | Yes                                | Decline                                 |
| 13 | Application not yet effective                                                         | TVR        | Yes                                | Decline                                 |
| 14 | Issuer Authentication failed on last transaction                                      | CVR        | Yes<br>Member <u>cannot</u> modify | Approve                                 |
| 15 | SDA failed                                                                            | TVR        | Yes<br>Member <u>cannot</u> modify | Decline                                 |
| 16 | Offline Data Authentication not performed<br>Note: Not applicable to ATM transactions | TVR        | Yes<br>Member <u>cannot</u> modify | Approve                                 |
| 17 | SDA failed on last transaction and was declined offline                               | CVR        | Yes<br>Member <u>cannot</u> modify | Approve                                 |
| 18 | Script update succeeded on last transaction                                           | CVR        | Yes<br>Member <u>cannot</u> modify | Approve<br>Member <u>cannot</u> modify  |
| 19 | Script update failed on last transaction                                              | CVR        | Yes<br>Member <u>cannot</u> modify | Approve                                 |
| 20 | Merchant forced transaction online                                                    | TVR        | Yes                                | Decline                                 |
| 21 | New card (first use)                                                                  | CVR        | Yes                                | Approve                                 |
| 22 | Magnetic stripe read of VSDC card at VSDC terminal                                    | *          | Yes<br>Member <u>cannot</u> modify | Approve                                 |
| 23 | Last online transaction not completed                                                 | CVR        | Yes                                | Approve                                 |
| 24 | Card Authentication failure and Card Authentication reliable                          | **         | Yes<br>Member <u>cannot</u> modify | Decline                                 |
| 25 | Card Authentication failure and Card Authentication unreliable                        | **         | Yes<br>Member <u>cannot</u> modify | Decline                                 |

|    | Stand-In Condition                                                   | Source | Route-to-Issuer Default            | Stand-in Authorization Response Default |
|----|----------------------------------------------------------------------|--------|------------------------------------|-----------------------------------------|
| 26 | Card Authentication not performed and Card Authentication unreliable | **     | Yes<br>Member <u>cannot</u> modify | Decline                                 |
| 27 | DDA failed                                                           | TVR    | Yes<br>Member <u>cannot</u> modify | Decline                                 |
| 28 | DDA failed on last transaction and was declined offline              | CVR    | Yes<br>Member <u>cannot</u> modify | Approve                                 |



## Appendix D: Compliance Report

After completing the tests outlined in Chapter 4: Test Cases, Acquirers must complete a Compliance Report and submit it to their Visa regional office. The one here is for guidance only and Acquirers must check with their Visa regional office for the procedures which apply for submission of reports. The Compliance Report collects information about the device as well as the results of the tests. Some regional offices may have online forms or other means of submitting results. The following form is for guidance only.

### D.1: Terminal Information

| PART I – Application Provider Identification                                               |  |
|--------------------------------------------------------------------------------------------|--|
|                                                                                            |  |
| Company Name:                                                                              |  |
| Contact Name:                                                                              |  |
| Address:                                                                                   |  |
| Acquirer BIN:                                                                              |  |
| Telephone:                                                                                 |  |
| Fax Number:                                                                                |  |
| Email Address:                                                                             |  |
| Version of the ADV Toolkit:<br>(this information is located on the user's guide and cards) |  |

## PART II – Payment Application and EMV Kernel Identification

|                                          |  |
|------------------------------------------|--|
| Terminal Name and Model Number:          |  |
| Payment Application Name and Version:    |  |
| IFM (Level 1) Approval Reference:        |  |
| EMV Kernel (Level 2) Approval Reference: |  |

## PART III – Terminal Resident Data Objects

|                             |  |
|-----------------------------|--|
| Terminal Type:              |  |
| Terminal Country Code:      |  |
| Application Version Number: |  |
| Terminal Currency Code:     |  |

## PART IV – EMV Specifications

|                                   |  |
|-----------------------------------|--|
| EMV Specification Date & Version: |  |
|-----------------------------------|--|

Note: When completing this section, please enter the Hex value equivalent of the terminal capabilities and additional terminal capabilities.

## Part V – Terminal Details

|   | Terminal Capabilities                                                    | (Yes / No)<br>or Value |
|---|--------------------------------------------------------------------------|------------------------|
|   | Card Data Input Capability                                               |                        |
| O | ▪ Manual Key Entry                                                       |                        |
| O | ▪ Magnetic Stripe                                                        |                        |
| M | ▪ IC with Contacts                                                       |                        |
|   | CVM Capability                                                           |                        |
| O | ▪ Plaintext PIN for ICC Verification                                     |                        |
| O | ▪ Enciphered PIN for online Verification                                 |                        |
| O | ▪ Signature (paper)                                                      |                        |
| O | ▪ Enciphered PIN for offline Verification                                |                        |
| M | ▪ No CVM Required                                                        |                        |
|   | Security Capability                                                      |                        |
| C | ▪ Static Data Authentication                                             |                        |
| O | ▪ Dynamic Data Authentication                                            |                        |
| O | ▪ Combined Dynamic Data Authentication/Application Cryptogram generation |                        |
| O | ▪ Card Capture                                                           |                        |
|   | ▪                                                                        |                        |

|   | Additional Terminal Capabilities | (Yes / No)<br>or Value |
|---|----------------------------------|------------------------|
|   | Transaction Type Capability      |                        |
| C | ▪ Cash                           |                        |
| C | ▪ Goods                          |                        |
| C | ▪ Services                       |                        |

|   |                                         |  |
|---|-----------------------------------------|--|
| C | ▪ Cash Back                             |  |
| C | ▪ Inquiry                               |  |
| C | ▪ Transfer                              |  |
| C | ▪ Payment                               |  |
| C | ▪ Administrative                        |  |
|   | Terminal Data Input Capability          |  |
| C | ▪ Numeric Keys                          |  |
| C | ▪ Alphabetic and Special Character Keys |  |
| C | ▪ Command Keys                          |  |
| C | ▪ Function Keys                         |  |
|   | Terminal Data Output Capability         |  |
| C | ▪ Print, Attendant                      |  |
| O | ▪ Print, Cardholder Name                |  |
| C | ▪ Display Attendant                     |  |
| C | ▪ Display Cardholder                    |  |
| C | ▪ Code Table 10                         |  |
| C | ▪ Code Table 9                          |  |
| C | ▪ Code Table 8                          |  |
| C | ▪ Code Table 7                          |  |
| C | ▪ Code Table 6                          |  |
| C | ▪ Code Table 5                          |  |
| C | ▪ Code Table 4                          |  |
| C | ▪ Code Table 3                          |  |
| C | ▪ Code Table 2                          |  |
| C | ▪ Code Table 1                          |  |

|   | Application Selection                                             | (Yes / No)<br>or Value |
|---|-------------------------------------------------------------------|------------------------|
| O | ▪ Support PSE selection Method                                    |                        |
| O | ▪ Support Cardholder Confirmation                                 |                        |
| O | ▪ Does Terminal have a preferred order of displaying applications |                        |
| O | ▪ Does terminal perform partial AID                               |                        |
| O | ▪ Does the terminal have multi language support                   |                        |

|  | Data Authentication | (Yes / No)<br>or Value |
|--|---------------------|------------------------|
|--|---------------------|------------------------|

|   |                                                                                                               |  |
|---|---------------------------------------------------------------------------------------------------------------|--|
| C | ▪ What is the maximum supported Certificate Authority Public Key Size                                         |  |
| C | ▪ What exponents does the terminal support                                                                    |  |
| O | ▪ During data authentication does the terminal check validity for revocation of Issuer Public Key Certificate |  |
| C | ▪ Does the terminal contain a DDOL                                                                            |  |

|   | Cardholder Verification Method                  | (Yes / No)<br>or Value |
|---|-------------------------------------------------|------------------------|
| O | ▪ Terminal support bypass PIN Entry             |                        |
| O | ▪ Terminal Support Get Data for PIN Try Counter |                        |

|   | Terminal Risk Management       | (Yes / No)<br>or Value |
|---|--------------------------------|------------------------|
| C | ▪ Floor Limit Checking         |                        |
| C | ▪ Random Transaction Selection |                        |
| C | ▪ Velocity Checking            |                        |
| O | ▪ Transaction Log              |                        |
| O | ▪ Exception File               |                        |

|   | Terminal Action Analysis | (Yes / No)<br>or Value |
|---|--------------------------|------------------------|
| O | ▪ Terminal Action Codes  |                        |

|   | Completion Processing                      | (Yes / No)<br>or Value |
|---|--------------------------------------------|------------------------|
| O | ▪ Transaction Forced Online Capability     |                        |
| O | ▪ Transaction Forced Acceptance Capability |                        |
| O | ▪ Does terminal Support Advices            |                        |
| C | ▪ Does Terminal support Referrals          |                        |

## D.2: ADVT Test Results

Please use the following table to record the test outcome:

- Pass—Place an X in the pass column if the terminal passes the test.
- Fail—Place an X in the fail column if the terminal fails the test.
- N/A—Place an X in the not applicable (N/A) column if the test does not apply. For example, some tests only apply to ATMs. If the terminal is a POS terminal, the test will not apply.
- Comments—Please provide additional information if the terminal fails the test (e.g., terminal error message, etc.).

NOTE: An asterisk (\*) precedes the test case if the test is for information gathering purposes only. The terminal is not considered out of compliance if it fails one of these tests.

| TEST CASE       | PASS | FAIL | N/A | COMMENTS                                                   |
|-----------------|------|------|-----|------------------------------------------------------------|
| Test Case 1a:   |      |      |     |                                                            |
| Test Case 1b:   |      |      |     |                                                            |
| Test Case 1c:   |      |      |     |                                                            |
| Test Case 1d:   |      |      |     |                                                            |
| Test Case 2:    |      |      | N/A | Moved from Test Case 21<br>For Information Only Test       |
| Test Case 3:    |      |      |     |                                                            |
| Test Case 4:    |      |      | N/A |                                                            |
| Test Case 5:    |      |      | N/A | Moved from Test Case 22                                    |
| Test Case 6:    |      |      |     |                                                            |
| Test Case 7:    |      |      | N/A | Moved from Test Case 23                                    |
| Test Case 8:    |      |      | N/A | Moved from Test Case 26<br>For Information Only Test       |
| Test Case 9:    |      |      | N/A | Moved from Test Case 30<br>For POS only (n/a for<br>ATMs)  |
| Test Case 10:   |      |      |     | Moved from Test Case 31<br>For POS only (n/a for<br>ATMs)  |
| * Test Case 11: |      |      |     | For Information Only Test                                  |
| Test Case 12:   |      |      |     |                                                            |
| Test Case 13:   |      |      |     |                                                            |
| Test Case 14:   |      |      |     |                                                            |
| Test Case 15:   |      |      |     |                                                            |
| Test Case 16:   |      |      |     |                                                            |
| Test Case 17:   |      |      |     |                                                            |
| Test Case 18:   |      |      |     | Moved from Test Case 49                                    |
| Test Case 19:   |      |      |     | Moved from Test Case 50                                    |
| Test Case 20:   |      |      |     |                                                            |
| * Test Case 21: |      |      |     | Moved from Test Case 32<br>For devices with Offline<br>PIN |

|                 |  |     |                                                            |
|-----------------|--|-----|------------------------------------------------------------|
| Test Case 22:   |  |     | Moved from Test Case 33<br>For devices with Offline<br>PIN |
| Test Case 23:   |  |     | Moved from Test Case 39                                    |
| Test Case 24:   |  |     | Moved from Test Case 41                                    |
| Test Case 25:   |  | N/A | Moved from Test Case 43                                    |
| * Test Case 26: |  |     | Moved from Test Case 44                                    |
| Test Case 27:   |  |     | Moved from Test Case 45                                    |
| Test Case 28:   |  |     | Moved from Test Case 46                                    |
| Test Case 29:   |  |     | Moved from Test Case 47<br>For Information Only Test       |
| Test Case 30:   |  |     | Moved from Test Case 48                                    |

After completing the compliance report, please sign it in the space provided below and submit it to the Visa regional office.

Print Name: \_\_\_\_\_

Title: \_\_\_\_\_

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

## D.3: ADVT Detailed Test Results Sheet (Optional)

The following table may be used to record detailed results of the outcome of each test case.

- Please place an “X” in the appropriate ‘Approval’ or ‘Decline’ column according to whether the transaction was completed Offline or sent Online for decision. If available, please also provide the Transaction Status Information (TSI) and the Terminal Verification Results (TVR) for each test case in the appropriate column. Additional information may also be provided in the ‘Comments’ section, such as the message displayed on the screen.

| Test Case | Offline |         | Online  |         | TSI | TVR | Comments                                                 |
|-----------|---------|---------|---------|---------|-----|-----|----------------------------------------------------------|
|           | Approve | Decline | Approve | Decline |     |     |                                                          |
| 1a        |         |         |         |         |     |     |                                                          |
| 1b        |         |         |         |         |     |     |                                                          |
| 1c        |         |         |         |         |     |     |                                                          |
| 1d        |         |         |         |         |     |     |                                                          |
| 2.        |         |         |         |         |     |     | For Information Only Test                                |
| 3.        |         |         |         |         |     |     |                                                          |
| 4.        |         |         |         |         |     |     |                                                          |
| 5.        |         |         |         |         |     |     |                                                          |
| 6.        |         |         |         |         |     |     |                                                          |
| 7.        |         |         |         |         |     |     |                                                          |
| 8.        |         |         |         |         |     |     | For POS only (n/a to ATMs)<br>For POS only (n/a to ATMs) |
| 9.        |         |         |         |         |     |     |                                                          |
| 10.       |         |         |         |         |     |     |                                                          |
| 11.       |         |         |         |         |     |     | For Information Only Test                                |
| 12.       |         |         |         |         |     |     |                                                          |
| 13.       |         |         |         |         |     |     |                                                          |
| 14.       |         |         |         |         |     |     |                                                          |
| 15.       |         |         |         |         |     |     |                                                          |
| 16.       |         |         |         |         |     |     |                                                          |
| 17.       |         |         |         |         |     |     |                                                          |
| 18.       |         |         |         |         |     |     |                                                          |
| 19.       |         |         |         |         |     |     |                                                          |
| 20.       |         |         |         |         |     |     |                                                          |
| 21.       |         |         |         |         |     |     | For devices with Offline PIN                             |
| 22.       |         |         |         |         |     |     | For devices with                                         |

| Test Case | Offline |         | Online  |         | TSI | TVR | Comments                  |
|-----------|---------|---------|---------|---------|-----|-----|---------------------------|
|           | Approve | Decline | Approve | Decline |     |     |                           |
|           |         |         |         |         |     |     | Offline PIN               |
| 23.       |         |         |         |         |     |     |                           |
| 24.       |         |         |         |         |     |     |                           |
| 25.       |         |         |         |         |     |     |                           |
| 26.       |         |         |         |         |     |     |                           |
| 27.       |         |         |         |         |     |     |                           |
| 28.       |         |         |         |         |     |     |                           |
| 29.       |         |         |         |         |     |     | For Information Only Test |
| 30.       |         |         |         |         |     |     |                           |

## Appendix E: List of Acronyms

| Acronym | Meaning                                      |
|---------|----------------------------------------------|
| a       | alpha                                        |
| AAC     | Application Authentication Cryptogram        |
| AAR     | Application Authentication Referral          |
| ADA     | Application Default Action                   |
| ADF     | Application Definition File                  |
| ADVT    | Acquirer Device Validation Toolkit           |
| AEF     | Application Elementary File                  |
| AFL     | Application File Locator                     |
| AID     | Application Identifier                       |
| AIP     | Application Interchange Profile              |
| an      | alphanumeric                                 |
| ans     | Alphanumeric special                         |
| APDU    | Application Protocol Data Unit               |
| API     | Application Priority Identifier              |
| ARPC    | Application Response Cryptogram              |
| ARQC    | Application Request Cryptogram               |
| ATC     | Application Transaction Counter              |
| ATM     | Automated Teller Machine                     |
| AUC     | Application Usage Control                    |
| b       | binary                                       |
| BIN     | BASE Identification Number                   |
| CA      | Certificate Authority                        |
| CAM     | Card Authentication Method                   |
| CDOL    | Card Risk Management Data Object List        |
| CID     | Cryptogram Information Data                  |
| cn      | compressed numeric                           |
| CVK     | Card Verification Key                        |
| CVM     | Cardholder Verification Method               |
| CVR     | Card Verification Result                     |
| CVV     | Card Verification Value                      |
| DDA     | Dynamic Data Authentication                  |
| DDF     | Directory Definition File                    |
| DDOL    | Dynamic Data Authentication Data Object List |
| DEA     | Data Encryption Algorithm                    |
| DES     | Data Encryption Standard                     |

|      |                                                            |
|------|------------------------------------------------------------|
| DGI  | Data Group Identifier (used by the Card Personalizer only) |
| DKI  | Derivation Key Index                                       |
| EMV  | Europay, MasterCard & Visa                                 |
| FCI  | File Control Information                                   |
| GPO  | Get Processing Options                                     |
| hex. | Hexadecimal                                                |
| IAC  | Issuer Action Code                                         |
| ICVV | Alternate Card Verification Value                          |
| IFM  | Interface Module                                           |
| MCC  | Merchant Category Code                                     |
| MDK  | Master Derivation Key                                      |
| N/A  | Not Applicable                                             |
| n    | numeric                                                    |
| PAN  | Primary Account Number                                     |
| PDOL | Processing Options Data Options List                       |
| PIN  | Personal Identification Number                             |
| PIX  | Proprietary Application Identifier Extension               |
| PK   | Public Key                                                 |
| PKI  | Certificate Authority Public Key Index                     |
| POS  | Point of Sale                                              |
| PSE  | Payment Systems Environment                                |
| PVK  | PIN Verification Key                                       |
| PVV  | PIN Verification Value                                     |
| RFU  | Reserved For Future Use                                    |
| RID  | Registered Application Provider Identifier                 |
| RSA  | Rivest, Shamir, Adleman                                    |
| SAD  | Signed Static Application Data                             |
| SAM  | Secure Access Method                                       |
| SDA  | Static Data Authentication                                 |
| STIP | Stand In Processing                                        |
| TAC  | Terminal Action Code                                       |
| TC   | Transaction Certificate                                    |
| TDOL | Transaction Certificate Data Object List                   |
| TLV  | Tag-Length-Value                                           |
| TSI  | Transaction Status Information                             |
| TVR  | Terminal Verification Result                               |
| UDK  | Unique Derived Key                                         |
| var. | variable                                                   |
| VCMS | Visa Certification Management System                       |
| VIP  | VisaNet Integrated Payment                                 |
| VLP  | Visa Low-value Payment                                     |
| VMTS | Visa Member Test System                                    |
| VSDC | Visa Smart Debit Credit                                    |
| VTs  | VisaNet Test System                                        |

